G

Command File Overview

This appendix presents an overview of the command file of Sentaurus Device.

Top Level of Command File on page 1708 presents the topmost level of the command file. Use this table to obtain a top-down overview of the command file. The remaining sections are ordered alphabetically. Headings of tables, therein, mostly start with a keyword. Use these tables to find details about keywords.

- Organization of Command File Overview
- Top Level of Command File
- CurrentPlot
- Device
- Electrode
- File
- GainPlot
- HydrogenBoundary
- IFM
- Math
- NoisePlot
- NonLocalPlot
- OpticalDevice
- Physics
- Plot
- RayTraceBC
- Solve
- System

- TensorPlot
- Thermode
- Various

Organization of Command File Overview

The tables in this appendix have two or three columns, and their rows are ordered alphabetically with respect to the first (and, as far as applicable, the middle) column. Placeholders (cross references, user-supplied values in angle brackets (<>)) precede explicit keywords, irrespective of alphabetic order.

The left column contains keywords that can appear in the command file. In the topmost rows of some tables, the left column references another table. This means that all keywords in the referenced table can appear in the referring table as well. Some keywords are followed by an opening parenthesis or an opening brace to indicate that the keyword expects a selection of options listed in the middle column of the current and the following rows.

The middle column contains options or values to the keyword in the left column, one per row. For a selection of options, the last row indicates the closing parenthesis or the closing brace. For some tables, the middle column would be empty and is omitted.

The right column contains the description of keywords, options, and values of the row. As far as applicable, the right column also provides the following additional information:

- The default value, which can be:
  - An explicit value. Those values are written in Courier font, as you would type them.
  - + or – to indicate the default (true or false, respectively) for keywords that support the – prefix.
  - *on* or *off* for keywords that set and reset flags, but do not support the – prefix.
  - An asterisk (*) to indicate the default among mutually exclusive alternatives.
  - An exclamation mark (!) if no default exists and you must specify a value.
- The unit assumed for the given quantity. In unit specifications, d denotes the dimension of the mesh. For dimensionless quantities, no unit is specified.

- The location for which keywords should be specified, given as characters in parentheses:
  - (g) stands for global parameters that must not appear in region-specific, interface-specific, or contact-specific sections.
  - (r) stands for region-specific (or material-specific) parameters that usually do not make sense when specified for interfaces or contacts.
  - (c) and (i) stand for contact-specific or interface-specific parameters.

Furthermore, the tables use the following conventions:

- An ellipsis (…) denotes that the preceding item can be repeated an arbitrary number of times.
- An asterisk (*) followed by an integer, both typeset in Times font, denotes that the preceding item must appear the given number of times.
- Optional components of specifications are enclosed in brackets, and the brackets are *not* part of the syntax.
- Angle brackets (<>) indicate user-supplied values. Table 207 summarizes the common types of specification. More specific notation is explained in the last column of the table where it is used.

Some tables use additional conventions as necessary. They are explained in the text that precedes the respective table.

Table 207 Notation for user-supplied values

Specification	Description
`<(x,y)> <[x,y]>` `<(x,y]> <[x,y)>`	Range of floating-point numbers from x to y. Parentheses are used when the limits are excluded from the range; brackets are used when they are included. To denote ranges unbound on one side, the corresponding limit is omitted.
`<carrier>`	A carrier type, `Electron` or `Hole`.
`<float>`	A floating-point number (integers are special cases thereof).
`<ident>`	An identifier. This is like a string, but is not enclosed in double quotation marks.
`<int>`	An integer.
`<n..m>`	A range of integers, from `n` to `m`, inclusively. Either `n` or `m` can be omitted, to denote ranges that are unbound on one side.

Table 207 Notation for user-supplied values (Continued)

Specification	Description
`<string>`	A sequence of characters (including digits and special characters) included in double quotation marks. Unlike most Sentaurus Device input, strings are case sensitive.
`<vector>`	A sequence of up to three floating-point numbers, enclosed in parentheses, and separated by spaces. For example: `(1.0 3 0)`
`<System_Coord>`	`CrystalSystem` or `SimulationSystem`.

Top Level of Command File

Table 208 lists the top level in the command file of Sentaurus Device.

Table 208 Top level of command file

`Table 212`		Specification for single devices.
`Device`	`<string>{Table 212}`	Device name and device specifications (mixed mode). Device Section on page 119
`OpticalDevice`	`<string>{Table 212}`	Optical device name and device specification (mixed mode and Composite Method on page 756). Device Section on page 119
`Solve{`	`Table 351}`	The problem to be solved.
`System{`	`Table 371}`	The circuit (mixed mode). System Section on page 112

CurrentPlot

Table 209 CurrentPlot{}Tracking Additional Data in the Current File

`<vector>`		µm Print data at coordinate `<vertex>`.
`Average(`	`Table 210)`	Print average over given domain.
`Device`	`=<string>`	Name of the device for which the parameter will be plotted.
`HBPlot`		- Generate time-domain or frequency-domain quantities in harmonic balance `.plt` files. Device Instance Plot Files With CurrentPlot on page 176

Table 209 CurrentPlot{}Tracking Additional Data in the Current File (Continued)

Integrate(	Table 210)	Print integral over given domain.
Material	=<string>	Name of material for which the parameter will be plotted if it has been specified for a particular material.
MaterialInterface	=<string>	Name of material interface for which the parameter will be plotted if it has been specified for a particular material interface.
Maximum(	Table 210)	Print maximum in given domain.
Minimum(	Table 210)	Print minimum in given domain.
Model	=<string>	Select a model for printing of parameters.
ModelParameter	=<string>	Select model parameter for plotting when using unified interface for optical generation computation. Parameter Ramping on page 772
OpticalDevice	=<string>	Name of the optical device for which the parameter will be plotted.
Parameter	=<string>	Print parameter <string>.
Polygon	BottomPlane=<string> TopPlane=<string> Vertices(<vector> <vector> <vector> ...)	Sample over the extruded polygon, defined by bottom and top planes and by the vertices of the polygon.
Region	=<string>	Name of the region for which the parameter will be plotted if it has been specified for a particular region.
RegionInterface	=<string>	Name of the region interface for which the parameter will be plotted if it has been specified for a particular region interface.
Tcl(	Table 211)	Evaluate a Tcl formula. Tcl Formulas on page 188

Table 210 Average(), Integrate(), Maximum(), and Minimum() Tracking Additional Data in the Current File

Table 378		Sample over location.

Table 210 Average(), Integrate(), Maximum(), and Minimum() Tracking Additional Data in the Current File (Continued)

Coordinates		Print coordinates where maximum, minimum, or average occurs.
DopingWell	<vector>	Sample over well, defined by point in the well.
Everywhere		Sample over entire device.
Insulator		Sample over all insulator regions of the device.
Name	=<string>	Name for sampled data to be used in output.
Semiconductor		Sample over all semiconductor regions of the device.
Window[	<vector> <vector>]	Sample over the window, defined by two specified corners.

Table 211 Tcl() Tcl Formulas

<ident>	=<string> | <float>	User-specific model parameters passed to an external Tcl current plot script. Current Plot File on page 1636
Dataset	=<string>	Dataset name that appears in the header section of the current plot file. Dataset Option on page 190
Finish	=<string>	Tcl code executed last for each plot time point. Finish Option on page 191
Formula	=<string>	Tcl code executed on each mesh vertex. Formula Option on page 190
Function	=<string>	Function name that appears in the header section of the current plot file. Function Option on page 190
Init	=<string>	Tcl code executed first for each plot time point. Init Option on page 190
Operation	=<string>	Sets the operation mode, that is, how the results of each vertex are combined. Operation Option on page 192
Tcl	="source <ident>"	Tcl code defined in external file for more complex tasks, where <ident> refers to the Tcl script file. Current Plot File on page 1636
Unit	=<string>	Unit of the current plot quantity. Unit Option on page 190

Feedback

Device

Table 212 Device{}, OpticalDevice{}, or single-device specification Specifying Physical Devices

CarrierPathPlot{	Table 202}	Carrier path output data. Plotting Results Along Carrier Paths
CurrentPlot{	<ident>	Use current plot PMI. Current Plot File on page 1582
	PMIModel (Table 336)	Use current plot PMI with parameters. Current Plot File on page 1582
	Table 202(Table 209)	Plot scalar data to current file. Tracking Additional Data in the Current File on page 183
	Table 203/Vector(Table 209)}	Plot vectorial data to current file. Tracking Additional Data in the Current File on page 183
Electrode{	Table 213}	Electrode specification. Specifying Electrical Boundary Conditions on page 125
eSHEDistributionPlot{	<vector>...}	Electron-energy distribution plot. Visualizing Spherical Harmonics Expansion Method on page 959
Extraction{	<ident> [=] <ident>...}	Process parameters. Extraction File on page 205
File{	Table 214}	Input and output files.
GainPlot{	Table 215}	Parameters for gain plot (LED).
hSHEDistributionPlot{	<vector>...}	Hole-energy distribution plot. Visualizing Spherical Harmonics Expansion Method on page 959
HydrogenBoundary{	Table 216}	Degradation model. Boundary Conditions on page 693
IFM{	Table 217}	Variations, noise models. IFM Section on page 901
Math	[(Table 375)]	Math specification, potentially restricted to a location.
{	Table 218}	

Table 212 Device{}, OpticalDevice{}, or single-device specification Specifying Physical Devices (Continued)

MonteCarlo{	<options>}	See the *Sentaurus™ Device Monte Carlo User Guide*
NoisePlot{	Table 244}	Noise output data. Noise Output Data on page 903
NonLocalPlot	(<vector>...) {Table 245}	Nonlocal plot. Visualizing Data Defined on Nonlocal Meshes on page 228
Physics	[(Table 375)]	Physical models, potentially restricted to a location.
{	Table 246}	
Plot{	Table 347}	Plot data. Device Plots on page 195
RayTraceBC{	Table 350}	Raytrace boundary conditions.
TensorPlot(	Table 373)	Tensor plot for beam propagation method. Visualizing Results on Native Tensor Grid on page 850
{	ComplexRefractiveIndex	
	OpticalField	
	OpticalIntensity}	
Thermode{	Table 374}	Thermode specification. Specifying Thermal Boundary Conditions on page 129
TrappedCarDistrPlot{	<vector>	Trapped carrier charge plot at position <vector>. Visualizing Traps on page 649
	Table 378={<vector>...}}	Trapped carrier charge plot restricted to location.

Electrode

Table 213 Electrode{}

AreaFactor	=<float>	1 Multiplier for electrode current. Reading a Structure on page 59
Barrier	=<float>	V Barrier voltage.

Table 213 Electrode{} (Continued)

Charge	=<float>	C Charge for floating electrode. Voltage must not be specified. Floating Contacts on page 312
	=(<float1> At <float2>[,] ...)	C, s Transient charge behavior. A list of pairs of charges <float1> at time <float2>, piecewise linearly interpolated.
CombineSemiAndContactCharges		off Combined treatment for semiconductor and contact charges for floating electrode. Floating Contact for Combined Semiconductor and Contact Charges on page 316
Current	=<float>	$A\ \mu\ m^{d-3}$ Current boundary condition. Voltage is used as initial guess only.
	=(<float1> At <float2>[,] ...)	$A\ \mu\ m^{d-3}$, s Transient current behavior. A list of pairs of currents <float1> at time <float2>, piecewise linearly interpolated.
CyclicNorm	=<string>	Output of cyclic extrapolation. File stores information about cyclic norm versus number of cyclic cycles and time.
DistResist	=<float>	$\Omega\ cm^2$ Distributed resistance. Resistive Contacts on page 302
	=SchottkyResist	Use Schottky contact resistance model. Resistive Contacts on page 302
EqOhmic		off Contact is Ohmic with a modified boundary condition. Modified Ohmic Contacts on page 291
eRecVelocity	=<[0,)>	2.573e6 $cm\ s^{-1}$ Electron recombination velocity. Schottky Contacts on page 292
Extraction{	bulk	Specify electrode as bulk for extraction purposes. Extraction File on page 205
	drain	Specify electrode as drain for extraction purposes. Extraction File on page 205
	gate	Specify electrode as gate for extraction purposes. Extraction File on page 205
	source}	Specify electrode as source for extraction purposes. Extraction File on page 205

Feedback

Table 213 Electrode{} (Continued)

FGcap=(	value=<float>	$F\,\mu\,m^{d-3}$ Additional capacitance for floating electrode. Floating Metal Contacts on page 313
	name=<string>)	Coupling to electrode <string>. Floating Metal Contacts on page 313
hRecVelocity	=<[0,)>	1.93e6 $cm\,s^{-1}$ Hole recombination velocity. Schottky Contacts on page 292
Material	=<string>	Electrode material. Contacts on Insulators on page 291
	=<string> (N=<(0,)>)	cm^{-3} Electrode material with n-doping. Contacts on Insulators on page 291
	=<string> (P=<(0,)>)	cm^{-3} Electrode material with p-doping. Contacts on Insulators on page 291
Name	=<string>	Name of electrode.
	=Regexp(<string>)	Regular expression for matching structure contacts.
Poisson	=Dirichlet	* Use Dirichlet-like boundary condition for Poisson equation.
	=Neumann	Use homogeneous Neumann boundary condition for Poisson equation at Ohmic contacts.
Resist	=<float>	$\Omega\,\mu\,m^{3-d}$ Contact resistance. Resistive Contacts on page 302
Schottky		off Contact is a Schottky contact. Schottky Contacts on page 292
SRDoping		Include the image force lowering of the potential energy barrier at Schottky contact. Resistive Contacts on page 302
Voltage	=<float>	V Contact voltage.
	=(<float1> At <float2>[,] ...)	V, s Transient voltage behavior. A list of pairs of voltages <float1> at time <float2>, piecewise linearly interpolated.

Feedback

Table 213 Electrode{} (Continued)

Workfunction	=<float>	eV Electrode workfunction. Contacts on Insulators on page 291

See Specifying Electrical Boundary Conditions

File

In the description of Table 214, it is indicated whether a file is input or output, as well as the extensions (in Courier font) that Sentaurus Device appends to the user-supplied name to obtain the full name.

The tags enclosed in at-signs (@) denote the default Sentaurus Workbench variables for the particular file name. (These defaults can be changed in tooldb.tcl, see *Sentaurus™ Workbench User Guide*, Global Configuration Files. (g) denotes keywords that must be used in global File sections only, while (d) denotes keywords that must be used in device-specific File sections only.

Table 214 File{}

ACExtract	=<string>	(g) Small-signal and noise analysis (output, _ac_des.plt, @acplot@). Small-Signal AC Analysis on page 162
Bandstructure	=<string>	(d) Base name for plotting local band structure data in an LED simulation (output, _kpbandstruc_vertexX_des.plt, _kpeigenfunc_vertexX_des.plt).
CarrierPath	=<string>	Carrier path and spatially interpolated plot variables (output .tdr). Save Carrier Paths as 1D Datasets in Separate TDR File on page 585
CMIPath	=<string>	(g) Search path for compact circuit files (extension .ccf) and the corresponding shared object files (extension .so.arch). The files are parsed and added to the System section of the command file. System Section on page 112
Current	=<string>	Device currents, voltages, charges, temperatures, and times (output, _des.plt, @plot@). Current File on page 180
DephasingRates	=<string>	Directory for saving dephasing rates using the second Born approximation.

Feedback

Table 214 File{} (Continued)

DevFields	=<string>	(d) Space distribution for trapped carrier charge density. It must match the grid file (input .tdr). Energetic and Spatial Distribution of Traps on page 610
	=<string>	(d) Space distribution for material resistivity. It must match the grid file (input .tdr). Transport in Metals on page 329
DevicePath	=<string>	(g) Load all files with the extension .device in the directory path <string>. The directory path has the format dir1:dir2:dir3. The devices found can be used in the System section. They are not overwritten by a definition with the same name in the command file. System Section on page 112
EmissionTable	=<string>	Tabulated emission data (output).
eSHEDistribution	=<string>	Electron distribution versus kinetic energy (output, _des.plt). Visualizing Spherical Harmonics Expansion Method on page 959
Extraction	=<string>	(d) Extraction file (output, extraction_des.xtr). Extraction File on page 205
Gain	=<string>	(d) Modal stimulated and spontaneous emission spectra (output, _gain_des.plt).
Grid	=<string>	(d) Device geometry and mesh (input, .tdr, or @tdr@).
hSHEDistribution	=<string>	Hole distribution versus kinetic energy (output, _des.plt). Visualizing Spherical Harmonics Expansion Method on page 959
IlluminationSpectrum	=<string>	Illumination spectrum. Illumination Spectrum on page 733
IonIntegral	=<string>	Breakdown path and spatially interpolated simulation results along the breakdown path (output, _des_ABA.tdr). Visualizing Breakdown Paths on page 574
LifeTime	=<string>	(d) Lifetime profiles (input, .tdr). Lifetime Profiles From Files on page 528
Load	=<string>	Old simulation results (input, .sav). Save and Load Statements on page 238
MesherInput	=<string>	Base name of boundary and command file to be used for calling Sentaurus Mesh. Specifying the Optical Solver on page 751

Feedback

Table 214 File{} (Continued)

MIMBand	=<string>	Metal–insulator–metal (MIM) band diagram (output, _mimband_des.tdr). Band Diagram With Traps on page 1268
MIMCurrent	=<string>	MIM steady-state currents (output, _mimcur_des.plt). Special MIM Current File (Currents and Capacitance) on page 1267
MIMSensitivity	=<string>	MIM sensitivity analysis (output, _<volt>_mimsa_des.plt). Sensitivity Analysis on page 1251
MobilityDoping	=<string>	(d) File from which donor and acceptor concentrations for mobility calculations are read. Mobility Doping File on page 517
NewtonPlot	=<string>	Convergence monitoring (output). NewtonPlot on page 235
NonLocalPlot	=<string>	Data defined on nonlocal line meshes (output). Visualizing Data Defined on Nonlocal Meshes on page 228
OpticalGenerationFile	=<string>	Load optical generation from a file. Optical AC Analysis on page 857
OpticalGenerationInput	=<string>	File from which optical generation rate is loaded. Loading and Saving Optical Generation From and to Files on page 738
OpticalGenerationOutput	=<string>	File to which optical generation rate is written. Loading and Saving Optical Generation From and to Files on page 738
OpticalGenerationOutput (collected)	=<string>	File to which optical generation rate is written. Consecutive plots are written into one file instead of several enumerated files. Loading and Saving Optical Generation From and to Files on page 738
OpticalSolverInput	=<string>	Command file of optical solver. Specifying the Optical Solver on page 751
OpticsOutput	=<string>	(g) Runtime log for optical solvers (output, _des.log, @optlog@).
Output	=<string>	"output" (g) Runtime log (output, _des.log, @log@).
ParameterPath	=<string>	(d) List of subdirectories in the $STROOT/tcad/$STRELEASE/lib/sdevice/MaterialDB directory that are added to the search path for parameter files. Physical Model Parameters on page 76

Feedback

Table 214 File{} (Continued)

Parameters	=<string>	(d) Device parameters (input, .par, @parameter@). Physical Model Parameters on page 76
Piezo	=<string>	(d) Stress data for piezoelectric model (input). Chapter 31, Mechanical Stress on page 1040
Plot	=<string>	Spatially distributed simulation results (output, _des.tdr, @tdrdat@). Device Plots on page 195
Plot(collected)	=<string>	Spatially distributed simulation results (output, _des.tdr, @tdrdat@). Consecutive plots are written into one file instead of several enumerated files. Device Plots on page 195
PMIPath	=<string>	(g) Search path to load shared object files (extension .so.arch) for PMI models. Command File of Sentaurus Device on page 1373
PMIUserFields	=<string>	(d) File containing any of the fields PMIUserField0 to PMIUserField299.
SpectralPlot	=<string>	Spatially distributed simulation results (output, _des.tdr, @spectralplot@) for each entry of the illumination spectrum as well as spectral curves (_des.plt) for the results of the optical solver. Consecutive plots are written into one file instead of several enumerated files, that is, SpectralPlot(collected) is the default. Illumination Spectrum on page 733
SpectralPlot(-collected)	=<string>	Spatially distributed simulation results (output, _des.tdr, @spectralplot@) for each entry of the illumination spectrum as well as spectral curves (_des.plt) for the results of the optical solver. Consecutive plots are written into several enumerated files. Illumination Spectrum on page 733
TensorPlot	=<string>	Base name for saving tensor plot files when using beam propagation method. Visualizing Results on Native Tensor Grid on page 850
Save	=<string>	Simulation results for retrieval with Load (output, _des.sav). Save and Load Statements on page 238
SaveOptField	=<string>	Base name for saving optical field data in an LED simulation.

Table 214 File{} (Continued)

SPICEPath	=<string>	Search path for SPICE circuit files (extension .scf). The files are parsed and added to the System section of the command file. The directory $STROOT/tcad/$STRELEASE/lib/sdevice/spice is automatically added to SPICEPath. File Section on page 120
TrappedCarPlotFile	=<string>	Trapped carrier charge density, trap occupancy probability, and trap density versus energy (output, _des.plt). Visualizing Traps on page 649

GainPlot

Table 215 GainPlot{}

Intervals	=<int>	Number of points to plot for each gain curve.
Range	=(<float>*2)	eV Energy range of the gain plot.
	=Auto	Automatically find the energy range.

HydrogenBoundary

Table 216 HydrogenBoundary{} Boundary Conditions

HydrogenAtom	=<float>	cm^{-3} Boundary concentration.
	=reflective	Homogeneous Neumann boundary condition
HydrogenIon	=<float>	cm^{-3} Boundary concentration.
	=reflective	Homogeneous Neumann boundary condition.
HydrogenMolecule	=<float>	cm^{-3} Boundary concentration.
	=reflective	Homogeneous Neumann boundary condition.
HydrogenSpeciesA	=<float>	cm^{-3} Boundary concentration.
	=reflective	Homogeneous Neumann boundary condition.

Feedback

Table 216 HydrogenBoundary{} Boundary Conditions (Continued)

HydrogenSpeciesB	=<float>	cm^{-3} Boundary concentration.
	=reflective	Homogeneous Neumann boundary condition.
HydrogenSpeciesC	=<float>	cm^{-3} Boundary concentration.
	=reflective	Homogeneous Neumann boundary condition.
Name	=<string>	Name of the hydrogen contact.

IFM

Table 217 IFM{} IFM Section

DeterministicVariation(	Table 301)	(g) Deterministic variations. Deterministic Doping Variations on page 897
Noise	[<string>] (Table 331)	(r) Noise sources. Noise Sources on page 875
RandomizedVariation	<string> (Table 337)	(g) Set sIFM models. Statistical Impedance Field Method on page 885

Math

Table 218 Math{}

Table 242		Transient time-step control.
AcceptNewtonParameter(	Table 219)	(g) Relaxed Newton method. Relaxed Newton Method on page 223
ACMethod	=Blocked	* Use block decomposition solver for ACCoupled.
ACSubMethod	=Table 231	Inner linear solver for Blocked method for ACCoupled.
AllowLayerThickness	=Everywhere	(g) Account for LayerThickness in all materials. Extracting Layer Thickness on page 422

Feedback

Table 218 Math{} (Continued)

	=Insulator	(g) Account for LayerThickness in insulators only.
	=Metal	(g) Account for LayerThickness in metals only.
	=Semiconductor	* (g) Account for LayerThickness in semiconductors only.
AnisoGradQF_formula	=0 | 1	0 Select formula for driving force. Driving Forces on page 991
AnisoSG		off (g) Use anisotropic Scharfetter–Gummel approximation for anisotropic models. Chapter 28, Anisotropic Properties on page 978
AnisoSG_DerivativeMinDen	=<float>	0.0 cm^{-3} (g) Lower limit of carrier densities for AnisoSG approximation. Corresponding derivatives are switched off if the node concentration is less than this value. Chapter 28, Anisotropic Properties on page 978
AnisoSG_MinDen	=<float>	10 cm^{-3} (g) Stabilization parameter for computation of x^* for AnisoSG approximation. Chapter 28, Anisotropic Properties on page 978
AutoCNPMinStepFactor		2.0 (g) Multiplier of MinStep for the automatic activation of CNormPrint. Automatic Activation of CNormPrint and NewtonPlot on page 235
AutomaticCircuitContact		on Poisson covers the circuit. Additional Equations Available in Mixed Mode on page 220
AutoNPMinStepFactor		2.0 (g) Multiplier of MinStep for the automatic activation of NewtonPlot. Automatic Activation of CNormPrint and NewtonPlot on page 235

Feedback

Table 218 Math{} (Continued)

AutoOrientation	=(<int>...)	Miller indices for surface orientations supported by AutoOrientation. Changing Orientations Used With Auto-Orientation on page 96
AutoOrientationSmoothing Distance	=<float>	0.0 µm (r) Auto-orientation smoothing distance. Auto-Orientation Smoothing on page 96
AvalDensGradQF		- Alternative driving force. Using Avalanche Generation on page 550
AvalDerivatives		+ Compute analytic derivatives of avalanche generation. Derivatives on page 218
AvalFlatElementExclusion	=<float>	0.0 degree (g) Exclude elements from contributing to avalanche generation. Using Avalanche Generation on page 550
AvalPostProcessing		off Impact ionization–generated carriers are not included self-consistently in the solution. Approximate Breakdown Analysis With Carriers on page 576
AverageAniso		(g) Use average-aniso approximation for anisotropic models. Chapter 28, Anisotropic Properties on page 978
AverageBoxMethod		+ Use element-oriented element intersection box method algorithm. If deactivated, use quadrilateral box method algorithm. Box Method Coefficients in the 3D Case on page 1309
AverageVolumeAndSurface		- Use element-face distribution algorithm with averaging correction for all obtuse elements. Element Intersection Box Method Algorithm on page 1313
BehavSrcNumericDeriv	=<float>	0 Compute the numeric Jacobian.

Feedback

Table 218 Math{} (Continued)

BM_ExtendedPrecision		+ Use “long double” precision to compute box method coefficients and control volumes. Extended Precision on page 1308
BM_StableCalculation		- Use modified algorithms to compute box method coefficients and control volumes. This feature is obsolete and will be removed in future releases.
BM_VoronoiRISurfaces3D		- Use improved discretization of region interfaces for distributed resistance simulations. Resistive Interfaces on page 309
BoxCoefficientsFromFile	[(GrdNumbering)]	Try to read sections of the geometry file. Saving and Restoring Box Method Coefficients on page 1320
BoxMeasureFromFile	[(GrdNumbering)]	Try to read sections of the geometry file. Saving and Restoring Box Method Coefficients on page 1320
BoxMethodFromFile		+ Read Voronoï surface from this file if the grid file has a VoronoiFaces section. Box Method Coefficients in the 3D Case on page 1309
BreakAtIonIntegral	(<int> <float>)	1 1 Terminate the quasistationary simulation when the <int> largest ionization integrals are greater than <float>. Approximate Breakdown Analysis on page 570
BreakAtOBA		off (g) Quasistationary ramp by OBA criteria. Using Off-State Breakdown Analysis on page 568
BreakCriteria{	Table 220}	Break criteria. Break Criteria: Conditionally Stopping the Simulation on page 130

Feedback

Table 218 Math{} (Continued)

BroadeningIntegration (GaussianQuadrature(Order	=<int>))	Use Gaussian quadrature to integrate the broadening spectrum of an LED simulation. Accelerating Gain Calculations and LED Simulations on page 1163
CarrierDensityInAltermatt		+ Use carrier density in the Altermatt ionization model. Physical Model Parameters on page 384
CDensityMin		0.0 A cm^{-2} (r) Current limit for parallel electric-field computation. Electric Field Parallel to the Interface on page 500
CheckRhsAfterUpdate		– Check whether the RHS can be reduced further after the update error has converged. Convergence and Error Control on page 215
CheckUndefinedModels		+ (g) Check for undefined physical parameters. Undefined Physical Models on page 91
ClipVolumeAndSurface		+ Use element-face intersection RobustBoxMethod algorithm with clipping correction for all obtuse elements. Element Intersection Box Method Algorithm on page 1313
CNormPrint		(g) Convergence monitoring. CNormPrint on page 234
CompatibleFEPolarization		Previous formulation of Ginzburg–Landau–Khalatnikov equation. Multidimensional on page 1017
ComputeDopingConcentration		– (g) Recompute net doping based on individual doping species. Specifying Doping Species on page 63

Feedback

Table 218 Math{} (Continued)

ComputeGradQuasiFermiAt Contacts	=UseElectrostatic Potential	* Use the electrostatic potential to compute the GradQuasiFermi driving force within elements touching a contact. Equation 378 on page 499
	=UseQuasiFermi	Use the quasi-Fermi potential to compute the GradQuasiFermi driving force within elements touching a contact. Equation 377 on page 499
ComputeIonizationIntegrals		off Compute ionization integrals using old discretization scheme. Approximate Breakdown Analysis on page 570
	(Table 222)	
ConstRefPot	=<float>	eV Value for φ_{ref}. Quasi-Fermi Potential With Boltzmann Statistics on page 258
ContactChargeIsWellCharge		+ (g) Compute the contact charge as charge in the contact's well. Electrode Charge Calculation
CoordinateSystem{	Table 1}	(g) Coordinate system of explicit coordinates in command file. Reading a Structure on page 59
cT_Range	=(<float*2)	10 80000 K (g) Lower and upper limit for carrier temperature. Numeric Parameters for Temperature Equations on page 287
CurrentPlot(	Digits=<float>	6 Number of digits in the names of quantities in the current plot file. CurrentPlot Options on page 187
	IntegrationUnit= <string>)	um Length unit for current plot integration. CurrentPlot Options on page 187
CurrentWeighting		off Compute contact currents using an optimal weighting scheme. Numeric Approaches for Contact Current Computation on page 271

Table 218 Math{} (Continued)

Cylindrical	(<float>)	off Use the 2D mesh to simulate a 3D cylindrical device. The device is assumed to be rotationally symmetric around the vertical axis given by *x* = <float> µm. <float> must be less than or equal to the smallest horizontal device coordinate, and defaults to 0. Reading a Structure on page 59
	(xAxis=<float>)	This is same as (<float>).
	(yAxis=<float>)	The device is assumed to be rotationally symmetric around the horizontal axis given by *y* = <float> µm. <float> must be less than or equal to the smallest vertical device coordinate. For UCS coordinates, the x-axis is the vertical axis, so you must specify the cylindrical axis using yAxis.
DeltaVolumeLimit	=<float>	1e-4 Limit for DeltaVolume. Region Non-Delaunay Elements on page 1322
DensityIntegral	(<int>)	30 (g) Defines a number of Gauss–Laguerre quadrature integration points. Using Multivalley Band Structure on page 368
DensLowLimit	=<float>	1e-100 cm^{-3} (g) Lower limit for carrier densities. Numeric Parameters for Current Densities on page 270
DensLowLimitPower	=<int>	-100 (g) Lower limit power for carrier densities ($10^{DensLowLimitPower}$ cm^{-3}). Numeric Parameters for Current Densities on page 270
Derivatives		+ Compute analytic derivatives of mobility and avalanche generation. Derivatives on page 218

Feedback

Table 218 Math{} (Continued)

Digits	=<float>	5 Approximate number of digits to which equations must be solved to be considered as converged. Convergence and Error Control on page 215
DirectCurrent		off Compute contact currents directly, using only contact nodes and their neighbors. Numeric Approaches for Contact Current Computation on page 271
DirectQuantumCorrection		off Use the quantum potential directly as the band edge. Notes on the Use of the Density Gradient Model on page 410
DualGridInterpolation	(Method=Conservative)	Use conservative element-based interpolation for dual-grid simulations. Dual-Grid Setup for Raytracing on page 817, Single-Grid Versus Dual-Grid LED Simulation on page 1155
	(Method=Simple)	Use bilinear vertex-based interpolation for dual-grid simulations. Dual-Grid Setup for Raytracing on page 817, Single-Grid Versus Dual-Grid LED Simulation on page 1155
eB2BGenWithinSelectedRegions		+ (g) Restrict nonlocal path band-to-band electron generation to regions where model is active. Dynamic Nonlocal Path Band-to-Band Tunneling Model on page 592
eDrForceRefDens	=<[0,)>	0 cm^{-3} (r) Damping parameter for high-field mobility driving force, alias for RefDens_eGradQuasiFermi_Zero. Interpolation of Driving Forces to Zero Field on page 502
ElementEdgeCurrent		– (g) Use an alternative element-edge approximation of the current density compared to the default edge approximation.

Feedback

Table 218 Math{} (Continued)

ElementVolumeAvalanche		– (g) Use truncated element-vertex volumes for avalanche generation. Using Avalanche Generation on page 550
eMobilityAveraging	=Element	* (r) Use element averaged electron mobility. Mobility Averaging on page 517
	=ElementEdge	(r) Use element-edge averaged electron mobility. Mobility Averaging on page 517
EnormalInterface(	MaterialInterface= [<string>...])	Material interfaces for normal electric field computation. Normal to the Interface on page 474
	RegionInterface= [<string>...])	Region interfaces for normal electric field computation. Normal to the Interface on page 474
EparallelToInterface(		Options for EparallelToInterface driving force. Electric Field Parallel to the Interface on page 500
	Direction=<vector>	! Direction of the current.
	Box=(<vector> <vector>))	Restrict Direction vector to vertices contained in box.
eQPBox(	QBoxForBC	Apply the box to select boundary condition on interfaces.
	Table 223)	Quantum box for eQuantumPotential. Density Gradient Model on page 404
EquilibriumSolution(	Table 224)	(g) Parameters for equilibrium solution used in specific models. Equilibrium Solution on page 257
Error	(Table 356)=<float>	Value of ε_A. Convergence and Error Control on page 215
ErrRef	(Table 356)=<float>	Value of x_{ref}. Convergence and Error Control on page 215
ExcludeTouchingContactParts		(g) off Allow simulation for device geometry with touching contacts.

Table 218 Math{} (Continued)

ExitOnFailure		(g) off Terminate the simulation as soon as a Solve command fails.
ExitOnUnknownParameterRegion		+ (g) Exit when unknown region, region interface, or electrode appears in parameter file.
-ExitOnUnknownParameterRegion		off (g) Print warning message when unknown region, region interface, or electrode appears in parameter file.
ExtendedPrecision	[(Table 28)]	(g) off Use extended precision floating-point arithmetic. Extended Precision on page 249
Extrapolate	(Table 225)	– Use extrapolation to obtain an initial guess for the next solution based on the previous solutions. Extrapolation on page 143, Extrapolation on page 155
FDW_SurfCondFactor	=<float>	1 Surface conductance factor in FDW model. Alternative Initial Guess for Quasi-Fermi Potentials From Linear Conduction Equation on page 262
FEPolarizationIP		Divergence-free polarization constraint. Using the Ginzburg–Landau–Khalatnikov Equation on page 1011
FEPolarizationIP	=<float>	0 Numeric parameter for imposing divergence-free polarization constraint. Chapter 29, Ferroelectric Materials on page 1003
FEPolarizationSolver2		off Use the improved solver for FEPolarization equation. Solver II on page 1017
FermiBandTailStatistics		+ Use Fermi–Dirac statistics for band tail density. Band Tails on page 361
GeometricDistances		+ (g) Use enhanced distance and normal to interface computation for mobility and MLDA. Normal to the Interface on page 474

Feedback

Table 218 Math{} (Continued)

GrainFieldGenerator	(Table 226)	(r) Generate grain field. Creating the Grain Field on page 1234
HarmonicBalance	(Table 227)	(g) Harmonic Balance on page 167
HB	{Table 228}	(g) Harmonic Balance on page 167
HBPlotFilePerNewton		- (g) Plot harmonic balance .plt files per converged solve. Circuit Currents and Voltages on page 175
hDrForceRefDens	=<[0,)>	0 cm^{-3} (r) Damping parameter for high-field mobility driving force, alias for RefDens_hGradQuasiFermi_Zero. Interpolation of Driving Forces to Zero Field on page 502
hMobilityAveraging	=Element	(r) Use element averaged hole mobility. Mobility Averaging on page 517
	=ElementEdge	* (r) Use element-edge averaged hole mobility. Mobility Averaging on page 517
hQPBox(	QBoxForBC	Apply the box to select boundary condition on interfaces.
	Table 229)	Quantum box for hQuantumPotential. Density Gradient Model on page 404
IgnoreTdrUnits		off (g) Ignore TDR units when loading data from a .tdr file. Reading a Structure on page 59
ILSrc	=<string>	ILS options. Linear Solvers on page 225
ImplicitACSystem		- (g) Construct implicit AC system. AC Analysis in Single-Device Mode on page 165
IncompleteNewton		– (g) Incomplete Newton. Incomplete Newton Algorithm on page 219

Feedback

Table 218 Math{} (Continued)

(	RhsFactor=<float>	10 Maximum change in RHS to allow old Jacobian to be reused.
	UpdateFactor=<float>)	0.1 Maximum change in update to allow old Jacobian to be reused.
Interrupt	=BreakRequest	Write .tdr or .sav file and terminate actual solve statement after signal INT occurs. Snapshots on page 198
	=PlotRequest	Write .tdr or .sav file and continue simulation after signal INT occurs. Snapshots on page 198
Iterations	=<0..>	50 Maximum number of Newton iterations. Convergence and Error Control on page 215
KMC	(Table 230)	KMC metal–insulator–metal (KMC-MIM) transport simulation. Chapter 36, Kinetic Monte Carlo MIM Transport on page 1222
LineSearchDamping	=<(0,1]>	1 Smallest allowed damping coefficient for line search damping. Damped Newton Iterations on page 218
lT_Range	=(<float*2)	50 5000 K (g) Lower and upper limit for lattice temperature. Numeric Parameters for Temperature Equations on page 287
MetalConductivity		+ Conductivity of metals. The thermal conductivity is simulated in any case. Transport in Metals on page 329
Method	=Table 231	Linear solver for Coupled.
	=Blocked	* Use block decomposition solver for Coupled. Linear Solvers on page 225
Method	=Bitlis	Use equation decomposition solver for electrostatic problems. Bitlis Solver on page 1018

Table 218 Math{} (Continued)

(	Iterations=<int>	200 Number of iterations between successive restarts.
	Restart=<int>	100 Maximum number of iterations.
	Tolerance=<float>)	1e-8 Convergence criterion.
MGPreconditioner		Set parameters to control multi-grid algorithm. Bitlis Solver on page 1018.
(	MaxCycles=<int>	Controls overall convergence of multi-grid algorithm.
	MaxIterations=<int>	Maximum number of smoother sweeps.
	MaxLevels=<int>	Maximum number of interpolation levels used to build multi-grid hierarchy.
	MinSize=<int>	Minimum size of the coarsest grid.
	RelaxationError=<float>	Controls smoothness of the error.
	SolutionError=<float>)	Controls overall convergence of multi-grid algorithm.
MixAverageBoxMethod		– Use mix average box method algorithm. Box Method Coefficients in the 3D Case on page 1309
MixedModeContactResistance	=<float>	Set a mixed-mode resistance to the circuit node. Ohmic Contacts on page 289
MLDAbox({	Table 233}...)	A box within which the interface is confined for the calculation of the MLDA distance function. Modified Local-Density Approximation Model on page 413
MLDAMinDistanceToContact	=<float>	75e-8 µm (g) Minimum distance to contact where MLDA model is switched off. Modified Local-Density Approximation Model on page 413

Feedback

Table 218 Math{} (Continued)

MobEnormalSignDependence		– (r) Mobility degradation only for carriers that are pulled to the interface by the electric field. Using Mobility Degradation at Interfaces on page 448
MobEnormalUniboRegularization Field	=<float>	0 V/cm (r) Regularization field to improve convergence. University of Bologna Surface Mobility Model on page 463
MVMLDAcontrols(	Table 234)	Multivalley MLDA controlling options. Using the MLDA Model on page 417
NaturalBoxMethod		– Use edge-oriented element intersection BM algorithm. Box Method Coefficients in the 3D Case on page 1309
NewHBFileNames		+ Use new output format. Circuit Currents and Voltages on page 175
NewtonPlot		(g) Convergence monitoring. NewtonPlot on page 235
(	Error	Write the error of all solution variables.
	MinError	Write file only if error decreases.
	NewtonPlotStep=<float>	Upper limit for step size for which to write files.
	Plot	Write everything specified in the Plot section. Device Plots on page 195
	Residual	Write the residuals (right-hand sides) of all equations.
	Update)	Write the updates of all solution variables from the previous step.
NoAutomaticCircuitContact		off (g) Poisson excludes the circuit. Additional Equations Available in Mixed Mode on page 220

Table 218 Math{} (Continued)

NonLocal	(Table 235)	(cir) Nonlocal mesh. Nonlocal Meshes on page 227
	<string> (Table 235)	(g) Named nonlocal mesh.
NonLocalLengthLimit	=<float>	1e-4 cm (g) Limit for nonlocal line length. Nonlocal Meshes on page 227
NonLocalPath	(Table 236)	Dynamic Nonlocal Path Trap-Assisted Tunneling on page 535, Handling Derivatives on page 599
NormalFieldCorrection	=<float>	(r) Normal field correction factor for mobility on interface points. Field Correction on Interface on page 476
NoSRHperPotential		off Omit potential derivatives of SRH recombination rate. Using Field Enhancement on page 531
NoSRHperT		off Omit temperature derivatives of SRH recombination rate. Using Field Enhancement on page 531
NotDamped	=<0..>	1000 (g) Number of iterations in each Newton iteration before Bank–Rose damping is activated. Damped Newton Iterations on page 218
NumberOfAssemblyThreads	=<int>	1 (g) Number of threads for assembly. Parallelization on page 247
NumberOfSolverThreads	=<int>	1 (g) Number of threads for linear solver. Parallelization on page 247
NumberOfThreads	=<int>	1 (g) Number of threads for linear solver and assembly. Parallelization on page 247
Numerically	[(Table 356...)]	off (g) Use numeric derivatives. The optional list restricts the numeric computation to specific Jacobian blocks. Derivatives on page 218

Feedback

Table 218 Math{} (Continued)

OBA_LeakageFactor	=<float>	1.0 (g) Fitting factor for leakage currents. Using Off-State Breakdown Analysis on page 568
OBA_InterfaceFactor	=<float>	0.0 (g) Fitting factor to account for interface effect. Using Off-State Breakdown Analysis on page 568
ParallelLicense	(Table 27)	(g) Determine the behavior if insufficient parallel licenses are available.
ParallelToInterfaceInBoundary Layer		+ (r) Controls the computation of driving forces for mobility and avalanche models along interfaces. Field Correction Close to Interfaces on page 504
(	ExternalBoundary	+ Include the external boundary in the interface for which this feature applies.
	ExternalXPlane	+ Include external x-planes in the interface for which this feature implies.
	ExternalYPlane	+ Include external y-planes in the interface for which this feature implies.
	ExternalZPlane	+ Include external z-planes in the interface for which this feature implies.
	FullLayer	off Apply switch to all elements that touch an interface either by a face, an edge, or a vertex.
	Interface	+ Include semiconductor–insulator region interfaces in the interface for which this feature applies.
	PartialLayer)	on Apply switch only to elements that touch an interface by an edge (2D) or a face (in 3D).
ParameterInheritance	=Flatten	* Region parameters inherit materialwise parameter settings. Combining Parameter Specifications on page 85

Table 218 Math{} (Continued)

	=None	Region parameter settings cause materialwise settings to be ignored.
PardisoSolver <name> (	Table 237	(g) Named section for pardiso parameters.
PeriodicBC ((		(g) Periodic boundary conditions (PBCs). Periodic Boundary Conditions on page 318
	Table 356	Equation for which to apply PBCs. If omitted, apply to all equations (RPBC only).
	Coordinates=(<float>*2)	µm Coordinate of Direction axis where the PBCs will be applied. Sentaurus Device replaces coordinates outside the device with coordinates at the outer boundary (RPBC only).
	Direction=<0..2>	Direction of periodicity: 0 for x-axis, 1 for y-axis, and 2 for z-axis.
	Factor=<float>	1e8 Tuning parameter α (RPBC only).
	MortarSide=<side>	XMin | YMin | ZMin Mortar side with <side> one of XMin, XMax, YMin, YMax, ZMin, or ZMax (MPBC only).
	Type=<type>)...)	RPBC Select PBC mode where <type> is one of RPBC or MPBC.
PlotExplicit		– (g) All Plot statements write as specified in Plot section only.
PlotLoadable		+ (g) All Plot statements write additional information required to load a simulation from a .tdr file.
PostProcess	(Transient=<ident> [(Table 380)])	(g) Use PMI <ident> to postprocess data. Postprocessing for Transient Simulations on page 1586

Feedback

Table 218 Math{} (Continued)

PrintLinearSolver		(g) Print additional information regarding the linear solver being used. Linear Solvers on page 225
PrintRhsPerEquation		(g) Convergence monitoring. PrintRhsPerEquation on page 234.
RandomizedVariation	<string> (Table 241)	(g) Use statistical impedance field method. Statistical Impedance Field Method on page 885
RecBoxIntegr	(<float> <int> <int>)	(1e-2 10 1000) Maximum relative deviation of covered volume, maximum number of levels, maximum number of total rectangles per box.
RecomputeQFP		Keep density variables constant, and recompute quasi-Fermi potentials when the electrostatic potential changes and carrier equations are not solved. Introduction to Carrier Transport Models on page 266
RefDens_eEparallel_ ElectricField	=<float>	0 cm^{-3} (r) Damping parameter for electron high-field mobility and avalanche driving forces. Interpolation of the Eparallel Driving Force on page 504 and Interpolation of Avalanche Driving Forces on page 565
RefDens_eEparallel_ ElectricField_Aval	=<float>	0 cm^{-3} (r) Damping parameter for electron avalanche driving force. Interpolation of Avalanche Driving Forces on page 565
RefDens_eEparallel_ ElectricField_HFS	=<float>	0 cm^{-3} (r) Damping parameter for the electron high-field mobility driving force. Interpolation of the Eparallel Driving Force on page 504
RefDens_eGradQuasiFermi_ ElectricField	=<float>	0 cm^{-3} (r) Damping parameter for electron high-field mobility and avalanche driving forces. Interpolation of the GradQuasiFermi Driving Force on page 502

Feedback

Table 218 Math{} (Continued)

RefDens_eGradQuasiFermi_ ElectricField_HFS	=<float>	0 cm^{-3} (r) Damping parameter for electron high-field mobility driving force. Interpolation of the GradQuasiFermi Driving Force on page 502
RefDens_eGradQuasiFermi_ EparallelToInterface_HFS	=<float>	0 cm^{-3} (r) Damping parameter for electron high-field mobility driving force. Interpolation of the GradQuasiFermi Driving Force on page 502
RefDens_eGradQuasiFermi_Zero	=<float>	0 cm^{-3} (r) Damping parameter for electron high-field mobility driving force, alias for eDrForceRefDens. Interpolation of Driving Forces to Zero Field on page 502
RefDens_Eparallel_ ElectricField	=<float>	0 cm^{-3} (r) Damping parameter for electron and hole high-field mobility and avalanche driving forces. Interpolation of the Eparallel Driving Force on page 504 and Interpolation of Avalanche Driving Forces on page 565
RefDens_Eparallel_ ElectricField_Aval	=<float>	0 cm^{-3} (r) Damping parameter for electron and hole avalanche driving forces. Interpolation of Avalanche Driving Forces on page 565
RefDens_Eparallel_ ElectricField_HFS	=<float>	0 cm^{-3} (r) Damping parameter for the electron and hole high-field mobility driving forces. Interpolation of the Eparallel Driving Force on page 504
RefDens_GradQuasiFermi_ ElectricField	=<float>	0 cm^{-3} (r) Damping parameter for electron and hole high-field mobility and avalanche driving forces. Interpolation of the GradQuasiFermi Driving Force on page 502

Table 218 Math{} (Continued)

RefDens_GradQuasiFermi_ ElectricField_HFS	=<float>	0 cm^{-3} (r) Damping parameter for electron and hole high-field mobility driving forces. Interpolation of the GradQuasiFermi Driving Force on page 502
RefDens_GradQuasiFermi_ EparallelToInterface_HFS	=<float>	0 cm^{-3} (r) Damping parameter for electron and hole high-field mobility driving forces. Interpolation of the GradQuasiFermi Driving Force on page 502
RefDens_GradQuasiFermi_Zero	=<float>	0 cm^{-3} (r) Damping parameter for electron and hole high-field mobility driving forces, alias for DrForceRefDens. Interpolation of Driving Forces to Zero Field on page 502
RefDens_hEparallel_ ElectricField	=<float>	0 cm^{-3} (r) Damping parameter for hole high-field mobility and avalanche driving forces. Interpolation of the Eparallel Driving Force on page 504 and Interpolation of Avalanche Driving Forces on page 565
RefDens_hEparallel_ ElectricField_Aval	=<float>	0 cm^{-3} (r) Damping parameter for hole avalanche driving force. Interpolation of Avalanche Driving Forces on page 565
RefDens_hEparallel_ ElectricField_HFS	=<float>	0 cm^{-3} (r) Damping parameter for the hole high-field mobility driving force. Interpolation of the Eparallel Driving Force on page 504
RefDens_hGradQuasiFermi_ ElectricField	=<float>	0 cm^{-3} (r) Damping parameter for hole high-field mobility and avalanche driving forces. Interpolation of the GradQuasiFermi Driving Force on page 502

Table 218 Math{} (Continued)

RefDens_hGradQuasiFermi_ ElectricField_HFS	=<float>	0 cm^{-3} (r) Damping parameter for hole high-field mobility driving force. Interpolation of the GradQuasiFermi Driving Force on page 502
RefDens_hGradQuasiFermi_ EparallelToInterface_HFS	=<float>	0 cm^{-3} (r) Damping parameter for hole high-field mobility driving force. Interpolation of the GradQuasiFermi Driving Force on page 502
RefDens_hGradQuasiFermi_Zero	=<float>	0 cm^{-3} (r) Damping parameter for hole high-field mobility driving force, alias for hDrForceRefDens. Interpolation of Driving Forces to Zero Field on page 502
RefNa_QuantumPotential	=<float>	1e25 cm^{-3} Semiconductor acceptor doping density for the application of the homogeneous Neumann boundary condition on semiconductor–insulator interfaces. Using the Density Gradient Model on page 406
RefNd_QuantumPotential	=<float>	1e25 cm^{-3} Semiconductor donor doping density for the application of the homogeneous Neumann boundary condition on semiconductor–insulator interfaces. Using the Density Gradient Model on page 406
RelErrControl		+ Use relative error control. Convergence and Error Control on page 215
RelTermMinDensity	=<(0,)>	1e3 cm^{-3} (g) Stabilization parameter for temperature relaxation term. Hydrodynamic Model Parameters on page 286
RelTermMinZeroDensity	=<(0,)>	2e8 cm^{-3} (g) Stabilization parameter for temperature relaxation term. Hydrodynamic Model Parameters on page 286

Table 218 Math{} (Continued)

RhsAndUpdateConvergence		– Newton converged if both RHS and update are converged. Convergence and Error Control on page 215
RhsFactor	=<float>	1e10 Maximum increase of the L_2-norm of the RHS between Newton iterations. Convergence and Error Control on page 215
RhsMax	=<float>	1e15 Maximum of L_2-norm of the RHS in each Newton iteration (transient simulations). Convergence and Error Control on page 215
RhsMaxQ	=<float>	1e100 Maximum of L_2-norm of the RHS in each Newton iteration (non-transient simulations). Convergence and Error Control on page 215
RhsMin	=<float>	1e-5 Minimum of L_2-norm of the RHS in each Newton iteration. Convergence and Error Control on page 215
RhsMinFactor	=<float>	1e-5 Factor for RhsMin affecting convergence. Convergence and Error Control on page 215
RobustBoxMethod		- Use alternative implementation of the box method calculations. Element Intersection Box Method Algorithm on page 1313
SaveWithinMinStep		- Add numeric guards to the Plot statement, which guarantees that all the specified points are saved. Saving and Plotting During a Quasistationary on page 143
SHECutoff	=<[0,)>	5.0 (g) Maximum kinetic energy to be plotted in the SHE distribution model. Using Spherical Harmonics Expansion Method on page 951

Feedback

Table 218 Math{} (Continued)

SHEIterations	=<0..>	20 (g) Number of iterations in the SHE distribution model. Using Spherical Harmonics Expansion Method on page 951
SHEMethod	=Table 231	super (g) Linear solver for the SHE distribution model. Using Spherical Harmonics Expansion Method on page 951
SHERefinement	=<1..>	1 (g) Number of energy grid intervals inside the phonon energy spacing in the SHE distribution model. Using Spherical Harmonics Expansion Method on page 951
SHESOR		+ Use the successive over-relaxation method in the SHE distribution model. Using Spherical Harmonics Expansion Method on page 951
SHESORError	=<float>	0.0 (g) Successive over-relaxation error criterion in the SHE distribution model. Using Spherical Harmonics Expansion Method on page 951
SHESORParameter	=<[1,2)>	1.1 (g) Successive over-relaxation parameter in the SHE distribution model. Using Spherical Harmonics Expansion Method on page 951
SHETopMargin	=<[0,)>	1.0 (g) Top energy margin in the SHE distribution model. Using Spherical Harmonics Expansion Method on page 951
SimpleBandTailIntegration		+ Use abrupt distribution function for band tail density. Band Tails on page 361
SimStats		off (g) Write simulation statistics to current plot file. Simulation Statistics for Plotting and Output on page 236
	(DOE_prefix=<string>	Use the specified prefix for DOE variables.

Table 218 Math{} (Continued)

	WriteDOE)	Write simulation statistics as DOE variables.
Smooth		off Keep mobility and recombination rates from the previous step to obtain better initial conditions for extreme nonlinear iterations.
SolvePrecision	=Table Table 29	off (g) Specify floating-point precision for Solve. Mixed Precision Solve
Spice_gmin	=<float>	1e-12 Ω^{-1} (g) SPICE minimum conductance. Math Section on page 121
Spice_Temperature	=<float>	300.15 K (g) Temperature of SPICE circuit. Math Section on page 121
SponEmissionIntegration (GaussianQuadrature(Order	=<int>))	Use Gaussian quadrature to integrate the spontaneous emission spectrum of an LED simulation. Accelerating Gain Calculations and LED Simulations on page 1163
StressLimit	=<float>	1e100 Pa (g) Upper limit on stress values read from files or specified. Values exceeding the limit will be truncated to the limit. Stress Limits on page 1045
StressMobilityDependence	=TensorFactor	Stress Tensor Applied to Low-Field Mobility on page 1111
StressSG		off (g) Use anisotropic Scharfetter–Gummel approximation for piezo mobility models. Chapter 28, Anisotropic Properties on page 978
SubMethod	=Table 231	(g) Inner linear solver for Blocked method. Linear Solvers on page 225

Feedback

Table 218 Math{} (Continued)

Surface	<string> (Table 377...)	(g) Define a surface. Mobility Degradation Components due to Coulomb Scattering on page 466, Random Geometric Fluctuations on page 879
TensorGridAniso		+ (g) Use tensor-grid approximation for anisotropic piezo mobility. Tensor Grid Option on page 1110
TensorGridAniso(	Aniso	+ (g) Use tensor-grid approximation for anisotropic models. Chapter 28, Anisotropic Properties on page 978
	Piezo)	+ (g) Same as TensorGridAniso.
ThinLayer(	Mirror=(<mirror>*3))	(g) Mirror planes for simulation system axes. <mirror> is Min, Max, None, or Both. Geometric Parameters of LayerThickness Command on page 424
TransferredElectronEffect2_eMinDerivativePerField	=<float>	-1e100 $cm^2 V^{-1} s^{-1}$ (r) Lower bound for velocity derivative $\partial v / \partial F_{hfs}$ (electrons only). Transferred Electron Model 2 on page 491
TransferredElectronEffect2_hMinDerivativePerField	=<float>	-1e100 $cm^2 V^{-1} s^{-1}$ (r) Lower bound for velocity derivative $\partial v / \partial F_{hfs}$ (holes only). Transferred Electron Model 2 on page 491
TransferredElectronEffect2_MinDerivativePerField	=<float>	-1e100 $cm^2 V^{-1} s^{-1}$ (r) Lower bound for velocity derivative $\partial v / \partial F_{hfs}$ (electrons and holes). Transferred Electron Model 2 on page 491
Transient	=BE	(g) Backward Euler method. Backward Euler Method on page 1340
	=TRBDF	* (g) TRBDF method. TRBDF Composite Method on page 1341

Table 218 Math{} (Continued)

TrapDLN	=<int>	13 (g) Levels to approximate trap energy distribution. Energetic and Spatial Distribution of Traps on page 610
TrapDLN(PeakSampling)	=<int>	13 Apply a peak sampling method to Gaussian and exponential distributions in all trap entries. Energetic and Spatial Distribution of Traps on page 610
Traps(	Damping=<[0,)>	10 Damping for traps for the nonlinear Poisson equation. A value of 0 deactivates damping. Chapter 17, Traps and Fixed Charges on page 608
	RegionWiseAssembly)	– Apply regionwise assembly for traps.
UpdateIncrease	=<float>	1.e100 (g) Maximum-allowed update error increase factor per Newton step.
UpdateMax	=<float>	1.e100 (g) Maximum-allowed update error in Newton algorithm.
UpdateQFPinTransientPoisson		+ (g) Update quasi-Fermi potentials in Transient, critical for Poisson-only solution. Numeric Control of Transient Analysis on page 152
UseSchurSolver		– (g) Replace Blocked by specialized MPBC solver. Specialized Linear Solver for MPBC on page 321
Volume	<string> (Table 376...)	(g) Define a volume. Random Band Edge Fluctuations on page 883
Wallclock		– (g) Report wallclock times rather than CPU times after each step in the simulation.
WeightedVoronoiBox		off (g) Compute-weighted coefficients and measure. Weighted Box Method Coefficients on page 1317

Feedback

Table 218 Math{} (Continued)

WithDistResistOnSemi		- Use improved discretization of contacts overlaying the oxide materials for distributed resistance simulations. Resistive Contacts on page 302

Table 219 AcceptNewtonParameter() Relaxed Newton Method

Digits	=<integer>	Number of relative error digits.
Error (	Electron=<float>	Absolute error parameter for electrons.
	eQuantumPotential=<float>	Absolute error parameter for electron quantum potential.
	eTemperature=<float>	Absolute error parameter for electron temperature.
	Hole=<float>	Absolute error parameter for holes.
	hQuantumPotential=<float>	Absolute error parameter for hole quantum potential.
	hTemperature=<float>	Absolute error parameter for hole temperature.
	Poisson=<float>	Absolute error parameter for potential.
	Temperature=<float>)	Absolute error parameter for lattice temperature.
ErrRef (	Electron=<float>	cm^{-3} Reference error parameter for electrons.
	eQuantumPotential=<float>	V Reference error parameter for electron quantum potential.
	eTemperature=<float>	K Reference error parameter for electron temperature.
	Hole=<float>	cm^{-3} Reference error parameter for holes.
	hQuantumPotential=<float>)	V Reference error parameter for hole quantum potential.
ErrRef (	hTemperature=<float>)	K Reference error parameter for hole temperature.

Table 219 AcceptNewtonParameter() Relaxed Newton Method (Continued)

`InvokeAtDivergence`		– Allow relaxed convergence to be considered for diverged solutions.
`RelErrControl`		Use relative error control.
`RhsAndUpdateConvergence`		Require both RHS and update error convergence.
`RhsMin`	`=<float>`	Minimum of RHS norm.
`UpdateScale`	`=<float>`	Additional factor for the update error.

Table 220 BreakCriteria{} Break Criteria: Conditionally Stopping the Simulation

`Current(`	`absval=<float>`	A μ m^{d-3} Upper limit for absolute current value.
	`contact=<string>`	! Name of contact with break criterion.
	`DevName=<ident>`	Identifier of circuit device name (mixed mode only).
	`maxval=<float>`	A μ m^{d-3} Upper limit for current.
	`minval=<float>`	A μ m^{d-3} Lower limit for current.
	`Node=<ident>)`	Identifier of circuit node name (mixed mode only).
`CurrentDensity(`	`DevName=<ident>`	Identifier of device (mixed mode only).
	`maxval=<float>)`	A cm^{-2} (r) Maximum current density.
`DevicePower | OuterDevicePower(`	`absval=<float>`	W μ m^{d-3} Upper limit for absolute power value.
	`DevName=<ident>`	Identifier of circuit device name (mixed mode only).
	`maxval=<float>`	W μ m^{d-3} Upper limit for power value.
	`minval=<float>)`	W μ m^{d-3} Lower limit for power value.
`ElectricField(`	`DevName=<ident>`	Identifier of circuit device name (mixed mode only).
	`maxval=<float>)`	V cm^{-1} (r) Maximum electric field.
`InnerDevicePower`	as `DevicePower`	W μ m^{d-3} Break criteria based on inner device power.
`LatticeTemperature(`	`DevName=<ident>`	Identifier of circuit device name (mixed mode only).
	`maxval=<float>)`	K (r) Maximum lattice temperature.

Feedback

Table 220 BreakCriteria{} Break Criteria: Conditionally Stopping the Simulation (Continued)

TDDBCurrent(	DevName=<ident>	Identifier of circuit device name (mixed mode only).
	maxval=<int>)	0 A Use maximum current as criterion to stop TDDB calculation. TDDB Time-Step Management on page 1256
TDDBCurrentRatio(	DevName=<ident>	Identifier of circuit device name (mixed mode only).
	maxval=<int>)	0 Use maximum current ratio as criterion to stop TDDB calculation. The current ratio is defined as I(t)/min(I(t') for t' < t). TDDB Time-Step Management on page 1256
TDDBTrapNumber(	DevName=<ident>	Identifier of circuit device name (mixed mode only).
	maxval=<int>)	0 Use maximum number of traps as criterion to stop TDDB calculation. TDDB Time-Step Management on page 1256
Voltage	as Current	V Voltage break criteria.

Table 221 ComputeCarrierPath() High-Field Entrance Position and Time

End(	DopingWell(point=<vector>))	Define end doping well using probe point point. End Condition on page 202
PathReflection(	closeToContact=<float>)	* 0.0 µm Path is reflected at device boundary. Path search stops when distance to a contact is smaller than specified value (in µm).
SaveEPaths		Save electron paths.
(	list=(<int>...)	List of path indices to be plotted.
	offset=<int>	0 Shift first plotted path.
	skip=<int>	0 Number of paths that are omitted for plotting.
	withPlotData)	+ Add interpolated plot variables.
SaveHPaths		Save hole paths.
(	list=(<int>...)	List of path indices to be plotted.
	offset=<int>	0 Shift first plotted path.
	skip=<int>	0 Number of paths that are omitted for plotting.
	withPlotData)	+ Add interpolated plot variables.

Table 221 ComputeCarrierPath() High-Field Entrance Position and Time (Continued)

Start		Start Condition on page 201
(	Cuboid(corner1=<vector> corner2=<vector>)	Define cuboid using corner points corner1 and corner2. Start From Vertex Positions: Cuboid on page 201
	Cuboid(corner1=<vector> corner2=<vector> grid=<vector>)	Define cuboid using corner points corner1 and corner2 and subgrid using grid. Start From Vertex Positions: Cuboid on page 201
	DopingWell(point=<vector>)	Define doping well using probe point point. Start From Vertex Positions: Doping Well on page 201
	DopingWell(point=<vector> grid=<vector>)	Define doping well using probe point point and subgrid using grid. Start From Vertex Positions: Doping Well on page 201
	Line(point1=<vector> point2=<vector>)	Define line using points point1 and point2. Start From Vertex Positions: Line on page 202
	Line(point1=<vector> point2=<vector> grid=<vector>))	Define line using points point1 and point2 and subgrid using grid. Start From Vertex Positions: Line on page 202
startFrom	="element" | "vertex"	"vertex" Path search starts from either the vertex position or the element center position.
thresholdField	=<float>	5e5 V/cm Minimum value of multiplication region.

Table 222 ComputeIonizationIntegrals() Approximate Breakdown Analysis

ComputeAll		All elements are taken into initial element list.
ComputeAtMaxElectricField		Element with maximal electric field is taken into initial element list.
ComputeAtMaxElectricField(	delta=<float>)	* 1e4 V/cm All elements where electric field is higher than maximal electric field minus delta are taken into initial element list.
Direction	=<string>	Electricfield Direction field used for the geometric path search.
MinElectricField	=<float>	1 V/cm All elements where electric field is higher than MinElectricField are taken into initial element list.
PathReflection	(Table 238)	Specification of path reflection options. Path is reflected at device boundary.

Table 222 ComputeIonizationIntegrals() Approximate Breakdown Analysis (Continued)

PathSearchStopDelta Potential	=<float> <float>	0 V 0 V Electrostatic potential shifts for the path search stop criterion using the electrostatic potential.
PathSearchStopField	=<float>	* 1000 V/cm Threshold value of the absolute value of the electric field where the path search stops.
PeriodicBC	(Table 239)	Specification of periodic boundary conditions.
SaveEIonIntegralPaths	=<int>	Number of paths to be stored based on electron ionization integral. Visualizing Breakdown Paths on page 574
SaveHIonIntegralPaths	=<int>	Number of paths to be stored based on hole ionization integral. Visualizing Breakdown Paths on page 574
Window	(Table 243)	Specification of geometric shapes for spatial restriction of initial element list.
WriteAll		Output information for all computed paths.

Table 223 eQPBox() Density Gradient Model

Attenuation_Length	=<float>	1e-3 µm Length over which the gamma (γ) parameter is effectively attenuated to zero using error function outside the box.
MaxX	=<float>	1e30 µm Maximum x-coordinate for quantum box.
MinX	=<float>	-1e30 µm Minimum x-coordinate for quantum box.
MaxY	=<float>	1e30 µm Maximum y-coordinate for quantum box.
MinY	=<float>	-1e30 µm Minimum y-coordinate for quantum box.
MaxZ	=<float>	1e30 µm Maximum z-coordinate for quantum box.
MinZ	=<float>	-1e30 µm Minimum z-coordinate for quantum box.

Table 224 EquilibriumSolution() Equilibrium Solution

Digits	=<float>	Relative error target.
Iterations	=<0..>	50 Maximum number of Newton iterations. Convergence and Error Control on page 215

Table 224 EquilibriumSolution() Equilibrium Solution (Continued)

LineSearchDamping	=<(0,1]>	1 Smallest allowed damping coefficient for line search damping. Damped Newton Iterations on page 218
NotDamped	=<0..>	1000 (g) Number of iterations in each Newton iteration before Bank–Rose damping is activated. Damped Newton Iterations on page 218
RelErrControl		+ Use relative error control. Convergence and Error Control on page 215

Table 225 Extrapolate() in Math{}, Quasistationary(), Transient() Extrapolation

Algorithm	=1 | 2	1 Switch Algorithm on page 147.
Exclude	(Table 356)	Exclude Equations on page 146
LowDensityLimit	=<float>	1e-100 cm^{-3} Lower limit for carrier density starting from which the appropriate density is extrapolated.
MinStep	=<float>	Minimum Step of Extrapolation on page 146
NumberOfFailures	=<int>	0 Number of Failed Extrapolations on page 147
Order	=<int>	1 Order of extrapolation in Quasistationary or Transient command.

Table 226 GrainFieldGenerator() Creating the Grain Field

AverageGrainSize	=<float>	µm Approximate average grain size in V_{reg}, or V_{box} if padding is used.
AverageGrainSizeVector	= (<ax>, <ay>, <az>)	µm Allow the grain average size to differ in different directions.
GrainLocation	= ((<x1>,<y1>,<z1>), (<x2>,<y2>,<z2>), ...)	µm Set specific locations where grains are placed in the region.
GrainLocationSize	= ((<sx1>,<sy1>,<sz1>), (<sx2>,<sy2>,<sz2>), ...)	µm Set elliptic radii of grains whose locations are specified with GrainLocation. There must be a one-to-one correspondence between values specified with GrainLocation and GrainLocationSize.
MinimumGrainSize	=<float>	µm Set the smallest-allowed distance between grain centers.
NumberOfGrains	=<int>	Number of grain centers located within V_{reg}.

Table 226 GrainFieldGenerator() Creating the Grain Field (Continued)

`Padding`	`=<float>`	Units of `AverageGrainSize`. Extend extremities of region by a distance of `Padding` × a_{gi} on all sides before randomizing the grain centers.
`PoissonDistribution`		Use the calculated number of grain centers as the expectation value for a Poisson distribution random number generator.
`RandomSeed`	`=<int>`	`0` Seed used for the random number generator. Using the same (nonzero) seed in subsequent simulations allows the grain generation to be repeated. The default means a new randomization for each simulation.
`SingleVertexBoundary`		+ Mark only one vertex of edges that cross a Voronoï boundary with a negative index. To mark both vertices on an edge, specify `-SingleVertexBoundary`.
`TwoDimensionalGrains`		Create grains in two dimensions and extrude in the thinnest region dimension.
`UniformSpatial Distribution`		Distribute grain centers randomly.

Table 227 HarmonicBalance() Harmonic Balance

`NewtonConvergence <name>`		Parameters impacting nonlinear convergence.

`(`	`Stagnation( Iterations=<int>`
	`[-]WithConvergence`
	`[-]WithDivergence))`
		Length of sequence required.
		+ Option to switch off convergence.
		+ Option to switch off divergence.

`NonlinearSolver <name>(`	`GeneralizedGaussSeidel( NumberOfFixedHarmonics=<int>)`	`8` Harmonics to be solved.
	`Newton()`	Parameters for Newton solver.

Feedback

Table 227 HarmonicBalance() Harmonic Balance (Continued)

	SolverMethod= GaussSeidel | GeneralizedGaussSeidel | Newton)	Select nonlinear solver.

Table 228 HB{} Harmonic Balance

LogDensity		– Use logarithm of scaled densities as solution variable.
LogTemperature		– Use logarithm of scaled temperatures as solution variable.
MDFT		– Use MDFT mode.
RhsScale	(Table 356)=<float>	Scaling of RHS in Newton (MDFT and SDFT modes).
sdft_update_norm_R		– Use update error norm (compatible with Version R-2020.09).
SolveSpectrum(	Name=<string>)	Solve spectrum with (mandatory) name.
{	(<int>*n)...}	List of spectrum multi-indices, where n is the number of tones. Solve Spectrum on page 170
Traps(	Solve = Decoupled | FastSteadyState | Frozen | Harmonic)	Select traps mode. Handling Traps on page 173
UpdateScale	(Table 356)=<float>	Scaling of update in Newton (MDFT and SDFT modes).
ValueMin	(Table 356)=<float>	Lower bound for quantity in time domain (MDFT mode only).
ValueVariation	(Table 356)=<float>	Allowed variation of quantity in time domain (MDFT mode only).

Table 229 hQPBox() Density Gradient Model

Attenuation_Length	=<float>	1e-3 µm Length over which the gamma (γ) parameter is effectively attenuated to zero using error function outside the box.
MaxX	=<float>	1e30 µm Maximum x-coordinate for quantum box.
MinX	=<float>	-1e30 µm Minimum x-coordinate for quantum box.

Feedback

Table 229 hQPBox() Density Gradient Model (Continued)

MaxY	=<float>	1e30 µm Maximum y-coordinate for quantum box.
MinY	=<float>	-1e30 µm Minimum y-coordinate for quantum box.
MaxZ	=<float>	1e30 µm Maximum z-coordinate for quantum box.
MinZ	=<float>	-1e30 µm Minimum z-coordinate for quantum box.

Table 230 KMC() KMC Simulation Space and Math Settings

AlignCellsToRefLocation		off Shift the KMC simulation space so that the center of a KMC cell will overlap with RefLocation.
ApproximateBessel		Specify approximate Bessel function for calculation of InelasticElectrodeToTrapTunneling, InelasticTrapToElectrodeTunneling, and TrapToTrapTunneling(MultiPhonon) tunneling models. Specifying the Approximate Bessel Function on page 1228
CellLength	=<float>	0.0005 µm KMC cell size (for TDDB only).
CurrentCompliance(		Use current compliance.
	Contact=<string>	Name of the contact applying current compliance. Default: empty string
	MaxVal=<float>	Maximum value accepted; it must be positive. Default: not applied
	MinVal=<float>)	Minimum value accepted; it must be negative. Default: not applied
HalfCellOffset		off Shift the KMC simulation space by a half-cell length relative to device structure (for TDDB only).
IntegrationPoints	=<int>	Set the number of integration points to use with Gauss– Legendre method. Setting the Integration Points on page 1228
MaxCurrent	=<float>	0 Use maximum current as criterion to stop TDDB calculation.
MaxCurrentRatio	=<float>	0 Use maximum current ratio as criterion to stop TDDB calculation. The current ratio is defined as I(t)/min(I(t') for t' < t).

Table 230 KMC() KMC Simulation Space and Math Settings (Continued)

MaxLocation	=(<float>, <float>, <float>)	µm Define KMC simulation space. KMC Simulation Space on page 1223
MaxTrapNumber	=<int>	0 Use maximum trap number as criterion to stop TDDB calculation.
MaxTrapNumberIncrease	=<int>	0 Each time step allows a maximum number of particles to increase if UseKMCTimeStep is provided.
MIMBackAndForthThreshold	=<int>	2000 Set number of back-and-forth events that occur before rate is adjusted. Adjusting the Calculated Tunneling Rates on page 1228
MIMEventCountCheckPoint	=<int>	10000 Specify how often the KMC simulator checks for the need to adjust tunneling rates. Adjusting the Calculated Tunneling Rates on page 1228
MinLocation	=(<float>, <float>, <float>)	µm Define KMC simulation space. KMC Simulation Space on page 1223
MoreFrequentRateResets		Reset all MIM tunneling rates after each time step. Resetting the Rate More Frequently on page 1227
PlotKMCStatistics		Write ReRAM particle and event statistics to the current plot file.
RandomSeed	=<int>	Set random seed. Specifying the Random Seed on page 1226
ReferenceElectrode	=<string>	Specify electrode to use as a reference for distances.
RefLocation	=(<float>, <float>, <float>)	µm Define the reference location to which to align a cell center.
ResetElectronFill		Reset the fraction of defects filled with electrons. Resetting the Fraction of Defects Filled With Electrons on page 1227
SkipKMC		Omit KMC simulation completely. Omitting the KMC Simulation Completely on page 1227
SkipKMCCurrentCheck	=<float>	Omit further calls to KMC simulator for remainder of Sentaurus Device transient ramp. Omitting KMC Simulations Near Steady State on page 1226

Feedback

Table 230 KMC() KMC Simulation Space and Math Settings (Continued)

SkipKMCNumber	=<int>	Set number of consecutive Sentaurus Device time-steps before omitting KMC. Omitting KMC Simulations Near Steady State on page 1226
TargetParticle(	Filament{i})	Particles used to determine the time step, which can include Filament1 to Filament3.
Trap2ElecMinDist	=<float>	0.0002 µm Set minimum-allowed defect-to-electrode distance. Setting the Minimum Defect Distances on page 1225
Trap2TrapMinDist	=<float>	0.0002 µm Set minimum-allowed defect-to-defect distance. Setting the Minimum Defect Distances on page 1225
UseAllCrossings		Sample potential across all crossings. Sampling Potential at All Crossings on page 1227
UseKMCTimeStep		– KMC simulator determines the time step for the transient calculation using MaxTrapNumberIncrease when the trap increases sufficiently fast.

Table 231 Linear solvers (Solvers User Guide)

GPU_DirectSolver		GPU accelerated direct solver. High accuracy for small and medium problems. Can be significantly faster than ParDiSo or ILS. The GPU solver options can be controlled through the GPUSolverSettings keyword as described in Table 232.
ILS(		Parallel, iterative linear solver. Customizable, high accuracy, and parallel performance for all problems. Default for 3D structures using set (1).
	MatrixPrescaling	– Performs a prescaling of the matrix entries before the matrix is passed to the linear solver. This option can have a beneficial impact on the quality of results.
	MultipleRHS	– Solve linear systems with multiple right-hand sides (only for AC analysis).
	Set=<int>)	Use ILS options from set <int>.
Pardiso(	Table 237	Select (subset of) parameters explicitly.

Feedback

Table 231 Linear solvers (Solvers User Guide) (Continued)

	MultipleRHS	– Solve linear systems with multiple right-hand sides (only for AC analysis).
	Name=<name>)	The name of the named PardisoSolver section where the parameters are taken from.
Super		Supernodal direct solver. Best accuracy for small problems, not parallelized. Default for 1D and 2D structures.

Table 232 GPUSolverSettings(). Options for the GPU direct solver.

Option		Description
ExternalPreprocessing	=<string>	"ILS" An additional preprocessing can be performed before the matrix is passed to the solver. For numerical challenging problems like device simulations, it is usually necessary to apply "ILS" preprocessing to achieve the required quality of results. The options are: • "None": No additional preprocessing. • "Scaling": Simple row and column prescaling is performed. • "ILS": The preprocessing options implemented for the iterative linear solver ILS can be applied (see Chapter 3, “ILS Solver”, in the *Solvers User Giuide*). • "Scaling_ILS": The scaling and ILS preprocessing is combined.
IterativeRefinement	=<int>	2 Performs the provided number of iterative refinement steps to improve the accuracy of the solution.
ILSParams(Ordering(Symmetric	=<string>))	"nd" Defines the symmetric ordering for the ILS preprocessing (see the Symmetric Ordering section in Chapter 3, “ILS Solver”, of the *Solvers User Guide*.) The options are: • "none" • "mmd" • "nd" • "ndrcm" • "rcm"

Feedback

Table 232 GPUSolverSettings(). Options for the GPU direct solver. (Continued)

Option		Description
ILSParams(Ordering(Nonsymmetric	=<string>))	"mpsilst" Defines the nonsymmetric ordering for the ILS preprocessing (see the Nonsymmetric Ordering section in Chapter 3, “ILS Solver”, of the Solvers User Guide). The options are: • "none" • "mpsils" • "mpsilst"
ILSParams(Prep	=<int>)	0 The option prep=1 forces a fixed initial ordering of the linear system for ILS preprocessing which helps in many cases to reduce the overall time spent in both symmetric and nonsymmetric ordering steps (see the Additional Options section in Chapter 3, “ILS Solver”, in the *Solvers User Guide*).
HybridMode	=<int>	-1 (off) The hybrid mode allows to overcome the single-GPU memory (device memory) limitation by also using the CPU memory (host memory) to store the factorization of the matrix. The hybrid mode comes with the extra cost of host-to-device and device-to-host memory transfers and thus diminishes the performance but allows to treat larger systems which cannot be processed on the device memory only. With HybridMode=<dev mem in kb>, the maximum amount of device memory in kB is defined that can be used to store the factorization of the matrix.
RecomputeNonsymmetricPermutation		off Recomputes the nonsymmetric matrix permutation before each factorization. It therefore has effect only if ILS preprocessing is enabled.
MultipleRHS		off Solving linear systems with multiple right-hand sides. This option applies to AC analysis only. It might produce minor performance improvements.

Table 233 MLDAbox({}...) Modified Local-Density Approximation Model

MaxX	=<float>	µm Upper x-coordinate of the box.
MaxY	=<float>	µm Upper y-coordinate of the box.
MaxZ	=<float>	µm Upper z-coordinate of the box.
MinX	=<float>	µm Lower x-coordinate of the box.

Table 233 MLDAbox({}...) Modified Local-Density Approximation Model (Continued)

MinY	=<float>	µm Lower y-coordinate of the box.
MinZ	=<float>	µm Lower z-coordinate of the box.

Table 234 MVMLDAcontrols() Using the MLDA Model

AveDistanceFactor	=<float>	0.05 Factor that controls computation of an averaged distance from a vertex to the interface for the multivalley MLDA model.
Load	=<string>	Name of file from which to load energy-dependent data.
LoadWithInterpolation	=<string>	Name of file from which to load energy-dependent data, possibly obtained for different mesh.
MaxDoping4Majority	=<float>	1e22 cm^{-3} Maximum doping concentration where the multivalley MLDA model is applied to majority carriers.
MaxIntDistance	=<float>	1e-6 cm Distance from the interface up to which the multivalley MLDA model is applied.
Save	=<string>	Name of file where to save energy-dependent data.

Table 235 Nonlocal() Nonlocal Meshes

Table 377		(g) Interface that is part of the reference surface.
Barrier	(Table 376...)	– (g) Regions that form the tunneling barrier.
Digits	=<float>	2 (cgi) Accuracy for nonlocal tunneling currents. Nonlocal Tunneling Parameters on page 920
Direction	=<vector>	(0 0 0) (cgi) If nonzero, suppress the construction of nonlocal mesh lines with a direction more than MaxAngle degrees different from <vector>.
Discretization	=<float>	1e100 cm (cgi) Maximum distance between nonlocal mesh points on a nonlocal line.
Electrode	=<string>	(g) Electrode that is part of the reference surface.

Feedback

Table 235 Nonlocal() Nonlocal Meshes (Continued)

Endpoint		(r) Allow nonlocal lines that end in the region. Default is Endpoint for semiconductors; otherwise, -Endpoint.
	(Table 376...)	(g) Regions where nonlocal lines can or cannot end.
EnergyResolution	=<(0,)>	0.005 eV (cgi) Minimum energy resolution for integrals in computation of nonlocal tunneling current. Nonlocal Tunneling Parameters on page 920
Length	=<float>	cm (cgi) Distance from the interface or contact up to which nonlocal mesh lines are constructed.
MaxAngle	=<float>	180 deg (cgi) Suppress construction of nonlocal mesh lines that enclose an angle of more than <float> degrees with the vector specified by Direction.
Outside		+ (cgi) Allow nonlocal mesh lines to leave the device.
Permeable		+ (r) Allow extension of nonlocal lines (as specified by the Permeation parameter) into or across the region.
	(Table 376...)	+ (g) Regions into or across which nonlocal lines can or cannot be extended.
Permeation	=<float>	0 cm (cgi) Length by which nonlocal mesh lines are extended across the interface or contact.
Refined		+ (r) Autorefine nonlocal lines inside the region.
	(Table 376...)	+ (g) Regions in which nonlocal lines are or are not autorefined.
Transparent		+ (r) Allow nonlocal lines crossing the region.
	(Table 376...)	+ (g) Regions that nonlocal lines can or cannot cross.

Feedback

Table 236 NonlocalPath() in Math{}, Quasistationary(), and Transient() Handling Derivatives

Derivative	=<int>	0: Without derivative computation (default) 1: With derivative computation. Handling Derivatives on page 599
Direction	=<vector>	Frozen tunneling direction.
MaxStep	<float>	(1.0) Higher boundary for step-size control.
MinStep	<float>	(1.0e-6) Lower boundary for step-size control.
N	=<int>	(3) Number of Newton iterations where Jacobian entries are collected.
Postprocessing		– Switch on and off postprocessing mode.
Strategy	=<int>	Strategy for filling of system Jacobian: 1: Collect all 2: Collect last N (default) 3: Collect last N with step size control

Table 237 PardisoSolver()

IterativeRefinement	= <int>	– Perform up to two iterative refinement steps to improve the accuracy of the solution.
MaxSupernodeSize	= <int>	0 Maximal size of supernodes to be constructed. Typical values are 128–512. Value 0 selects a value in the range 64–128 (depending |on matrix type and number of threads).
ND_InitialPartitioningSize (		The initial partitioning size s requested from nested dissection (ND) is given by $s = i_f n_t + i_s$ and bounded into interval [0,128].
	Factor = <int>	1 The value i_f in the formaula above
	Shift = <int>	0 The value i_s in the formaula above
MatrixPrescaling		– Performs a prescaling of the matrix entries before the matrix is passed to the linear solver. This option can have a beneficial impact on the quality of results.
MultipleRHS		– Solve linear systems with multiple right-hand sides (only for AC analysis).

Table 237 PardisoSolver() (Continued)

NonsymmetricPermutation		+ Compute an initial nonsymmetric matrix permutation and scaling, which places large matrix entries on the diagonal.
RecomputeNonsymmetricPermutation)		– Compute a nonsymmetric matrix permutation and scaling before each factorization.

Table 238 PathReflection() Approximate Breakdown Analysis

CloseToContact	=<float>	0.1 Path search stops when distance to a contact is smaller than specified value (in µm).
ReflectAtMaterial	=<string>	Path is reflected at specified material.
ReflectAtRegion	=<string>	Path is reflected at specified region.

Table 239 PeriodicBC() Approximate Breakdown Analysis

Direction	=0 | 1 | 2	Direction of periodicity: 0 for x-axis 1 for y-axis 2 for z-axis
Coordinates	=(<float>*2))	µm Coordinate of left and right periodic boundary plane.
MaxReflections	=<int>	5000 Maximum number of periodic path reflections until the path search stops.

Table 240 RandomField() Spatial Correlations and Random Fields

AverageGrainSize	=<vector>	µm Average grain size along main axes.
CorrelationFunction	=Exponential	Use exponential correlations.
	=Gaussian	Use Gaussian correlations.
	=Grain	* Use grain-based correlations.
Lambda	=<vector>	µm Correlation length for main axes
MaxInternalPoints	=<int>	2147483647 Maximum-allowed number of points for Fourier transform.

Feedback

Table 240 RandomField() Spatial Correlations and Random Fields (Continued)

Resolution	=<vector>	(0.25 0.25 0.25) Spatial resolution of exponential and Gaussian randomization.

Table 241 RandomizedVariation() in Math{} Statistical Impedance Field Method

ExtrudeTo3D		– For correlated variations, internally extrude 2D structures to three dimensions.
NumberOfSamples	=<0..>	! Number of samples.
RandomField(	Table 240)	(g) Declare random field.Spatial Correlations and Random Fields on page 887
Randomize	=<int>	0 Seed for random number generator.

Table 242 Transient time-step control Numeric Control of Transient Analysis

CheckTransientError		off Error control of transient integration method.
NoCheckTransientError		on No error control of transient integration method.
TransientDigits	=<float>	3 Relative accuracy for time-step control.
TransientError	(Table 356)=<float>	Absolute error for time-step control.
TransientErrRef	(Table 356)=<float>	Error reference for time-step control.
TrStepRejectionFactor	=<float>	Factor f_{rej}. Controlling Transient Simulations on page 1342

Table 243 Window() Approximate Breakdown Analysis

Cuboid(	corner1=<vector>	First corner of cuboid.
	corner2=<vector>)	Diagonally opposite corner of cuboid.
Cylinder(	center1=<vector>	Center of first circular base of cylinder.
	center2=<vector>	Center of second circular base of cylinder.
	radius=<float>)	Radius of cylinder.
Sphere(	center=<vector>	Center of sphere.

Feedback

Table 243 Window() Approximate Breakdown Analysis (Continued)

	radius=<float>)	Radius of sphere.

NoisePlot

Table 244 NoisePlot{} Noise Output Data

Table 202	Scalar plot data.
Table 203/Vector	Vector plot data.
AllLNS	All used local noise sources.
AllLNVSD	All used local noise voltage spectral densities.
AllLNVXVSD	All use local noise voltage cross-correlation spectral densities.
GreenFunctions	All used Green's functions and their gradients.

NonLocalPlot

Table 245 NonLocalPlot{} Visualizing Data Defined on Nonlocal Meshes

Table 202		Scalar data.
EigenEnergy(		Eigenenergies. 1D Schrödinger Equation on page 393
	Electron[(Number=<int>)]	Restrict output to [<int> lowest] electron eigensolutions.
	Hole[(Number=<int>)])	Restrict output to [<int> lowest] hole eigensolutions.
OverlapIntegral		Overlap integrals. 1D Schrödinger Equation on page 393
WaveFunction(		Wavefunctions. 1D Schrödinger Equation on page 393
	Electron[(Number=<int>)]	Restrict output to [<int> lowest] electron eigensolutions.

Feedback

Table 245 NonLocalPlot{} Visualizing Data Defined on Nonlocal Meshes (Continued)

	Hole[(Number=<int>)])	Restrict output to [<int> lowest] hole eigensolutions.

OpticalDevice

See Table 212 on page 1711.

Physics

Table 246 Physics{} Physics in Sentaurus Device

Active(	Type=QuantumWell)	Activate the localized QW model or the nonlocal QW model to use in conjunction with QWLocal or Schroedinger, respectively.
Affinity	(<ident> [(Table 380)])	(r) Use PMI model <ident> for electron affinity. Electron Affinity on page 1463
AlphaParticle(	Table 293)	(g) Generation by alpha particles. Carrier Generation by Alpha Particles on page 860
AnalyticTEP		(g) Analytic expression for thermoelectric power. Thermoelectric Power on page 1146
Aniso(	Table 296)	Anisotropic properties. Chapter 28, Anisotropic Properties on page 978
AreaFactor	=<float>	1 (g) Multiplier for current and heat flux densities at electrodes and thermodes. Reading a Structure on page 59
BarrierLowering		off (c) Use barrier lowering for Schottky contact. Barrier Lowering at Schottky Contacts on page 296
BreakdownProbability(	Table 251)	Avalanche breakdown probability. Avalanche Breakdown Probability on page 577
ComplexRefractive Index(	Table 299)	off (g) Use complex refractive index model. Complex Refractive Index Model on page 777
CondInsulator		(r) Turn an insulator into a conductive insulator. Conductive Insulators on page 335

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

DefaultParametersFromFile		Initialize default parameters from parameter files instead of using built-in values. Default Parameters on page 93
Deterministic Variation(	Table 301)	(g) Deterministic variations. Deterministic Variations on page 897
Dipole(		(i) Use dipole interface model. Dipole Layer on page 256
	Reference=<string>)	! Reference side <string> (either region or material name).
Discontinuity		Create discontinuous interface or interfaces. Discontinuous Interfaces on page 322
DistResist	=<float>	$\Omega\,cm^2$ (i) Distributed resistance at metal–semiconductor or metal–metal interfaces. Resistive Interfaces on page 309
	=SchottkyResist	Emulate a Schottky interface. Resistive Interfaces on page 309
eBandTailDOS(	Exponent	off (r) Exponential shape band tails. Band Tails on page 361
	Gaussian	off (r) Gaussian shape band tails. Band Tails on page 361
	GaussianFromOrganic	off (r) Gaussian shape band tails as implemented for organic materials. Band Tails on page 361
	TwoPopulation)	off (r) Two-population model. Band Tails on page 361
eBarrierTunneling	<string> [(Table 253)]	off (g) Nonlocal tunneling from and to conduction band. Nonlocal Tunneling at Interfaces, Contacts, and Junctions on page 916
EffectiveIntrinsicDensity	(Table 306)	(r) Band gap and bandgap narrowing. Band Gap and Electron Affinity on page 340
EffectiveMass	(GaussianDOS)	(r) Use simplified Gaussian density-of-states model for organic semiconductors. Gaussian Density-of-States for Organic Semiconductors on page 358
	(<ident> [(Table 380)])	(r) Use PMI model <ident> for effective mass. Effective Mass on page 1460

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

eMLDA	[()]	– (r) MLDA model for electrons. Modified Local-Density Approximation Model on page 413
(	LambdaTemp)	+ (r) Use temperature-dependent λ for both electrons and holes.
eMobility(	Table 289)	(r) Electron mobility model. Chapter 15, Mobility on page 429
eMultiValley		off (r) Multivalley statistics. Multivalley Band Structure on page 363, Multivalley Band Structure on page 1058
eMultiValley(	DensityIntegral	off (r) Numeric density computation. Using Multivalley Band Structure on page 368
	kpDOS	off (r) Two-band k · p models for electron Δ_2 valleys. Using Multivalley Band Structure on page 368
	mcDOS	off (r) Monte Carlo DOS. Using Multivalley Band Structure on page 368, Using the MLDA Model on page 417
	MLDA	off (r) Multivalley MLDA quantization model. Modified Local-Density Approximation Model on page 413, Using Multivalley Band Structure on page 1059, Inversion Layer on page 1067
	MLDA (-Nonparabolicity)	off (r) Exclude band nonparabolicity in MLDA model. Using the MLDA Model on page 417
	Nonparabolicity	off (r) Band nonparabolicity. Nonparabolic Band Structure on page 364
	parfile	on (r) With kpDOS, it adds the parameter file valleys. Using Multivalley Band Structure on page 368
	RelativeToDOSMass	off (r) Multivalley effective DOS. Using Multivalley Band Structure on page 368
	ThinLayer)	off (r) Geometric quantization model. Using Multivalley Band Structure on page 368, Using the MLDA Model on page 417
EnergyRelaxationTimes	(<ident> [(Table 380)])	Use PMI model <ident> to compute energy relaxation times. Energy Relaxation Times on page 1595

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

	(constant)	Use constant energy relaxation times.
	(formula)	Use the value of formula in the parameter file.
	(irrational)	Use the ratio of two irrational polynomials. Energy-Dependent Energy Relaxation Time on page 971
eNMP(	Table 309)	(i) Extended nonradiative multiphonon (eNMP) model. Extended Nonradiative Multiphonon Model on page 708
eQCvanDort		off (r) van Dort model for electrons. The van Dort Model on page 392
eQuantumPotential	[(Table 311)]	– (r) Activate electron density gradient quantum correction. Density Gradient Model on page 404
eQuasiFermi	=<float>	V (r) Initial quasi-Fermi potential specification for electrons. Regionwise Specification of Initial Quasi-Fermi Potentials on page 262
eRecVelocity	=<[0,)>	2.573e6 cm s^{-1} (i) Electron recombination velocity. Electric Boundary Conditions for Metals on page 330
eSHEDistribution	(Table 312)	off (r) Specify the region where the electron SHE distribution is calculated. Using Spherical Harmonics Expansion Method on page 951
eThermionic		(i) Thermionic emission model for electrons. Conductive Insulators on page 335, Thermionic Emission Current on page 964
	(HCI)	(i) Inject hot electrons locally. Destination of Injected Current on page 936
	(Organic_Gaussian)	(i) Thermionic-like Gaussian emission model at organic heterointerfaces for electrons. Gaussian Transport Across Organic Heterointerfaces on page 968
eTransitionRate(	Transition [<string>] (Table 342))	off (r) Define the transitions between the different electron valleys. Intervalley Transitions on page 373
ExternalBoltzmann Solver	<string> ()	(g) Connection to external Boltzmann solver. External Boltzmann Solver on page 402

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

ExternalSchroedinger	<string> (Table 314)	(g) Connection to external 2D Schrödinger solver. External 2D Schrödinger Solver on page 400
FEPolarization	(Table 162)	Ferroelectric polarization model. Ginzburg–Landau–Khalatnikov Equation on page 1009
FEPolarization(	(Table 162)	Landau coefficients. Using the Ginzburg–Landau–Khalatnikov Equation on page 1011
	Direction	Direction of the ferroelectric polarization component being solved. Default on page 1015
	PMIModel	Landau expansion PMI. Ferroelectrics on page 1555
	SFactor	Dataset name for initial polarization. Using the Ginzburg–Landau–Khalatnikov Equation on page 1011
	Tensor)	Anisotropic formulation of the model. Using the Ginzburg–Landau–Khalatnikov Equation on page 1011
Fermi		off (g) Fermi statistics. Fermi Statistics on page 259
	(-WithJoyceDixon)	(g) Fermi statistics with old Wuensche approximation for Fermi integrals. Fermi Statistics on page 259
	WithFukushima	(g) Fermi statistics with new Fukushima approximation for Fermi integrals.Fermi Statistics on page 259
FermiForTEPAnalytic		off (g) Enables the Fermi correction for the analytic thermoelectric power coefficient computation. Using Thermoelectric Power
FloatCoef	=<float>	0 (g) Interpolation coefficient for initial guess in floating wells. Initial Guess for Electrostatic Potential and Quasi-Fermi Potentials in Doping Wells on page 261
GateCurrent(	Table 315)	(i) Gate currents (hot-carrier injection, some of the tunneling models).

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

GaussianDOS_full		(r) Use Gaussian density-of-states model for organic semiconductors. Gaussian Density-of-States for Organic Semiconductors on page 358
hBandTailDOS(	Exponent	off (r) Exponential shape band tails. Band Tails on page 361
	Gaussian	off (r) Gaussian shape band tails. Band Tails on page 361
	GaussianFromOrganic	off (r) Gaussian shape band tails as implemented for organic materials. Band Tails on page 361
	TwoPopulation)	off (r) Two-population model. Band Tails on page 361
hBarrierTunneling	<string> [(Table 253)]	off (g) Nonlocal tunneling from and to valence band. Nonlocal Tunneling at Interfaces, Contacts, and Junctions on page 916
HeatCapacity(	Table 320)	(r) Heat capacity.
HeatPreFactor	=<float>	1. (r) Scaling factor for lattice heat generation. Scaling of Lattice Heat Generation on page 287
HeatSource(	<ident>)	Name of the PMI model.
HeavyIon(	Table 294)	off (g) Generation by heavy ions. Carrier Generation by Heavy Ions on page 862
HeteroInterface		(i) Double points at interfaces. Abrupt and Graded Heterojunctions on page 62
hMLDA(	[()]	– (r) MLDA model for holes. Modified Local-Density Approximation Model on page 413
	LambdaTemp)	+ (r) Use temperature-dependent λ for both electrons and holes.
hMobility(	Table 290)	(r) Hole mobility model. Chapter 15, Mobility on page 429
hMultiValley		off (r) Multivalley statistics. Multivalley Band Structure on page 363, Multivalley Band Structure on page 1058

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

hMultiValley(	DensityIntegral	off (r) Numeric density computation. Using Multivalley Band Structure on page 368
	kpDOS	off (r) Six-band k · p model for hole bands. Multivalley Band Structure on page 363
	mcDOS	off (r) Monte Carlo DOS. Using Multivalley Band Structure on page 368, Using the MLDA Model on page 417
	MLDA	off (r) Multivalley MLDA quantization model. Modified Local-Density Approximation Model on page 413, Using Multivalley Band Structure on page 1059
	MLDA (-Nonparabolicity)	off (r) Exclude band nonparabolicity in MLDA. Using the MLDA Model on page 417
	Nonparabolicity	off (r) Band nonparabolicity. Nonparabolic Band Structure on page 364
	parfile	on (r) With kpDOS, it adds the parameter file valleys. Using Multivalley Band Structure on page 368
	RelativeToDOSMass	off (r) Multivalley effective DOS. Using Multivalley Band Structure on page 368
	ThinLayer)	off (r) Geometric quantization model. Using Multivalley Band Structure on page 368, Using the MLDA Model on page 417
hQCvanDort		off (r) van Dort model for holes. The van Dort Model on page 392
hQuantumPotential	[(Table 311)]	– (r) Activate hole density gradient quantum correction. Density Gradient Model on page 404
hQuasiFermi	=<float>	V (r) Initial quasi-Fermi potential specification for holes. Regionwise Specification of Initial Quasi-Fermi Potentials on page 262
hRecVelocity	=<[0,)>	1.93e6 cm s^{-1} (i) Hole recombination velocity. Electric Boundary Conditions for Metals on page 330
hSHEDistribution	(Table 312)	off (r) Specify the region where the hole SHE distribution is calculated. Using Spherical Harmonics Expansion Method on page 951

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

hThermionic		(i) Thermionic emission model for holes. Conductive Insulators on page 335, Thermionic Emission Current on page 964
	(HCI)	(i) Inject hot holes locally.Destination of Injected Current on page 936
	(Organic_Gaussian)	(i) Thermionic-like Gaussian emission model at organic heterointerfaces for holes. Gaussian Transport Across Organic Heterointerfaces on page 968
hTransitionRate(	Transition [<string>] (Table 342))	off (r) Define the transitions between the different hole valleys. Intervalley Transitions on page 373
Hydrodynamic	[()]	(g) Hydrodynamic model for electrons and holes. Hydrodynamic Model for Current Densities on page 269, Hydrodynamic Model for Temperatures on page 282
	(eTemperature)	(g) Hydrodynamic model for electrons only. Hydrodynamic Model for Current Densities on page 269, Hydrodynamic Model for Temperatures on page 282
	(hTemperature)	(g) Hydrodynamic model for holes only. Hydrodynamic Model for Current Densities on page 269, Hydrodynamic Model for Temperatures on page 282
HydrogenDiffusion		off (ri) Hydrogen transport model without reaction or hydrogen type specification. Using MSC–Hydrogen Transport Degradation Model on page 701
	(Table 322)	off (ri) Hydrogen transport model with reaction or hydrogen type specification. Using MSC–Hydrogen Transport Degradation Model on page 701
IncompleteIonization(	Table 324)	Incomplete ionization. Chapter 13, Incomplete Ionization on page 380
IniteQuasiFermi InithQuasiFermi		off. Alternative Initial Guess for Quasi-Fermi Potentials From Linear Conduction Equation on page 262

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

	(TotalConductance)	Use total linear conductance equation. Alternative Initial Guess for Quasi-Fermi Potentials From Linear Conduction Equation on page 262
	(FDW)	Use floating doping well model. Alternative Initial Guess for Quasi-Fermi Potentials From Linear Conduction Equation on page 262
KMC_MIM_Transport(	Table 260)	Tunneling processes for KMC-MIM simulations. Tunneling Processes on page 1235
LatticeTemperature Limit	=<float>	K Maximum lattice temperature. Break Criteria: Conditionally Stopping the Simulation on page 130
LayerThickness		off Thickness extraction command. Extracting Layer Thickness on page 422
(	ChordWeight=<float>	0 Weight of chord length. Thickness Extraction on page 425
	MaxFitWeight=<float>	0 Thickness Extraction on page 425
	MinAngle=(<float>*2)	0 0 Angle constraints. Thickness Extraction on page 425
	Thickness=<float>)	µm Explicit thickness value.
LED(	Table 282)	Activate LED framework.
MagneticField	=<vector>	T (g) Magnetic field. Chapter 32, Galvanic Transport on page 1127
Mechanics(	Table 325)	(g) Mechanical stress solver. Mechanics Solver on page 1119
MetalResistivity(	<ident> [(Table 380)])	PMI for metal resistivity. Metal Resistivity on page 1573
MetalWorkfunction(	Table 326)	(r) Metal workfunction. Metal Workfunction on page 332
Mobility(	Table 289)	(r) Mobility model. Chapter 15, Mobility on page 429
MoleFraction(	Table 328)	Mole fractions. Mole-Fraction Specification on page 68

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

MSConfigs(	MSConfig (Table 329...)...)	(r) Multistate configurations. Specifying Multistate Configurations on page 657
MSDistResist	=<float>	Ω cm^2 (i) Distributed resistance at metal–semiconductor interface. Resistive Contacts on page 302
	=SchottkyResist	Emulate a Schottky interface. Resistive Contacts on page 302
MSPeltierHeat		(i) Peltier heat at metal–semiconductor interfaces or semiconductor contacts. Heating at Contacts, Metal–Semiconductor and Conductive Insulator–Semiconductor Interfaces on page 1150
MultiValley		off (r) Multivalley statistics. Multivalley Band Structure on page 363, Multivalley Band Structure on page 1058
MultiValley(	DensityIntegral	off (r) Numeric density computation. Using Multivalley Band Structure on page 368
	kpDOS	off (r) k · p model for electrons and holes. Using Multivalley Band Structure on page 368
	MLDA	off (r) Multivalley MLDA quantization model. Modified Local-Density Approximation Model on page 413, Using Multivalley Band Structure on page 1059, Inversion Layer on page 1067
	MLDA (-Nonparabolicity)	off (r) Exclude band nonparabolicity in MLDA. Using the MLDA Model on page 417
	Nonparabolicity	off (r) Band nonparabolicity. Nonparabolic Band Structure on page 364
	parfile	on (r) With kpDOS, it adds the parameter file valleys. Using Multivalley Band Structure on page 368
	RelativeToDOSMass)	off (r) Multivalley effective DOS. Using Multivalley Band Structure on page 368
NBTI(	Table 330)	(i) NBTI degradation model. Two-Stage NBTI Degradation Model on page 703
Noise	[<string>] (Table 331)	(r) Noise sources. Noise Sources on page 875
OpticalAbsorptionHeat(	ScalingFactor=<float>	1 Optical Absorption Heat on page 1161

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

	Table 276)	Optical Absorption Heat on page 1161
Optics(	Table 333)	Specifying the Type of Optical Generation Computation on page 731
Piezo(	Table 335)	(r) Stress and strain models. Using Stress and Strain on page 1043
Piezoelectric_ Polarization	(<ident> [(Table 380)])	Use PMI model <ident> to compute piezoelectric polarization. Piezoelectric Polarization on page 1525
	(strain)	Use strain model to compute piezoelectric polarization. Strain Model on page 1112
	(stress)	Use stress model to compute piezoelectric polarization. Stress Model on page 1114
Polarization		off (r) Use ferroelectric model. Chapter 29, Ferroelectric Materials on page 1003
	(initialvector= <vector>)	(0,0,0) Initial polarization vector for starting the simulation. Using Ferroelectrics on page 1003
	(Memory=<2..>)	10 Maximum nesting of minor loops. Using Ferroelectrics on page 1003
	(Surface Reconstruction)	off (r) Improved algorithm for calculating the directional polarization. Using Ferroelectrics on page 1003
PostTemperature		(g) Use simplified self-heating model. Uniform Self-Heating on page 275
		* Compute dissipated power as integral of Joule heat over entire device.
	(IV_diss)	(g) Compute dissipated power as $\sum IV$ over all electrodes.
	(IV_diss(<string> ...))	(g) Compute dissipated power as $\sum IV$ over user-selected electrodes.
QWLocal(	Table 267)	Localized QW model parameters. Localized Quantum-Well Model on page 1214
Radiation(	Table 295)	Radiation model. Chapter 22, Radiation on page 859

Table 246 Physics{} Physics in Sentaurus Device (Continued)

RandomizedVariation	<string> (Table 337)	(g) Set sIFM models. Statistical Impedance Field Method on page 885
RayTraceBC(	Table 269)	Physics material or region interface–based definition of boundary conditions. Boundary Condition for Raytracing on page 798
RecGenHeat		(g) Generation–recombination processes act as heat sources. Hydrodynamic Model for Temperatures on page 282
(	OptGenOffset=<float>	0.5 Divide contribution of optical generation rate into energy gain/loss terms H_n and H_p. Hydrodynamic Model for Temperatures on page 282
	OptGenWavelength= <float>)	µm Wavelength if optical generation is loaded from file. Hydrodynamic Model for Temperatures on page 282
Recombination(	Table 247)	Generation–recombination model. Chapter 16, Generation–Recombination on page 525
Schottky		off (i) Use Schottky boundary conditions. Electric Boundary Conditions for Metals on page 330
Schroedinger	(Table 338)	(i) Schrödinger solver. 1D Schrödinger Equation on page 393
	<string> (Table 338)	(g) Schrödinger solver on named nonlocal mesh.
SingletExciton		off (ri) Activate region or interface for singlet exciton equation.
(	BarrierType (Table 298)	(i) Set the barrier type at organic heterointerface. Using the Singlet Exciton Equation on page 327
	FluxBC	off (i) Impose a zero flux boundary condition on specified interface. Using the Singlet Exciton Equation on page 327
	Recombination (Table 271))	off (r) Switch on generation and recombination terms in singlet exciton equation. Using the Singlet Exciton Equation on page 327

Feedback

Table 246 Physics{} Physics in Sentaurus Device (Continued)

TATNonlocalPathNC	=<float>	cm^{-3} (i) Room temperature effective density-of-states N_C for electron trap-assisted tunneling from Schottky contacts or Schottky metal–semiconductor interfaces. Using the Dynamic Nonlocal Path Trap-Assisted Tunneling Model on page 537
Temperature	=<float>	300 K (g) Device (lattice) temperature.
TEPower	(<string>)	(gr) Use PMI to compute thermoelectric power. Using Thermoelectric Power on page 1149
	(Analytic)	(gr) Use analytic thermoelectric power. Using Thermoelectric Power on page 1149
	(Tabulated_Si)	(gr) Use a tabulated set of experimental values for silicon to compute the thermoelectric power. Using Thermoelectric Power on page 1149
ThermalConductivity(	Table 341)	(r) Thermal conductivity.
Thermionic		(i) Thermionic emission model for electrons and holes. Conductive Insulators on page 335, Thermionic Emission Current on page 964
	(HCI)	(i) Inject hot carriers locally. Destination of Injected Current on page 936
	(Organic_Gaussian)	(i) Thermionic-like Gaussian emission model at organic heterointerfaces. Gaussian Transport Across Organic Heterointerfaces on page 968
Thermodynamic		off (g) Thermodynamic transport model. Thermodynamic Model for Current Densities on page 269, Thermodynamic Model for Lattice Temperature on page 279
(	NLLTunnelingRecGen Heat)	(g) Add also the nonlocal line tunneling heating mechanisms to the right-hand side of Equation 76. Thermodynamic Model for Lattice Temperature on page 279
Traps((	Table 344)...)	Traps. Chapter 17, Traps and Fixed Charges on page 608
WithConsistentStatistics		+ (g) Use the approximation for Fermi integrals specified in the command file throughout.

Feedback

Generation and Recombination

Table 247 Recombination() Generation–Recombination

<ident>	[(Table 380)]	Use PMI model <ident>. Generation–Recombination on page 1409
Auger		off (r) Auger recombination. Auger Recombination Model on page 544
	(WithGeneration)	off (r) Auger recombination and generation.
Avalanche(	Table 248)	off (r) Impact ionization. Avalanche Generation on page 549
Band2Band		Use Schenk model with density correction. Schenk Band-to-Band Tunneling Model on page 588
	(Table 249)	off (r) Band-to-Band Tunneling Models on page 586
CDL(	Table 275)	off (r) Coupled defect level recombination. Coupled Defect Level Recombination on page 541
ConstantCarrierGeneration	(value=<float>)	off (r) Constant carrier generation. Constant Carrier Generation Model on page 548
eAvalanche(	Table 248)	off (r) Electron impact ionization. Avalanche Generation on page 549
hAvalanche(	Table 248)	off (r) Hole impact ionization. Avalanche Generation on page 549
intrinsicRichter		off (r) Extended recombination model for silicon, covering Auger and radiative recombination. Intrinsic Recombination Model for Silicon on page 545
Radiative		– (r) Radiative recombination. Radiative Recombination Model on page 543
SRH(	Table 275)	off (r) Shockley–Read–Hall recombination. Shockley–Read–Hall Recombination on page 526
SurfaceSRH		off (i) Interface Shockley–Read–Hall recombination. Surface SRH Recombination Model on page 540

Table 247 Recombination() Generation–Recombination (Continued)

TrapAssistedAuger		off (r) Trap-assisted Auger recombination. Trap-Assisted Auger Recombination Model on page 539

Table 248 Avalanche() Generation–Recombination

<ident> [(Table 380)]	Use PMI model <ident>. Avalanche Generation on page 1403
BandgapDependence	– Include a dependency on the energy bandgap in the avalanche generation models. Using Avalanche Generation on page 550
CarrierTempDrive	Use temperature driving force (default for hydrodynamic simulation). Avalanche Generation With Hydrodynamic Transport on page 566
ElectricField	Use plain electric field driving force. Approximate Breakdown Analysis on page 570
Eparallel	Use current-parallel electric field driving force. Driving Force on page 565
GradQuasiFermi	* Use gradient quasi-Fermi potential driving force. For hydrodynamic simulation, default is CarrierTempDrive. Driving Force on page 565
Hatakeyama	Use Hatakeyama model. Hatakeyama Avalanche Model on page 560
Lackner	Use Lackner model. Lackner Model on page 555
Okuto	Use Okuto–Crowell model. Okuto–Crowell Model on page 553
UniBo	Use University of Bologna model. University of Bologna Impact Ionization Model on page 556
UniBo2	Use new University of Bologna impact ionization model. New University of Bologna Impact Ionization Model on page 558
vanOverstraeten	* Use van Overstraeten model. van Overstraeten – de Man Model on page 552

Table 249 Band2Band() Band-to-Band Tunneling Models

DensityCorrection	=Local	Use density correction. Schenk Density Correction on page 589
	=None	* Use plain densities. Schenk Density Correction on page 589

Table 249 Band2Band() Band-to-Band Tunneling Models (Continued)

FranzDispersion		– Use Franz dispersion in the direct nonlocal path model. Using the Nonlocal Path Band-to-Band Tunneling Model on page 596
InterfaceReflection		+ Consider interface reflection in the nonlocal path model. Using Band-to-Band Tunneling on page 586
Model	=E1	Use simple model with *P* = 1. Simple Band-to-Band Tunneling Models on page 592
	=E1_5	Use simple model with *P* = 1.5. Simple Band-to-Band Tunneling Models on page 592
	=E2	Use simple model with *P* = 2. Simple Band-to-Band Tunneling Models on page 592
	=Hurkx	Use Hurkx model. Hurkx Band-to-Band Tunneling Model on page 590
	=modifiedHurkx	Use modified Hurkx model. Modified Hurkx Band-to-Band Tunneling Model on page 591
	=NonlocalPath	Use nonlocal path model. Dynamic Nonlocal Path Band-to-Band Tunneling Model on page 592
	=Schenk	Use Schenk model. Schenk Band-to-Band Tunneling Model on page 588
ParameterSetName	=(<string>...)	Named parameter sets to be used. Using Band-to-Band Tunneling on page 586

Table 250 BPM() Beam Propagation Method

Bidirectional(	Error=<float>	! Relative error used as a break criterion for iterative algorithm in bidirectional beam propagation method.
	Iterations=<int>)	! Maximum number of iterations used as a break criterion for iterative algorithm in bidirectional beam propagation method.
Boundary(	GridNodes=<float>*2	! Number of PML boundary grid nodes to be inserted at left and right sides.
	Order=<1..2>	! Order of spatial variation of complex stretching parameter.
	Side=<string>	! "X", "Y", or "Z".

Feedback

Table 250 BPM() Beam Propagation Method (Continued)

	`StretchingParameter Imag=(<float>*2)`	! Minimum and maximum values of imaginary part of stretching parameter.
	`StretchingParameter Real=(<float>*2)`	! Minimum and maximum values of real part of stretching parameter.
	`Type="PML"`	! Use PML boundary conditions.
	`VacuumGridNodes= (<float>*2))`	! Number of vacuum grid nodes to be inserted at left and right sides.
`Excitation(`	`CenterGauss=<vector>`	µm Gaussian center. Size of vector is $d-1$.
	`SigmaGauss=<vector>`	µm Gaussian half width. Size of vector is $d-1$.
	`TruncationPositionX= (<float>*2)`	Left and right truncation positions of plane wave for x-axis.
	`TruncationPositionY= (<float>*2)`	Left and right truncation positions of plane wave for y-axis.
	`TruncationSlope= <vector>`	Plane-wave truncation slope. Size of vector is $d-1$.
	`Type="Gaussian"`	Use Gaussian excitation.
	`Type="PlaneWave")`	Use truncated plane wave excitation.
`GridNodes`	`=<vector>`	! Number of grid nodes in each spatial dimension.
`ReferenceRefractiveIndex`	`=<float>`	! Value for reference refractive index. General on page 844
	`=average`	Use average refractive index in propagation plane as reference refractive index.
	`=fieldweighted`	Use field-weighted refractive index in propagation plane as reference refractive index.
	`=maximum`	Use maximum refractive index in propagation plane as reference refractive index.

Table 250 BPM() Beam Propagation Method (Continued)

ReferenceRefractive IndexDelta	=<float>	ReferenceRefractiveIndexDelta is added to ReferenceRefractiveIndex in the calculation. To be used for fine-tuning the numerics. General on page 844

Table 251 BreakdownProbability() Avalanche Breakdown Probability

InterpolatedDiscretization		Use enhanced path search approach.
MinElectricField	=<float>	1.0e5 V/cm Minimum value of multiplication region.
Nonlocal()	Table 252	Use nonlocal McIntyre equations.
	lambda_e	20 nm Electron mean free path.
	lambda_h	20 nm Hole mean free path.

Table 252 vanOverstraetendeMan()

vanOverstraetendeMan()	gamma	Model parameter.
	E0_e	4.0e5 Vcm^{-1} Threshold field for electron model parameter.
	aLow_e	7.03e5 cm^{-1} Low field electron model parameter.
	bLow_e	1.231e6 Vcm^{-1} Low field electron model parameter.
	aHigh_e	7.03e5 cm^{-1} High field electron model parameter.
	bHigh_e	1.231e6 Vcm^{-1} High field electron model parameter.
	E0_h	4.0e5 Vcm^{-1} Threshold field for hole model parameter.
	aLow_h	1.582e6 Vcm^{-1} Low field hole model parameter.
	bLow_h	2.036e6 Vcm^{-1} Low field hole model parameter.
	aHigh_h	6.71e5 cm^{-1} High field hole model parameter.
	bHigh_h	1.693e6 Vcm^{-1} High field hole model parameter.

Table 253 eBarrierTunneling() and hBarrierTunneling() Nonlocal Tunneling at Interfaces, Contacts, and Junctions

Band2Band	=None	* No band-to-band tunneling.
	=Full	Include band-to-band tunneling with consistent parallel momentum integral with the direct nonlocal path band-to-band tunneling model.
	=Simple	Include band-to-band tunneling with simple parallel momentum integral.
	=UpsideDown	Include band-to-band tunneling with nonphysical parallel momentum integral.
BandGap		– Allow tunneling into the band gap at the interface for which tunneling is specified. Use this option for backward compatibility only.
BarrierLowering		– Apply a position-dependent barrier lowering for tunneling barrier.
ModifiedWKB		– ModifiedWKB model instead of WKB for tunneling probabilities. Nonlocal Tunneling at Interfaces, Contacts, and Junctions This feature does not work with the TwoBand or Band2Band option.
Multivalley		– Use multivalley band structure. Band-to-Band Contributions to Nonlocal Tunneling Current on page 932
PeltierHeat		– Include Peltier heating terms for tunneling carriers. Equation 872 on page 934
Schroedinger		– Schrödinger equation–based model instead of WKB for tunneling probabilities. This option does not work with the TwoBand or Band2Band option. Schrödinger Equation–Based Tunneling Probability
Transmission		– Use additional interface transmission coefficients. Equation 861 on page 924
TwoBand		– Use two-band dispersion relation. Equation 860 on page 923

Table 254 ElectricField() in SRH() and CDL() SRH Field Enhancement

DensityCorrection	=Local	(r) Use density correction. Density Correction for Schenk and Hurkx Trap-Assisted Tunneling Models on page 534
	=None	* (r) Use local densities.
Lifetime	=Constant	* (r) No field-enhanced lifetime.

Table 254 ElectricField() in SRH() and CDL() SRH Field Enhancement (Continued)

	=Hurkx	(r) Use Hurkx model for lifetime enhancement. Hurkx Trap-Assisted Tunneling Model on page 533
	=Schenk	(r) Use Schenk model for lifetime enhancement. Schenk Trap-Assisted Tunneling Model on page 532

Table 255 Farfield(...) in Physics{Optics(OpticalSolver(RayTracing(...)))} Far Field and Sensors for Raytracing

Origin	=Auto	Automatically compute origin as the center of the device. The default is Auto.
	=<vector>	User-specified origin as a vector in micrometers.
Discretization	=<int>	Default is 360 for two dimensions, and 36 for three dimensions.
ObservationRadius	=<float>	Radius in micrometers. The default is 1e6 µm, that is, 1 m.
Sensor(	Table 273)	Specify a sensor detector.
SensorSweep(	Table 274)	Specify sensor sweep.

Table 256 FDTD() Specifying the Optical Solver

GenerateMesh		Control of tensor mesh generation using Sentaurus Mesh.
(	ForEachWavelength	Generate tensor mesh whenever the wavelength changes compared with the previous FDTD solution.
	Once	Generate tensor mesh once at the beginning of the simulation.
	Wavelength=(<float>*<0..m>))	Specify list of strictly monotonically increasing wavelengths between which the tensor mesh is generated only once.

Table 257 FromFile() Loading Solution of Optical Problem From File and Controlling Interpolation When Loading Optical Generation Profiles

DatasetName	=AbsorbedPhotonDensity | OpticalGeneration	AbsorbedPhotonDensity Name of dataset to be loaded from file.
GridInterpolation	=Conservative | Simple	Interpolation algorithm to be used if source and destination grid are different.

Feedback

Table 257 FromFile() Loading Solution of Optical Problem From File and Controlling Interpolation When Loading Optical Generation Profiles (Continued)

IdentifyingParameter	=(<string>...)	! Name of identifying parameters or parameter paths.
ImportDomain(	Table 258)	Specify source and destination regions as well as domain truncation for interpolation.
ProfileIndex	=<int>	0 ID of loaded profiles after sorting with respect to leading IdentifyingParameter.
ShiftVector	=<vector>	(0,0,0) Displacement of source grid.
SpectralInterpolation	=Linear | Off | PiecewiseConstant	Off Type of interpolation between loaded profiles.
WeightedAPDintegration layers	=<int>	Default is the number of regions with a limit of 20. However, you can choose a higher number.Accurate Absorbed Photon Density for 1D Optical Solvers on page 774
WeightedAPDIntegration SetBoundaries	=(<float>...)	Specifies the boundaries of layers measured from the illumination window in propagation direction.

Table 258 ImportDomain() Controlling Interpolation When Loading Optical Generation Profiles

DestinationBoxCorner1	=<vector>	(-Inf, -Inf, -Inf) Lower-left corner of box in destination grid used to restrict interpolation domain.
DestinationBoxCorner2	=<vector>	(Inf, Inf, Inf) Upper-right corner of box in destination grid used to restrict interpolation domain.
DestinationRegions	=(<string>...)	Regions in destination grid selected for interpolation.
SourceBoxCorner1	=<vector>	(-Inf, -Inf, -Inf) Lower-left corner of box in source grid used to restrict interpolation domain.
SourceBoxCorner2	=<vector>	(Inf, Inf, Inf) Upper-right corner of box in source grid used to restrict interpolation domain.
SourceRegions	=(<string>...)	Regions in source grid selected for interpolation.

Feedback

Table 259 KMCDefects() Kinetic Monte Carlo ReRAM

Diffusion(Particle{i}(		Diffusion event for specified particle.
	Dipole=<float>	Dipole.
	Ea=<float>	Activation energy.
	Frequency=<float>))	Maximum rate of diffusion.
Filament{i}(		Particles defined by keywords Filament1 to Filament3 are conductive particles and are considered in the conductance equation.
	Exclude=<float>	List of excluded particles.
	Name=<string>	Name of the particle.
	ShareCapacity=<float>)	List of particle sharing capacities.
FilamentGrowth(Filament{i}(		Filament growth of Filament{i} from Particle{j}.
	Dipole=<float>	Dipole.
	Ea=<float>	Activation energy.
	Frequency=<float>))	Maximum rate of diffusion.
	Particle{j}))	Specify particle.
FilamentRecession(Filament{i}(		Filament recession of Filament{i} to Particle{j}.
	Dipole=<float>	Dipole.
	Ea=<float>	Activation energy.
	Frequency=<float>))	Maximum rate of diffusion.
	Particle{j}))	Specify particle.
FilamentSeeds(		Seeds of filament. Filament growth starts at seeds.
	Filament{i})	Filament seeds for Filament{iWW}.
FrenkelPair{i}(		Frenkel pair involved in the generation or recombination process.

Table 259 KMCDefects() Kinetic Monte Carlo ReRAM (Continued)

	`Filament{i}( region=<string>)`	Filament in Frenkel pair; region can be specified if at interface.
	`Particle{i}( region=<string>)`	Particle in Frenkel pair; region can be specified if at interface.
	`SameSite)`	Two particles are at the same location (for bulk generation or recombination only).
`Generation( FrenkelPair{i}(`		Generation event for `FrenkelPair{i}`.
	`Dipole=<float>`	Dipole.
	`Ea=<float>`	Activation energy.
	`Frequency=<float>))`	Maximum rate of diffusion.
`Particle{i}(`		Definition of particle type. Sentaurus Device supports three different types of particle and filament.
	`Exclude=<float>`	List of excluded particles.
	`Name=<string>`	Name of the particle.
	`ShareCapacity=<float>)`	List of particle sharing capacities.
`Recombination( FrenkelPair{i}(`		Recombination event for `FrenkelPair{i}`.
	`Dipole=<float>`	Dipole.
	`Ea=<float>`	Activation energy.
	`Frequency=<float>))`	Maximum rate of diffusion.

Table 260 KMC_MIM_Transport() Tunneling Processes

`AvgOccupancyInMIMCharge`		Use average occupancy associated with a defect for electron charge. Inelastic Electrode-to-Trap and Trap-to-Electrode Tunneling on page 1240

Feedback

Table 260 KMC_MIM_Transport() Tunneling Processes (Continued)

`ConductanceEquation`		Switched on only when `TDDB` specified. Time-Dependent Dielectric Breakdown Simulations on page 1253
`ConductivePathCurrent`		Specify conductive path current (CPC) analysis. Conductive Path Current Analysis on page 1250
`DeltaTrapPotential`		Use a field-independent capture coefficient c_0 in calculations. Inelastic Electrode-to-Trap and Trap-to-Electrode Tunneling on page 1240
`DirectTunneling`		Select direct tunneling. Direct Tunneling on page 1236
`ElasticElectrodeToTrap Tunneling`		Select elastic electrode-to-trap tunneling. Elastic Electrode-to-Trap and Trap-to-Electrode Tunneling on page 1239
`ElasticTrapToElectrode Tunneling`		Select elastic trap-to-electrode tunneling. Elastic Electrode-to-Trap and Trap-to-Electrode Tunneling on page 1239
`ElasticUsePZeroSZero`		Use elastic tunneling rates based on inelastic expressions. Option for Elastic Tunneling on page 1240
`EnergyDependentMass`		Select to account for energy-dependent mass. Energy-Dependent Mass on page 1247
`HeatEquation`		Switched on only when `TDDB` specified. Time-Dependent Dielectric Breakdown Simulations on page 1253
`ImageChargeBarrierLowering`		Select to account for image charge barrier lowering. Image Charge Barrier Lowering on page 1246
`IncludeMIMChargeInPoisson`		Include defect and electron charge in solution of Poisson equation. Including the MIM Charge in the Poisson Equation on page 1248
`InelasticElectrodeToTrap Tunneling`		Select inelastic electrode-to-trap tunneling. Image Charge Barrier Lowering on page 1246
`InelasticIgnorePMinus`		Ignore $p < 0$ terms in inelastic tunneling expressions. Options for Inelastic Tunneling on page 1243
`InelasticIncludePZero`		Include p = 0 term in inelastic tunneling expressions. Options for Inelastic Tunneling on page 1243

Feedback

Table 260 KMC_MIM_Transport() Tunneling Processes (Continued)

`InelasticPZeroOnly`		Use only p = 0 term in inelastic tunneling expressions. Options for Inelastic Tunneling on page 1243
`InelasticTrapToElectrode Tunneling`		Select inelastic trap-to-electrode tunneling. Inelastic Electrode-to-Trap and Trap-to-Electrode Tunneling on page 1240
`PooleFrenkelEmission`		Select Poole–Frenkel emission.Poole–Frenkel Emission on page 1238
`PrintBandDiagram`		Write separate band diagram files. Band Diagram With Traps on page 1268
`PrintCPCPaths`		Print information about paths used in CPC calculation. Conductive Path Current Analysis on page 1250
`PrintCPCRates`		Print information about rate calculations used in CPC calculation. Conductive Path Current Analysis on page 1250
`PrintDefectOccupancy`		Print a defect summary, including occupancy and energy to log file and standard output. Occupancy and Energy on page 1264
`PrintMIMEventStatistics`		Print MIM event statistics to standard output. Event Statistics on page 1265
`PrintMIMRates`		Print calculated tunneling rates to log file and standard output. Tunneling Rates on page 1266
`PrintRECRates`		Print tunnel rates calculated by specifying `RateEquationCurrent`. Rate Equation Current Calculation on page 1249
`RateEquationCurrent`		Invoke solution of rate equations for a system of defects and selected tunneling models. Rate Equation Current Calculation on page 1249
`SensitivityAnalysis`		Specify sensitivity analysis. Sensitivity Analysis on page 1251
`(`	`EtRange= (<float>, <float>, <float>)`	eV Specify start, end, and step size for defect energy levels.

Table 260 KMC_MIM_Transport() Tunneling Processes (Continued)

	`PlotType=1 | 2 | 3`	`1` Set type of plot for Sentaurus Visual to generate. Options are: • `1`: Plot absolute energy versus X_t (flat band). • `2`: Plot absolute energy versus X_t (bias included). • `3`: Plot E_t versus X_t.
	`Point=(<float>,<float>,<float>)`	`(0, 0, 0)` µm Specify path for defect locations.
	`PrintAll`	Print leakage values for all (X_t, E_t) pairs to log file.
	`XtRange=(<float>,<float>,<float>))`	µm Specify start, end, and step size for defect locations.
`TrapToTrapTunneling`	`[(Inelastic Phonon) | (MultiPhonon)]`	Select trap-to-trap tunneling model, both elastic and inelastic. Alternatively, select either inelastic phonon or multiphonon trap-to-trap tunneling model. Trap-to-Trap Tunneling on page 1243

Table 261 Medium() Transfer Matrix Method Solver

`ExtinctionCoefficient`	`=<float>`	`1`
`Location`	`=bottom`	Position of medium with respect to extracted layer structure.
	`=top`	! Position of medium with respect to extracted layer structure.
`Material`	`=<string>`	Name of material.
`RefractiveIndex`	`=<float>`	`1`

Table 262 NonlocalPath() in SRH() Dynamic Nonlocal Path Trap-Assisted Tunneling

`Fermi`		– Use Fermi statistics.
`Lifetime`	`=Hurkx`	* Use Hurkx model for lifetime enhancement.

Feedback

Table 262 NonlocalPath() in SRH() Dynamic Nonlocal Path Trap-Assisted Tunneling (Continued)

	=Schenk	Use Schenk model for lifetime enhancement.
TwoBand		– Use Two-band dispersion for the transmission coefficient.

Table 263 OptBeam(()...) Using Optical Beam Absorption Method

LayerStackExtraction(	ComplexRefractiveIndex Threshold=<float>	0 Choose threshold value for layer creation when using ElementWise extraction mode.
	Mode=RegionWise | ElementWise	RegionWise Specify whether layer stack is created on a per-element or per-region basis.
	Position=(<float>*3)	Specify starting point of extraction line.
	WindowName=<string>	Reference to illumination window in Excitation section.
	WindowPosition=<ident>)	Center Specify starting point of extraction line in terms of a cardinal direction (North, South, East, West, NorthEast, SouthEast, NorthWest, SouthWest).

Table 264 OpticalGeneration{} Specifying the Type of Optical Generation Computation and Controlling Interpolation When Loading Optical Generation Profiles

AutomaticUpdate		+ Controls whether optical generation is recomputed if quantities on which it depends are not up-to-date.
ComputeFrom MonochromaticSource		Optical Generation From Monochromatic Source on page 733
(	Scaling=<float>	1
	TimeDependence(Table 278))	
ComputeFromSpectrum		Illumination Spectrum on page 733
(	KeepSpectralData	Keep spectral data (for example, for plotting or to avoid recomputation of optics) when computing the optical generation resulting from an illumination spectrum.

Table 264 OpticalGeneration{} Specifying the Type of Optical Generation Computation and Controlling Interpolation When Loading Optical Generation Profiles (Continued)

	RefreshEveryTime	Force recomputation of respective optical generation contribution at every occasion.
	Scaling=<float>	1
	Select(Table 272)	Active parameters of multidimensional spectrum file.
	TimeDependence(Table 278))	
QuantumYield	=<float>	1 Quantum Yield Models on page 740
QuantumYield(	EffectiveAbsorption	Quantum Yield Models on page 740
	Factor=<float>	1 Quantum Yield Models on page 740
	pmiModel=<string>	Quantum Yield Models on page 740
	StepFunction(Table 277)	Quantum Yield Models on page 740
	Unity)	Quantum Yield Models on page 740
ReadFromFile		Loading and Saving Optical Generation From and to Files on page 738
(	DatasetName= AbsorbedPhotonDensity | OpticalGeneration	
	TimeDependence(Table 278)	
	Scaling=<float>	1
	RefreshEveryTime	Force recomputation of respective optical generation contribution at every occasion.
	GridInterpolation= Simple | Conservative	Interpolation algorithm to be used if source and destination grid are different.
	ShiftVector=<vector>	(0,0,0) Displacement of source grid.
	ImportDomain(Table 258))	Specify source and destination regions as well as domain truncation for interpolation.
Scaling	=<float>	1
SetConstant		Constant Optical Generation on page 739

Table 264 OpticalGeneration{} Specifying the Type of Optical Generation Computation and Controlling Interpolation When Loading Optical Generation Profiles (Continued)

(	RefreshEveryTime	Force recomputation of respective optical generation contribution at every occasion.
	TimeDependence(Table 278)	
	Value=<float>)	s^{-1} cm^{-3} Value for optical generation rate.
TimeDependence(	Table 278)	Specification of type of time dependency.

Table 265 OpticalTurningPoints() Optical Turning Points

Dt	=<float>	Limiting time step for range $[t_0 t_3]$ for time-dependence FromFile only. The default value is given by the smallest time interval specified in the OptGenTransientScaling file defined in the File section. For all other types of time dependence, use Dt to set DtRise, DtPlateau, and DtFall to the same value instead of setting each keyword separately. If any of these keywords is set, it overwrites the value of Dt.
DtFall	=<float>	0 s Limiting time step for range $[t_2 t_3]$. Does not apply to time-dependence FromFile.
DtFallEnd	=<float>	$(t_1 - t_0) / 10$ s Limiting time step at t_3.
DtFallStart	=<float>	$(t_1 - t_0) / 10$ s Limiting time step at t_2. Does not apply to time-dependence FromFile.
DtPlateau	=<float>	0 s Limiting time step for range $[t_1 t_2]$. Does not apply to time-dependence FromFile.
DtRise	=<float>	0 s Limiting time step for range $[t_0 t_1]$. Does not apply to time-dependence FromFile.
DtRiseEnd	=<float>	$(t_1 - t_0) / 10$ s Limiting time step at t_1. Does not apply to time-dependence FromFile.
DtRiseStart	=<float>	$(t_1 - t_0) / 10$ s Limiting time step at t_0.
MinAmplitude	=<float>	0.01 Signal amplitude used to define time points t_0 and t_3 for analytic signals with asymptotic tails.

Table 266 Physics(...){ RayTraceBC(TMM(...)) }

LayerStructure{	<float> <string>; <float> <string>; ... <float> <string>;	Definition of multilayer structure used for TMM calculation. First column contains thickness of layer in µm. Second column contains material name of layer.
MapOptGenToRegions(	<string> <string> ...)	Set list of regions to which the lumped optical generation of the TMM BC is mapped.
QuantumEfficiency	=<float>	Specify the quantum efficiency of the TMM BC optical generation.
ReferenceMaterial	=<string>	Definition of LayerStructure orientation. The topmost layer in the LayerStructure specification is connected to the region with material ReferenceMaterial.
ReferenceRegion	=<string>	Definition of LayerStructure orientation. The topmost layer in the LayerStructure specification is connected to the region with name ReferenceRegion.

Table 267 QWLocal() Localized Quantum-Well Model

eDensityCorrection		– Activate electron quantization model in the quantum well. Quantum-Well Quantization Model on page 421
ElectricFieldDep		+ Switch on electric field dependency for the localized QW model.
hDensityCorrection		– Activate hole quantization model in the quantum well. Quantum-Well Quantization Model on page 421
MaxElectricField	=<float>	1e6 V/cm Cutoff value for electric field.
NumberOfCrystalFieldSplitHoleSubbands	=<int>	Set the maximum number of crystal-field split-hole subbands.
NumberOfElectronSubbands	=<int>	Set the maximum number of electron subbands.
NumberOfHeavyHoleSubbands	=<int>	Set the maximum number of heavy-hole subbands.
NumberOfLightHoleSubbands	=<int>	Set the maximum number of light-hole subbands.
NumberOfValenceBands	=<int>	2 Set the number of valence bands.

Table 267 QWLocal() Localized Quantum-Well Model (Continued)

Polarization	= TE | TM | Mixed	TE Set the polarization used for the computation of the optical transition matrix element. Optical Transition Matrix Element for Wurtzite Crystals on page 1206
PolarizationFactor	=<[0,1]>	1 Set the polarization factor used for the computation of the optical transition matrix element in mixed polarization simulations. Optical Transition Matrix Element for Wurtzite Crystals on page 1206
WidthExtraction(	Table 281)	Set QW width extraction parameters.

Table 268 RayDistribution() Distribution Window of Rays

Dx	=<float>	Specify discretized x-size for Mode=Equidistant.
Dy	=<float>	Specify discretized y-size for Mode=Equidistant.
Mode	=AutoPopulate	Automatically populate rays within the excitation shape.
	=Equidistant	Equidistant distribution of rays within the excitation shape.
	=MonteCarlo	Monte Carlo distribution of rays within the excitation shape.
NumberOfRays	=<int>	Set number of rays for Mode=MonteCarlo or Mode=AutoPopulate.
Scaling	=<float>	Multiply the rays in this window by a scaling factor.
WindowName	=<string>	Specify name of the ray distribution window. No name sets the RayDistribution section as global.

Table 269 RayTraceBC() in Physics material or region interface–based definition of boundary condition

Fresnel		Fresnel boundary condition.
PMIModel	=<ident> [(Table 380)]	Name of the PMI model associated with this BC contact.
Reflectivity	=<[0,1]>	0 Reflectivity.
TMM(	Table 266)	Specification of TMM multilayer structure.

Table 269 RayTraceBC() in Physics material or region interface–based definition of boundary condition (Continued)

Transmittivity	=<[0,1]>	0 Transmittivity.

Table 270 RayTracing(()...) in Physics{Optics(OpticalSolver(...))}, unified raytracing interface Raytracing

CompactMemoryOption		Activate the compact memory model of raytracing.
DepthLimit	=<int>	Stop tracing a ray after passing through more than <int> material boundaries.
ExternalMaterialCRIFile	=<string>	Include a CRI file to define the external media. External Material in Raytracer on page 808
Farfield(	Table 255)	Activate the far field and sensor feature. Far Field and Sensors for Raytracing on page 814
IntensityAbsCoeff	=<float>	Set nonzero absorption coefficient. Computing Optical Intensity on page 810
MinIntensity	=<float>	Stop tracing a ray when its intensity becomes less than <float> times the original intensity. RelativeMinIntensity is an equivalent keyword.
MonteCarlo		Activate Monte Carlo raytracing.
NonSemiconductorAbsorption		Include optical generation calculation in nonsemiconductor region or material.
OmitReflectedRays		Discard all reflected rays from raytracing process.
OmitWeakerRays		Discard the weaker ray at a material interface. This is chosen by comparing the reflectivity and transmittivity.
PlotInterfaceFlux		Activate plotting of interface fluxes on all BCs. Plotting Interface Flux on page 812
PolarizationVector	=Random	Default. Automatically choose a random polarization vector that is perpendicular to the starting ray direction.
	=<vector>	Set a fixed polarization vector for the starting ray.
Print		Create the raytree in the output .tdr file.
Print(Skip(	<int>))	Create a reduced raytree by omitting every user-defined subtree count.

Table 270 RayTracing(()...) in Physics{Optics(OpticalSolver(...))}, unified raytracing interface Raytracing (Continued)

RayDistribution(	Table 268)	Create a ray distribution to be used in conjunction with the excitation illumination window.
RedistributeStoppedRays		Distribute the total power accumulated at terminated rays back into the raytree.
RetraceCRIchange	=<float>	Fractional change of the complex refractive index to force retracing of rays.
UserWindow(	Table 280)	Input a set of starting rays from a file that is specified by the user. User-Defined Window of Rays on page 793
VirtualRegions{	<string> <string> ...}	Specify virtual regions whereby the raytracer will ignore their existence.
WeightedOpticalGeneration		Switch on weighted method to distribute optical generation from element to vertices. Weighted Interpolation for Raytrace Optical Generation on page 810

Table 271 Recombination() in SingletExciton()

Bimolecular	off (r) Switch on bimolecular recombination in continuity and singlet exciton equations. Singlet Exciton Equation on page 325, Bimolecular Recombination Model on page 602
Dissociation	off (i) Switch on interface exciton dissociation in continuity and singlet exciton equations. Singlet Exciton Equation on page 325, Exciton Dissociation Model on page 603
eQuenching	off (r) Switch on quenching of singlet exciton due to free electrons. Singlet Exciton Equation on page 325
hQuenching	off (r) Switch on quenching of singlet exciton due to free holes. Singlet Exciton Equation on page 325
radiative	off (r) Switch on directly radiative decay of singlet exciton associated with light emission.
trappedradiative	off (r) Switch on trap-assisted radiative decay of singlet exciton associated with light emission.

Table 272 Select() Enhanced Spectrum Control

AllowDuplicates		off (g) Control whether duplicate entries of the spectrum are ignored.
Condition	=<string>	"true" Define a Boolean Tcl expression to select a subset of a multidimensional spectrum.
Parameter	=(<string>...)	Active parameters of multidimensional spectrum file.
Var	=<float>	0 Auxiliary parameter that is used to provide more flexibility when specifying a selection condition for the multidimensional spectrum.

Table 273 Sensor(...) in unified raytracing far field Far Field and Sensors for Raytracing

Angular(	Phi=(<float>*2)	Define the range of φ for the angular sensor.
	Theta=(<float>*2))	Define the range of θ for the angular sensor.
Line(	Corner1=<vector>	Define first point of line sensor.
	Corner2=<vector>	Define second point of line sensor.
	UseNormalFlux)	Compute the projected-to-normal flux.
Name	=<string>	Name of the sensor.
Rectangle(	AxisAligned	Take Corner1 and Corner2 as opposing corners of the rectangle sensor.
	Corner1=<vector>	Define first corner of rectangle sensor.
	Corner2=<vector>	Define second corner of rectangle sensor.
	Corner3=<vector>	Define third corner of rectangle sensor.
	UseNormalFlux)	Compute the projected-to-normal flux.

Table 274 SensorSweep in unified raytracing far field Far Field and Sensors for Raytracing

Name	=<string>	Specify name of sensor sweep.
Ndivisions	=<int>	Set number of subdivisions for the collection ring.
Phi	=(<float>*2)	Specify range of φ for the sensor ring.

Table 274 SensorSweep in unified raytracing far field Far Field and Sensors for Raytracing (Continued)

Theta	=(<float>*2)	Specify range of θ for the sensor ring.
VaryPhi		Set the sensor sweep as a latitude ring.
VaryTheta		Set the sensor sweep as a longitudinal ring.

Table 275 SRH() and CDL() Shockley–Read–Hall Recombination, Coupled Defect Level Recombination

<ident>	[(Table 380)]	Use PMI model <ident> to compute lifetimes. Lifetimes on page 1412
DopingDependence		Doping dependence. SRH Doping Dependence on page 527
ElectricField(	Table 254)	(r) Field enhancement. SRH Field Enhancement on page 531
ExpTempDependence		Exponential temperature dependence. SRH Temperature Dependence on page 529
NonlocalPath(	Table 262)	(r) Nonlocal trap-assisted tunneling enhancement. Dynamic Nonlocal Path Trap-Assisted Tunneling on page 535
TempDependence		Temperature dependence. SRH Temperature Dependence on page 529

Table 276 StepFunction() Optical Absorption Heat

Bandgap		Use $E_g - E_{bgn}$ as the cutoff energy for interband absorption.
EffectiveBandgap		Use $(E_g - E_{bgn} + 2 \cdot (3/2)kT)$ as the cutoff energy for interband absorption.
Energy	=<float>	eV Use specified value as the cutoff energy for interband absorption.
Wavelength	=<float>	μm Use specified value as the cutoff energy for interband absorption.

Table 277 StepFunction() Quantum Yield Models

Bandgap		Use E_g as the cutoff energy for interband absorption.
EffectiveBandgap		Use $E_g - E_{bgn}$ as the cutoff energy for interband absorption.
Energy	=<float>	eV Use specified value as the cutoff energy for interband absorption.

Table 277 StepFunction() Quantum Yield Models (Continued)

Wavelength	=<float>	µm Use specified value as the cutoff energy for interband absorption.

Table 278 TimeDependence() Specifying Time Dependency for Transient Simulations

FromFile		off Reads the time dependency as a table from file.
OpticalTurningPoints(	Table 265)	+ (g) Specification of optical turning points. Optical Turning Points on page 749.
Scaling	=<float>	1 Scaling factor for optical generation.
WavePeriods	=<int>	Number of periods of the periodic signal.
WaveTime	=(<float>*2)	s Time interval $(t_{\min}, t_{\max})$ when the optical generation rate is constant.
WaveTPeriod	=<float>	s Period of periodic signal.
WaveTPeriodOffset	=<float>	s Offset of periodic signal (only applies to linear and Gaussian time dependency).
WaveTSigma	=<float>	s Standard deviation σ_t of the temporal Gaussian distribution that describes the decay of the optical generation rate outside the time interval WaveTime.
WaveTSlope	=<float>	s^{-1} Slope that characterizes the linear decay of the optical generation rate outside the time interval WaveTime.

Table 279 TMM() Transfer Matrix Method

IntensityPattern		(r) Choose type of intensity pattern.
	=Envelope	Compute the envelope of the optical intensity instead of the regular optical intensity.
	=StandingWave	* Compute the regular optical intensity without applying any algorithm that filters out oscillations on the wavelength scale.
LayerStackExtraction(	ComplexRefractive IndexThreshold=<float>	0 Choose threshold value for layer creation when using ElementWise extraction mode.
	Medium(Table 261))	Specification of media surrounding the extracted layer structure.

Table 279 TMM() Transfer Matrix Method (Continued)

	Mode=RegionWise | ElementWise	RegionWise Specify whether layer stack is created on a per-element or per-region basis.
	Position=(<float>*3)	Specify starting point of extraction line.
	WindowName=<string>	Reference to illumination window in Excitation section.
	WindowPosition=<ident>)	Center Specify starting point of extraction line in terms of a cardinal direction (North, South, East, West, NorthEast, SouthEast, NorthWest, SouthWest).
NodesPerWavelength	=<float>	Number of nodes per wavelength used for computation of optical field.
PropagationDirection	=Perpendicular | Refractive	Refractive Choose interpolation mode for 1D TMM solution.
RoughInterface		+ Flag interfaces as rough, that is, physics of rough interface scattering is applied.
Scattering(	AngularDiscretization=<int>	91 Angular discretization of interval –π/2, π/2 used in scattering solver.
	MaxNumberOfIterations=<int>	10 Break criterion for iterative scattering solver.
	Tolerance=<float>)	1e-3 Break criterion for iterative scattering solver.

Table 280 UserWindow in Physics{Optics(OpticalSolver(Raytracing(...)))}, User-Defined Window of Rays

NumberOfRays	=<int>	Set number of rays in file.
PolarizationVector	=Random	Generate random polarization for the user input rays.
	=ReadFromExcitation	Read the polarization from the Excitation section.
	=ReadFromFile	Read the polarization vectors from the user input ray file.
RaysFromFile	=<string>	Set file name of the user input rays.

Feedback

Table 281 WidthExtraction() Accelerating Gain Calculations and LED Simulations

ChordWeight	=<float>	Specify chord weight for width extraction. Thickness Extraction on page 425
MinAngle	=(<float>, <float>)	Specify minimum angles for width extraction. Thickness Extraction on page 425
SideMaterial	=("mat1", ..., "matn")	Specify the materials adjoining the QW.
SideRegion	=("reg1", ..., "regn")	Specify the regions adjoining the QW.

LED

Note:

LED simulations present unique challenges that require problem-specific model and numerics setups. Contact TCAD Support for advice if you are interested in simulating LEDs (see Contacting Your Local TCAD Support Team Directly).

Table 282 LED() Light-Emitting Diodes

Bandstructure(	CrystalType= Zincblende | Wurtzite)	Zincblende Crystal structure of active region. Electronic Band Structure for Wurtzite Crystals on page 1201
Broadening(	Table 283)	Activate gain-broadening models or nonlinear gain saturation model. Gain-Broadening Models on page 1199, Simple Quantum-Well Subband Model on page 1208
Optics(	Table 285)	Optics.
QWExtension	=AutoDetect	Autodetect width of quantum wells. The quantum-well region must be specified by the keyword Active. Radiative Recombination and Gain Coefficients on page 1195
QWTransport		Use ‘three-point’ QW model with thermionic emission. Radiative Recombination and Gain Coefficients on page 1195
SplitOff	=<float>	eV Spin-orbit split-off energy. Strain Effects on page 1212

Table 282 LED() Light-Emitting Diodes (Continued)

SponEmissionCoeff	(<ident> [(Table 380)])	Use PMI model <ident> for spontaneous emission. Importing Gain and Spontaneous Emission Data With PMI on page 1218
SponIntegration	(<float>,<int>)	eV Energy integration range and number of discretization intervals for numeric integration (for LED simulations only). Spontaneous Emission Rate and Power on page 1156
SponScaling	=<float>	Scaling factor for matrix element of spontaneous emission. Radiative Recombination and Gain Coefficients on page 1195
StimEmissionCoeff	(<ident> [(Table 380)])	Use PMI model <ident> for stimulated emission. Importing Gain and Spontaneous Emission Data With PMI on page 1218
StimScaling	=<float>	Scaling factor for matrix element of stimulated emission. Radiative Recombination and Gain Coefficients on page 1195
Strain	(RefLatticeConst= <float>)	m Use strain model for quantum well with given reference lattice constant. Strain parameters are input in the parameter file. Strain Effects on page 1212

Table 283 Broadening() Gain-Broadening Models

Gamma	=<float>	eV Broadening coefficient Γ.
Type	=CosHyper	Use hyperbolic-cosine broadening. Hyperbolic-Cosine Broadening on page 1200
	=Landsberg	Use Landsberg broadening. Landsberg Broadening on page 1200
	=Lorentzian	Use Lorentzian broadening. Landsberg Broadening on page 1200

Table 284 ClusterActive() in LED(Optics(RayTrace())) Clustering Active Vertices

ClusterQuantity	=Nodes	Clustering by recursive grouping of active vertices.
	=OpticalGridElement	Clustering by grouping active vertices in each active optical element.
	=PlaneArea	Clustering by evenly dividing up QW plane area and grouping active vertices in each area element.

Table 284 ClusterActive() in LED(Optics(RayTrace())) Clustering Active Vertices (Continued)

NumberOfClusters	=<int>	Specify number of clusters, only for Nodes or PlaneArea clustering.

Table 285 Optics() in LED() LED Optics: Raytracing

RayTrace(	Table 288)	Raytracing.

Table 286 OutputLightToolsFarfieldRays() in LED(Optics(RayTrace())) Interfacing Far-Field Rays to LightTools

Filename	=<string>	Base file name of the LightTools® ray data file to be output.
SaveType	=Ascii	Choose ASCII format for LightTools ray data file.
	=Binary	Choose binary format for LightTools ray data file.
WavelengthDiscretization	=<int>	Number of discretization for the spectrum. The span of the spectrum is determined automatically.

Table 287 OutputLightToolsRays() in LED(Optics(RayTrace(Disable()))) Interfacing LED Starting Rays to LightTools

IsotropyType	=InBuilt	Use the internal geodesic distribution of starting rays from each active emission vertex cluster.
	=Random	Use a random distribution of starting rays.
	=UserRays	Use the user input set of starting rays defined by the keyword SourceRaysFromFile (see Table 288).
RaysPerCluster	=<int>	Set number of starting rays in each active cluster.
SaveType	=Ascii	Choose ASCII format for LightTools ray data file.
	=Binary	Choose binary format for LightTools ray data file.
WavelengthDiscretization	=<int>	Number of discretization for the spectrum. The span of the spectrum is determined automatically.

Table 288 RayTrace() in LED(Optics()) LED Optics: Raytracing

ClusterActive(	Table 284)	Activate the clustering option. Clustering Active Vertices on page 1170
CompactMemoryOption		Activate the compact memory model for LED raytracing. Will not work with the full photon-recycling model.
Coordinates	=Cartesian	*
	=Cylindrical	
DebugLEDRadiation	(<string> <float1> <float2> <float3>)	off Trace the origin of the output rays that are within the angles [<float1>,<float2>] (in degrees) and of minimum intensity specified by the <float3> parameter. In two dimensions, the angle is defined in the regular polar coordinates. In three dimensions, the angles are defined from the z-axis, as in θ in regular spherical coordinates. The results are output to the file specified by <string>.
DepthLimit	=<int>	5 Stop tracing the ray after passing through more than <int> material boundaries.
Disable		off Deactivate raytracing but still run LED simulation.
Disable(OutputLightTools Rays(	Table 287))	Output starting rays from active vertices to a LightTools ray data file. Interfacing LED Starting Rays to LightTools on page 1174
EmissionType	(Anisotropic(Sine(<float>*3) Cosine(<float>*3)))	
	(Isotropic)	*
ExcludeHorizontalSource	(<float>)	off Omit the source rays that are emitted within the horizontal angular zone specified by the <float> parameter (in degrees).
LEDRadiationPara	(<float>, <int>)	µm Observation radius and discretization of the observation circle or sphere.
LEDSpectrum	(<float>*2 <int>)	eV Starting and ending energy range, and number of subdivisions in that energy range.
MinIntensity	=<float>	1e-5 Stop tracing the ray when the intensity of a ray becomes less than <float> times the original intensity.

Feedback

Table 288 RayTrace() in LED(Optics()) LED Optics: Raytracing (Continued)

MonteCarlo		Activate Monte Carlo raytracing.
MoveBoundaryStartRays	=<float>	0 nm Shift the starting ray position at device boundary inwards. Recommended values are 1 to 5 nm.
NonActiveAbsorptionOff		Do not add nonactive region absorption as generation rate to continuity equation. Nonactive Region Absorption (Photon Recycling) on page 1186
ObservationCenter	=<vector>	Fixed observation center for LED radiation.By default, center of device.
OmitReflectedRays		Discard all reflected rays in raytracing process.
OmitWeakerRays		Discard the weaker ray at a material interface. This is decided by comparing the reflectivity and transmittivity.
OptGenScaling	=<float>	Set a multiplication factor to the nonactive optical generation computed.
OutputLightToolsFarfield Rays(	Table 286)	Output far-field rays with embedded spectrum information into a LightTools ray data file. Interfacing Far-Field Rays to LightTools on page 1184
PolarizationVector	=Random	Default. Automatically choose a random polarization vector that is perpendicular to the starting ray direction.
	=<vector>	Set a user-defined polarization vector.
Print		off Output all rays to the output .tdr file.
	(ActiveVertex (<int>))	off Output only ray paths emitted from this active vertex.
	(Skip(<int>))	1 Output every other <int> ray to the output .tdr file.
PrintRayInfo	(<string>)	off Print all indices and positions of starting rays into the file specified by <string>.
PrintSourceVertices	(<string>)	off Print the index and coordinates of all active vertices into the file specified by <string>.
ProgressMarkers	=<int>	5 Completion meter for raytracing.

Feedback

Table 288 RayTrace() in LED(Optics()) LED Optics: Raytracing (Continued)

RaysPerVertex	=<int>	10 Number of rays starting from each active source vertex. For 3D, the number of starting rays are constrained by 6, 18, 68, and so on. The number in the sequence is chosen such that RaysPerVertex is slightly larger or equal to it.
RaysRandomOffset		Randomize the angular shift of the starting rays.
	(RandomSeed=<int>)	Set a fixed random seed so that repeated simulations will yield the exact pseudorandom results.
RedistributeStoppedRays		Distribute the total accumulated power in terminated rays back into the raytree.
RetraceCRIchange	=<float>	Fractional change of the complex refractive index to force retracing of rays.
SourceRaysFromFile(	<string>)	Import a set of starting ray directions from the file name specified by <string>. The number of imported rays are specified by RaysPerVertex.
Staggered3DFarfieldGrid		Use the staggered 3D far-field collection sphere. Staggered 3D Grid LED Radiation Pattern on page 1180
TraceSource	()	Retrace the source of the output rays to produce a map of the source regions that give the strongest ray output.
TurnOffTreeNodeCount		Deactivate counting the total number of nodes in the raytree.
Wavelength	=<float>	nm Wavelength.
	=AutoPeak	Take wavelength at the peak of the spontaneous emission rate spectrum. LED Wavelength on page 1160
	=AutoPeakPower	Take wavelength at the peak of the spontaneous emission power spectrum.
	=Effective	Wavelength computed such that total power = total rate × effective photon energy.

Feedback

Mobility

Table 289 Mobility(), eMobility(), hMobility() Mobility

BalMob(		off Use ballistic mobility model. Ballistic Mobility Model on page 510
	Fermi	off Use Fermi–Dirac correction in the ballistic mobility. Using the Ballistic Mobility Model on page 513
	Frensley	off Use Frensley rule for final mobility. Using the Ballistic Mobility Model on page 513
	KineticEnergy	off Use kinetic ballistic mobility model. Using the Ballistic Mobility Model on page 513
	Lch=<float>	10^7 nm Use explicit channel length in simple ballistic mobility. Using the Ballistic Mobility Model on page 513
	TempDep)	off Use temperature dependency in simple ballistic mobility. Using the Ballistic Mobility Model on page 513
BandTailHighField Saturation(	Table 290)	off Use band tail mobility. Band Tail Mobility on page 506
BTCSMob(	Nonlocal	off Account for scattering by remote traps. General Model for Mobility Degradation by Traps in the Bulk and at Semiconductor Interfaces on page 515
	NonlocalLengthLimit= <float>)	0 cm Distance up to which to account for remote traps, when Nonlocal option is active.
CarrierCarrier Scattering	(BrooksHerring)	off Use Brooks–Herring carrier–carrier scattering model. Carrier–Carrier Scattering on page 441
	(ConwellWeisskopf)	off Use Conwell–Weisskopf carrier–carrier scattering model. Carrier–Carrier Scattering on page 441
CNTMob(	Lch=<float>)	off Use CNT mobility model. Carbon Nanotube Mobility Model on page 481
ConstantMobility		+ Use constant mobility if neither PhuMob nor DopingDependence is specified. Mobility due to Phonon Scattering on page 431

Table 289 Mobility(), eMobility(), hMobility() Mobility (Continued)

Diffusivity(	Table 290)	off Use non-Einstein diffusivity. Non-Einstein Diffusivity on page 505
DopingDependence(	<ident> [(Table 380)]	off PMI model <ident>. Doping-Dependent Mobility on page 1422
	Arora	off Use Arora doping-dependent mobility model. Doping-Dependent Mobility Degradation on page 431
	BalMob([Lch=<float>])	off Use low-field ballistic mobility model. Low-Field Ballistic Mobility Models on page 439
	Masetti	off Use Masetti doping-dependent mobility model. Doping-Dependent Mobility Degradation on page 431
	PhuMob [see PhuMob options in Table 289]	off Use Philips unified mobility model. Philips Unified Mobility Model on page 443
	PhuMob2	off Use an alternative Philips model. Using an Alternative Philips Unified Mobility Model on page 444
	PMIModel(Table 336)	off Use PMI for MSC-dependent mobility. Multistate Configuration–Dependent Apparent Band-Edge Shift on page 1466
	UniBo)	off Use University of Bologna doping-dependent mobility model. Doping-Dependent Mobility Degradation on page 431
eBandTailHighField Saturation(	Table 290)	off Use band tail mobility for electrons. Band Tail Mobility on page 506
eDiffusivity(	Table 290)	off Use non-Einstein electron diffusivity. Non-Einstein Diffusivity on page 505
eHighFieldSaturation(	Table 290)	off Electron high-field saturation. High-Field Saturation Models on page 487
Enormal(	<ident> [(Table 380)]	off PMI model <ident>. Mobility Degradation at Interfaces on page 1443
	Coulomb2D	off ‘Two-dimensional’ ionized impurity mobility degradation. Mobility Degradation Components due to Coulomb Scattering on page 466

Table 289 Mobility(), eMobility(), hMobility() Mobility (Continued)

	IALMob(Table 291)	off Inversion and accumulation layer mobility model. Mobility Degradation at Interfaces on page 448
	InterfaceCharge [(SurfaceName=<string>)]	off Negative and positive interface charge mobility degradation. Mobility Degradation Components due to Coulomb Scattering on page 466
	Lombardi(Table 292)	off Enhanced Lombardi model. Mobility Degradation at Interfaces on page 448
	Lombardi_highk	off Enhanced Lombardi model with high-k degradation. Mobility Degradation at Interfaces on page 448
	NegInterfaceCharge [(SurfaceName=<string>)]	off Negative interface charge mobility degradation. Mobility Degradation Components due to Coulomb Scattering on page 466
	PosInterfaceCharge [(SurfaceName=<string>)]	off Positive interface charge mobility degradation. Mobility Degradation Components due to Coulomb Scattering on page 466
	RCS	off Remote Coulomb scattering mobility degradation. Remote Coulomb Scattering Model on page 470
	RPS	off Remote phonon scattering mobility degradation. Remote Phonon Scattering Model on page 473
	UniBo)	off University of Bologna surface mobility model. Mobility Degradation at Interfaces on page 448
hBandTailHighField Saturation(	Table 290)	off Use band tail mobility for electrons. Band Tail Mobility on page 506
hDiffusivity(	Table 290)	off Use non-Einstein hole diffusivity. Non-Einstein Diffusivity on page 505
hHighFieldSaturation(	Table 290)	off Hole high-field saturation. High-Field Saturation Models on page 487
HighFieldSaturation(	Table 290)	off High-field saturation. High-Field Saturation Models on page 487

Table 289 Mobility(), eMobility(), hMobility() Mobility (Continued)

IncompleteIonization		off Incomplete ionization-dependent mobility. Incomplete Ionization–Dependent Mobility Models on page 516
MultiValley		off Use the explicit multivalley mobility model. Explicit Multivalley Mobility Model on page 493
PhuMob		off Philips unified mobility model. Philips Unified Mobility Model on page 443
(	Arsenic	* Use arsenic parameters. Table 62 on page 447
	Phosphorus)	Use phosphorus parameters. Table 62 on page 447
ThinLayer		off Thin-layer mobility model. Thin-Layer Mobility Model on page 476
(	ChordWeight=<float>	0 Weight of chord length. Thickness Extraction on page 425
	IALMob(Table 291)	Use the inversion and accumulation layer mobility model with the thin-layer mobility model. Thin-Layer Mobility Model on page 476
	Lombardi(Table 292)	* Use the enhanced Lombardi model with the thin-layer mobility model. Thin-Layer Mobility Model on page 476
	MinAngle=(<float>*2)	0 0 Angle constraints. Thickness Extraction on page 425
	Thickness=<float>)	µm Explicit thickness value.
ToCurrentEnormal	as Enormal	off Dependence on electric field normal to current. Mobility Degradation at Interfaces on page 448
Tunneling		– Tunneling correction to mobility. Equation 251 on page 406

Table 290 HighFieldSaturation(), Diffusivity() High-Field Saturation Models

<ident>	[(Table 380)]	PMI model <ident>. High-Field Saturation on page 1430

Feedback

Table 290 HighFieldSaturation(), Diffusivity() High-Field Saturation Models (Continued)

AutoOrientation		– Use a parameter set based on the orientation of the nearest interface. Auto-Orientation for the High-Field Saturation Models on page 489
CarrierTempDrive		Temperature driving force (default for hydrodynamic simulation). Hydrodynamic Driving Force on page 502
CarrierTempDriveBasic		Basic temperature driving force. Basic Model on page 495
CarrierTempDriveME		Meinerzhagen–Engl temperature driving force. Meinerzhagen–Engl Model on page 495
CarrierTempDrivePolynomial		Energy-dependent mobility model using an irrational polynomial. Energy-Dependent Mobility on page 973
CarrierTempDriveSpline		Energy-dependent mobility model using spline interpolation. Spline Interpolation on page 974
CaugheyThomas		* Use Canali–Hänsch model. Extended Canali Model on page 489
DensityDependentVsat		Carrier density-dependent velocity saturation model. Density-Dependent Velocity Saturation Model on page 497
ElectricField		Electric field. Electric Field on page 502
Eparallel		Use electric field parallel to current as driving force. Electric Field Parallel to the Current on page 499
EparallelToInterface		Use electric field parallel to the closest semiconductor–insulator interface as driving force. Electric Field Parallel to the Interface on page 500
GradQuasiFermi		* Use gradient of quasi-Fermi potential as driving force. For hydrodynamic simulations, default is CarrierTempDrive. Gradient of Quasi-Fermi Potential on page 499
ParameterSetName	=<string>	Name of parameter set to use. Named Parameter Sets for the High-Field Saturation Models on page 489
PFMob		Use Poole–Frenkel mobility model. Poole–Frenkel Mobility on page 507

Feedback

Table 290 HighFieldSaturation(), Diffusivity() High-Field Saturation Models (Continued)

PMIModel(	Table 336)	PMI model dependent on two fields. High-Field Saturation With Two Driving Forces on page 1439
TransferredElectronEffect		Use transferred electron model. Transferred Electron Model on page 491
TransferredElectronEffect2		Use an alternative transferred electron model. Transferred Electron Model 2 on page 491
VRHMob		Use variable range hopping mobility model. Variable Range Hopping Transport Mobility on page 509

Table 291 IALMob() Mobility Degradation at Interfaces

AutoOrientation		– Use a parameter set based on the orientation of the nearest interface. Auto-Orientation for the IALMob Model on page 463
ClusteringEverywhere		– Use clustering formulas for all occurrences of N_{A} and N_{D}. Inversion and Accumulation Layer Mobility Model on page 454
FullPhuMob		+ Use the full PhuMob mobility expressions. Inversion and Accumulation Layer Mobility Model on page 454
ParameterSetName	=<string>	Name of IALMob parameter set. Named Parameter Sets for the IALMob Model on page 463
PhononCombination	=<0..2>	1 Selects how 2D and 3D phonon mobility are combined. Inversion and Accumulation Layer Mobility Model on page 454

Table 292 Lombardi() Mobility Degradation at Interfaces

AutoOrientation		– Use a parameter set based on the orientation of the nearest interface. Auto-Orientation for the Lombardi Model on page 453
ParameterSetName	=<string>	Name of EnormalDependence parameter set. Named Parameter Sets for the Lombardi Model on page 453

Radiation Models

Table 293 AlphaParticle() Carrier Generation by Alpha Particles

Direction	=<vector>	! Direction of motion of the particle.

Table 293 AlphaParticle() Carrier Generation by Alpha Particles (Continued)

Energy	=<float>	! eV Energy of the alpha particle.
Location	=<vector>	! µm Point where the alpha particle enters the device (bidirectional track).
StartPoint	=<vector>	! µm Point where the alpha particle enters the device (one-directional track).
Time	=<float>	! s Time at which the charge generation peaks.

Table 294 HeavyIon() Carrier Generation by Heavy Ions

Direction	=<vector>	Direction of motion of the ion.
Exponential		* Use exponential shape for the spatial distribution $R(w)$.
Gaussian		Use Gaussian shape for the spatial distribution $R(w)$.
Length	=[<float>...]	Length l where Wt_hi and LET_f are specified (in cm by default or µm if the PicoCoulomb option is selected).
LET_f	=[<float>...]	Linear energy transfer (LET) function (in pairs cm^{-3} by default or pC μm^{-1} if the PicoCoulomb option is selected). Entries in the list match those in Length.
Location	=<vector>	µm Point where the heavy ion enters the device (bidirectional track).
PicoCoulomb		Switch units for LET_f, Wt_hi, and Length.
SpatialShape	=<ident> [(Table 380)]	PMI for spatial distribution function. Heavy Ion Spatial Distribution on page 1604
StartPoint	=<vector>	µm Point where the heavy ion enters the device (one-directional track).
Time	=<float>	s Time at which the ion penetrates the device.
Wt_hi	=[<float>...]	Characteristic distance, $w_t(l)$ (in cm by default or µm if the PicoCoulomb option is selected). Entries in the list match those in Length.

Table 295 Radiation() Radiation

Dose	=<float>	rad Total dose over exposure time.
DoseRate	=<float>	rad s^{-1} Dose rate.

Feedback

Table 295 Radiation() Radiation (Continued)

DoseTime	=(<float>*2)	! s Exposure time.
DoseTSigma	=<float>	! s Standard deviation of rise and fall of dose rate over time.

Various

Table 296 Aniso() Anisotropic Properties

Avalanche		Use anisotropic avalanche generation. Anisotropic Avalanche Generation on page 996
direction	[(<System_Coord>)]	Crystal or simulation system coordinate.
	=(<float>*3)	Anisotropic direction. Anisotropic Direction in 3D Cases on page 983
	=xAxis	Equivalent to direction=(1 0 0).
	=yAxis	Equivalent to direction=(0 1 0).
	=zAxis	Equivalent to direction=(0 0 1).
eAvalanche		Use anisotropic electron avalanche generation. Anisotropic Avalanche Generation on page 996
eMobility		Use self-consistent anisotropic mobility for electrons. Self-Consistent Anisotropic Mobility on page 993
eMobilityFactor	(Total)=<float>	Total anisotropic mobility factor for electrons. Total Anisotropic Mobility on page 992
eQuantumPotential		Use electron anisotropic density-gradient equation with tensor grid approximation. Density Gradient Model on page 404
hAvalanche		Use anisotropic hole avalanche generation. Anisotropic Avalanche Generation on page 996
hMobility		Use self-consistent anisotropic mobility for holes. Self-Consistent Anisotropic Mobility on page 993
hMobilityFactor	(Total)=<float>	Total anisotropic mobility factor for holes. Total Anisotropic Mobility on page 992

Feedback

Table 296 Aniso() Anisotropic Properties (Continued)

hQuantumPotential		Use hole anisotropic density-gradient equation with tensor grid approximation. Using the Density Gradient Model on page 406
MetalResistivity		Use self-consistent anisotropic metal resistivity. Self-Consistent Anisotropic Metal Resistivity on page 996
MetalResistivity Factor	(Total)=<float>	Total anisotropic metal resistivity factor. Total Anisotropic Metal Resistivity Factor on page 995
Mobility		Use self-consistent anisotropic mobility. Self-Consistent Anisotropic Mobility on page 993
Poisson		Use anisotropic electrical permittivity. Anisotropic Electrical Permittivity on page 998
Temperature		Use anisotropic thermal conductivity. Anisotropic Thermal Conductivity on page 1000

Table 297 BandEdge() in RandomizedVariation() Band Edge Variations

Table 240		Statistical properties of the correlation.
Table 339		Spatial restriction of variation.
Amplitude_Chi	=<float>	eV Amplitude of affinity variation for exponential and Gaussian correlation.
Amplitude_Eg	=<float>	eV Amplitude of bandgap variation for exponential and Gaussian correlation.
Chi2Eg	=<float>	0 Correlation between affinity and band gap.
ChiComponent	=<0..1>	0 Named random field component for affinity randomization.
EgComponent	=<0..1>	1 Named random field component for bandgap randomization.
GrainChi	=(<float>...)	eV Grain affinity values.
GrainEg	=(<float>...)	eV Grain bandgap values.
GrainProbability	=(<float>...)	Probability for grain types.
Volume	=<string>	Named volume in which to activate the variation.

Table 298 BarrierType() in SingletExciton() Singlet Exciton Equation

Bandgap		* (i) Use bandgap difference as barrier.
CondBand		(i) Use conduction band-edge difference as barrier.
ValBand		(i) Use valence band-edge difference as barrier.

Table 299 ComplexRefractiveIndex() Complex Refractive Index Model

CarrierDep(	imag	Use carrier dependency of extinction coefficient. Carrier Dependency on page 779
	real)	Use carrier dependency of refractive index. Carrier Dependency on page 779
CRIModel(	Name=<string>)	Activate user-defined complex refractive index model. Complex Refractive Index Model Interface on page 1502
GainDep	(real(lin))	Use linear gain dependency of refractive index. Gain Dependency on page 780
	(real(log))	Use logarithmic gain dependency of refractive index. Gain Dependency on page 780
TemperatureDep(	real)	Use temperature dependency of refractive index. Temperature Dependency on page 778
WavelengthDep(	imag	Use wavelength dependency of extinction coefficient. Wavelength Dependency on page 778
	real)	Use wavelength dependency of refractive index. Wavelength Dependency on page 778

Table 300 Conductivity() in RandomizedVariation() Metal Conductivity Variations

Table 240		Statistical properties of the correlation.
Table 339		Spatial restriction of variation.
Amplitude	=<float>	A/cm V Amplitude of conductivity variation for exponential and Gaussian correlation.
GrainConductivity	=(<float>...)	A/cm V Grain conductivity values.
GrainProbability	=(<float>...)	Probability for grain types.

Table 300 Conductivity() in RandomizedVariation() Metal Conductivity Variations (Continued)

Volume	=<string>	Named volume in which to activate the variation.

Table 301 DeterministicVariation() Deterministic Variations

DopingVariation	<string>(Table 304)	Deterministic Doping Variations on page 897
GeometricVariation	<string>(Table 319)	Deterministic Geometric Variations on page 899
ParameterVariation	<string>(Table 334)	Parameter Variations on page 900

Table 302 Doping() in Noise() Random Dopant Fluctuations

Table 339		Spatial restriction of fluctuations.
BandgapNarrowing		Include effect on bandgap narrowing.
Mobility		Include effect on mobility.
Type	=Acceptor	Only acceptors fluctuate.
	=Donor	Only donors fluctuate.
	=Doping	* All dopants fluctuate.

Table 303 Doping() in RandomizedVariation() Doping Variations

Table 339		Spatial restriction of variation.
BandgapNarrowing		+ Account for doping dependency of bandgap narrowing.
Mobility		+ Account for doping dependency of mobility.
Type	=Acceptor	Randomize acceptors only.
	=Donor	Randomize donors only.
	=Doping	* Randomize all dopants.

Table 304 DopingVariation() in DeterministicVariation() Deterministic Doping Variations

Table 339		Spatial restriction of variation.

Table 304 DopingVariation() in DeterministicVariation() Deterministic Doping Variations (Continued)

Amplitude	=<vector>	µm Vectorial amplitude for gradient specification.
Amplitude_Abs	=<vector>	µm Vectorial amplitude for absolute gradient specification.
Amplitude_Iso	=<float>	µm Isotropic amplitude for gradient specification.
BandgapNarrowing		+ Account for doping dependency of bandgap narrowing.
Conc	=<float>	cm^{-3} Normalization concentration.
Factor	=<float>	1 Multiplier for variation.
Mobility		+ Account for doping dependency of mobility.
SFactor	=<string>	Dataset name for variation.
Type	=Acceptor	Apply variation to acceptor concentration.
	=Donor	Apply variation to donor concentration.
	=Doping	* Apply variation according to sign to acceptor or donors.

Table 305 DopingVariation() in RandomizedVariation() Doping Profile Variations

Table 240		Statistical properties of the correlation.
Table 339		Spatial restriction of variation.
Amplitude	=<float>	0 µm Vectorial amplitude.
Amplitude_Iso	=<float>	0 µm Isotropic amplitude.
BandgapNarrowing		+ Account for doping dependency of bandgap narrowing.
Conc	=<float>	0 cm^{-3} Normalization concentration.
Mobility		+ Account for doping dependency of mobility.
SFactor	=<string>	! Dataset name for variation.
Type	=Acceptor	Apply variation to acceptor concentration.
	=Donor	Apply variation to donor concentration.
	=Doping	* Apply variation according to sign to acceptor or donors.

Table 306 EffectiveIntrinsicDensity() Band Gap and Electron Affinity

BandGap	(<ident> [(Table 380)])	Use PMI model <ident> to compute band gap E_g. Band Gap on page 1454
BandGapNarrowing	(<ident> [(Table 380)])	Use PMI model <ident> for bandgap narrowing. Bandgap Narrowing on page 1457
	(BennettWilson)	* Bennett–Wilson model.
	(delAlamo)	del Alamo model.
	(JainRoulston)	Jain–Roulston model.
	(oldSlotboom)	Old Slotboom model.
	(Slotboom)	Slotboom model.
	(TableBGN)	Table-based model.
NoBandGapNarrowing		No bandgap narrowing.
NoFermi		off Omit correction Equation 177 even when using Fermi statistics. Bandgap Narrowing With Incomplete Ionization on page 350

Table 307 eLucky(), Lucky(), eFiegna(), Fiegna() Effective Field

CarrierTempDrive		Use field derived from temperature.
CarrierTempPost		Use field derived from postprocessed temperature.
DynamicBarrierHeight		- Compute dynamic barrier height for each injection path. Injection Barrier and Image Potential on page 939
Eparallel		* Use field parallel to interface.

Table 308 hLucky(), hFiegna() Effective Field

CarrierTempDrive		Use field derived from temperature.
CarrierTempPost		Use field derived from postprocessed temperature.
Eparallel		* Use field parallel to interface.

Table 309 eNMP() Extended Nonradiative Multiphonon Model

Conc	=<float>	0 cm^{-2} Precursor concentration for the eNMP model.
NumberOfSamples	=<int>	0 Number of random samples.
SFactor	=<ident> [(Table 380)]	Name of the PMI model to compute the eNMP precursor concentration. Using the eNMP Model on page 711 and Space Factor on page 1624
	=<string>	Dataset for the spatial distribution of the eNMP precursor concentration. Using the eNMP Model on page 711

Table 310 Epsilon() in RandomizedVariation() Dielectric Constant Variations

Table 240		Statistical properties of the correlation.
Table 339		Spatial restriction of variation.
Amplitude	=<float>	1 Amplitude of relative dielectric constant variation for exponential and Gaussian correlation.
GrainEpsilon	=(<float>...)	1 Grain relative dielectric constant values.
GrainProbability	=(<float>...)	Probability for grain types.
Volume	=<string>	Named volume in which to activate the variation.

Table 311 eQuantumPotential() and hQuantumPotential() Density Gradient Model

AnisoAxes	[(<System_Coord>)]	Crystal or simulation system coordinate.
	={<vector> <vector>}	Two directions of anisotropy. Anisotropic Direction in 3D Cases on page 983
AutoOrientation		off Use parameter set based on orientation of nearest interface. Auto-Orientation for the Density Gradient Model on page 410
BoundaryCondition	=Dirichlet	(ci) Enforce homogeneous Dirichlet boundary condition.
	=HomogeneousNeumann	(ci) Enforce homogeneous Neumann boundary condition on region interfaces.

Table 311 eQuantumPotential() and hQuantumPotential() Density Gradient Model (Continued)

	=Neumann	(ci) Enforce homogeneous Neumann boundary condition.
Density		– (r) Use Equation 246 instead of Equation 247.
direction	[(<System_Coord>)]	Crystal or simulation system coordinate.
	=(<float>*3)	Anisotropic direction. Anisotropic Direction in 3D Cases on page 983
	=xAxis	Equivalent to direction=(1 0 0).
	=yAxis	Equivalent to direction=(0 1 0).
	=zAxis	Equivalent to direction=(0 0 1).
Formula	=<0..1>	0 (r) Select between formulation with mass *before* or *between* differential operators.
Gamma(	EffectiveMass	+ Quantization mass is density of states mass. Gamma Factor for Density Gradient Model on page 1599
	name=<string> [(Table 380)])	PMI for γ. Gamma Factor for Density Gradient Model on page 1599
Ignore		– (r) Compute, but do not apply quantum correction.
LocalModel	=<ident> [(Table 380)]	(r) Name of PMI for apparent band-edge shift. Chapter 14, Quantization on page 390, Apparent Band-Edge Shift on page 1451
	=SchenkBGN_elec	(r) Schenk bandgap narrowing model for electrons. Schenk Bandgap Narrowing Model on page 346
	=SchenkBGN_hole	(r) Schenk bandgap narrowing model for holes. Schenk Bandgap Narrowing Model on page 346
LocalModelOnly		– (r) Use with the LocalModel specification. Automatically sets γ = 0 in density gradient specification.
ParameterSetName	=<string>	Name of QuantumPotentialParameters parameter set. Named Parameter Sets for the Density Gradient Model on page 410

Table 311 eQuantumPotential() and hQuantumPotential() Density Gradient Model (Continued)

Resolve		– (r) Use more accurate discretization of non-heterointerfaces.

Table 312 eSHEDistributionl(), hSHEDistribution() Spherical Harmonics Expansion Method

AdjustImpurityScattering		+ Use an option to adjust impurity scattering to match the low-field mobility specified in the Physics section.
FullBand		– Use the full band structure with the default band-structure file.
	=<string>	! File name of the user-defined band-structure file.
RTA		– Use relaxation time approximation.

Table 313 ExternalBoltzmannSolver() External Boltzmann Solver

Carriers=(	Electron | Hole)	Use quantum correction for electrons or holes.
ContactName	=<string>	Name of contact for which Sentaurus Device sends the value of the current to the Subband-BTE solver of Sentaurus Device QTX for outputting current–voltage characteristics.
DampingLength	=<float>	0.005 µm Distance from volume enclosed by slices beyond which quantum correction and effective mobility, if activated, is deactivated.
MaxMismatch	=<float>	1e-4 µm Tolerance in matching of slice meshes and device mesh.
Mobility(	<carrier>	Activate extraction of effective mobility by Subband-BTE solver of Sentaurus Device QTX, which is applied to 3D domain enclosed by slices in Sentaurus Device. The specification of exactly one carrier type is mandatory.

Table 313 ExternalBoltzmannSolver() External Boltzmann Solver (Continued)

	(-DisableInternalMobilityModels))	Activate Sentaurus Device mobility models so that the effective mobility extracted by the Subband-BTE solver of Sentaurus Device QTX can complement certain mobility models specified in the Sentaurus Device command file.
NumberOfSlices	=<2..>	* Number of slices.
SBTECommandFile	=<string>	Start the Subband-BTE solver automatically using this command file.
Volume	=<string>	Domain in which to restrict quantum corrections and effective mobility, if activated.

Table 314 ExternalSchroedinger() External 2D Schrödinger Solver

Carriers=(	Electron	Use quantum correction for electrons.
	Hole)	Use quantum correction for holes.
DampingLength	=<float>	0.005 µm Distance from volume enclosed by slices beyond which quantum correction is deactivated.
MaxMismatch	=<float>	1e-4 µm Tolerance in matching of slice meshes and device mesh.
NumberOfSlices	=<2..>	* Number of slices.
SBandCommandFile	=<string>	Start Sentaurus Band Structure automatically, using this command file.
Volume	=<string>	Domain to which to restrict quantum corrections.

Table 315 GateCurrent() Tunneling, Hot-Carrier Injection

<ident>(	<carrier>... [Table 380])	(i) Use PMI model <ident> to compute hot-carrier injection. Hot-Carrier Injection on page 1607
DirectTunneling		(i) Schenk direct tunneling model. Direct Tunneling on page 911
eFiegna	[(Table 307)]	(i) Fiegna model for electrons. Fiegna Hot-Carrier Injection on page 943

Feedback

Table 315 GateCurrent() Tunneling, Hot-Carrier Injection (Continued)

eLucky	[(Table 307)]	(i) Lucky electron model. Classical Lucky Electron Injection on page 942
eSHEDistribution		(i) SHE distribution model for electrons. SHE Distribution Hot-Carrier Injection on page 945
Fiegna	[(Table 307)]	(i) Fiegna model for electrons and holes. Fiegna Hot-Carrier Injection on page 943
Fowler		(i) Fowler–Nordheim tunneling. Fowler–Nordheim Tunneling on page 909
	(EVB)	(i) Fowler–Nordheim valence band tunneling of electrons. Fowler–Nordheim Tunneling on page 909
GateName	=<string>	(i) Electrode for monitoring gate current. Overview of Hot-Carrier Injection Models on page 935
hFiegna	[(Table 308)]	(i) Fiegna model for holes. Fiegna Hot-Carrier Injection on page 943
hLucky	[(Table 308)]	(i) Lucky hole model. Classical Lucky Electron Injection on page 942
hSHEDistribution		(i) SHE distribution model for holes. SHE Distribution Hot-Carrier Injection on page 945
InjectionRegion	=<string>	Regions into which hot carriers are injected. Destination of Injected Current on page 936
InjectionTargetMaterials	=(<string>...)	Materials into which hot carriers are injected, for specifying injection current destination. Destination of Injected Current on page 936
InjectionTargetRegions	=(<string>...)	Regions into which hot carriers are injected, for specifying injection current destination. Destination of Injected Current on page 936
Interface		(i) Activate carrier injection with explicitly evaluated boundary conditions for continuity equations during a transient (Carrier Injection With Explicitly Evaluated Boundary Conditions for Continuity Equations on page 961) or if not in transient monitor interface current. Overview of Hot-Carrier Injection Models on page 935

Table 315 GateCurrent() Tunneling, Hot-Carrier Injection (Continued)

Lucky	[(Table 307)]	(i) Lucky model for electrons and holes. Classical Lucky Electron Injection on page 942

Table 316 Gaussian() in IntensityDistribution() Spatial Intensity Function Excitation

Length	=(<float>,<float>)	Length of Gaussian decay, cannot specify together with Sigma.
PeakPosition	=(<float>,<float>)	Peak position of the Gaussian profile, in local coordinate of the defined shape function.
PeakWidth	=(<float>,<float>)	Width of the plateau of the modified Gaussian profile.
Scaling	=<float>	Scaling factor for the Gaussian profile.
Sigma	=(<float>,<float>)	Sigma of Gaussian decay, cannot specify together with Length.

Table 317 Geometric() in RandomizedVariation() Geometric Variations

Table 240		Statistical properties of the correlation.
Table 339		Spatial restriction of variation.
Amplitude	=<vector>	0 µm Vectorial amplitude for displacement.
Amplitude_Iso	=<float>	0 µm Isotropic amplitude for displacement.
Options	=<0..1>	0 Select approximation specific to insulator position variations.
Surface	=<string>	! Name of varying surface.
WeightDielectric	=<float>	0 Interpolation coefficient between two approximations of dielectric term.
WeightQuantumPotential	=<float>	0.5 Interpolation coefficient between two approximations for density-gradient quantum-correction term.

Table 318 GeometricFluctuations in Noise() Random Geometric Fluctuations

Table 339		Spatial restriction of variation.

Table 318 GeometricFluctuations in Noise() Random Geometric Fluctuations (Continued)

Options	=<0..1>	0 Select approximation specific to insulator position variations.
WeightDielectric	=<float>	0 Interpolation coefficient between two approximations of dielectric term.
WeightQuantumPotential	=<float>	0.5 Interpolation coefficient between two approximations for density-gradient quantum-correction term.

Table 319 GeometricVariation() in DeterministicVariation() Deterministic Geometric Variations

Table 339		Spatial restriction of variation.
Amplitude	=<vector>	µm Vectorial amplitude for displacement.
Amplitude_Iso	=<float>	µm Isotropic amplitude for displacement.
Options	=<0..1>	0 Select approximation specific to insulator position variations.
Surface	=<string>	! Name of varying surface.
WeightDielectric	=<float>	0 Interpolation coefficient between two approximations of dielectric term.
WeightQuantumPotential	=<float>	0.5 Interpolation coefficient between two approximations for density-gradient quantum-correction term.

Table 320 HeatCapacity() Heat Capacity

<ident>	[(Table 380)]	PMI model <ident> for lattice heat capacity. Heat Capacity on page 1481
Constant		Constant lattice heat capacity.
PMIModel(	Table 336)	Use multistate configuration–dependent model.
TempDep		* Temperature-dependent lattice heat capacity.

Table 321 HydrogenAtom(), HydrogenIon(), HydrogenMolecule(), HydrogenSpeciesA(), HydrogenSpeciesB(), HydrogenSpeciesC() Hydrogen Transport

Alpha	=<ident>	Use PMI model for prefactor of the thermal diffusion term α_{td}.

Table 321 HydrogenAtom(), HydrogenIon(), HydrogenMolecule(), HydrogenSpeciesA(), HydrogenSpeciesB(), HydrogenSpeciesC() Hydrogen Transport (Continued)

Diffusivity	=<ident>	Use PMI model for $D_i \exp(-E_{di}/(kT))$.

Table 322 HydrogenDiffusion() Hydrogen Transport

HydrogenAtom(	Table 321)	Hydrogen atom specification.
HydrogenIon(	Table 321)	Hydrogen ion specification.
HydrogenMolecule(	Table 321)	Hydrogen molecule specification.
HydrogenReaction(	Table 323)	Reactions Between Mobile Elements on page 694
HydrogenSpeciesA(	Table 321)	Hydrogen species A specification.
HydrogenSpeciesB(	Table 321)	Hydrogen species B specification.
HydrogenSpeciesC(	Table 321)	Hydrogen species C specification.

Note:

In Table 323, d is 3 for bulk reactions and 2 for interface reactions.

Table 323 HydrogenReaction() Reactions Between Mobile Elements

FieldFromMaterial	=<string>	(i) Material where the electric field is obtained.
FieldFromRegion	=<string>	(i) Region where the electric field is obtained.
ForwardReactionCoef	=<float>	0 $/\mathrm{cm}^d\mathrm{s}$ Forward reaction coefficient.
ForwardReactionEnergy	=<float>	0 eV Forward reaction activation energy.
ForwardReactionFieldCoef	=<float>	0 cm/V Forward reaction field coefficient.
LHSCoef		Define particle numbers to be removed.
(	Electron=<int>	0 Number of electrons to be removed.
	Hole=<int>	0 Number of holes to be removed.
	HydrogenAtom=<int>	0 Number of hydrogen atoms to be removed.
	HydrogenIon=<int>	0 Number of hydrogen ions to be removed.

Feedback

Table 323 HydrogenReaction() Reactions Between Mobile Elements (Continued)

	HydrogenMolecule=<int>	0 Number of hydrogen molecules to be removed.
	HydrogenSpeciesA=<int>	0 Number of hydrogen species A to be removed.
	HydrogenSpeciesB=<int>	0 Number of hydrogen species B to be removed.
	HydrogenSpeciesC=<int>)	0 Number of hydrogen species C to be removed.
Material	=<string>	(i) Material where the recombination rate enters.
Region	=<string>	(i) Region where the recombination rate enters.
ReverseReactionCoef	=<float>	0 $/\mathrm{cm}^d\mathrm{s}$ Reverse reaction coefficient.
ReverseReactionEnergy	=<float>	0 eV Reverse reaction activation energy.
ReverseReactionField Coef	=<float>	0 cm/V Reverse reaction field coefficient.
RHSCoef		Define particle numbers to be created.
(	Electron=<int>	0 Number of electrons to be created.
	Hole=<int>	0 Number of holes to be created.
	HydrogenAtom=<int>	0 Number of hydrogen atoms to be created.
	HydrogenIon=<int>	0 Number of hydrogen ions to be created.
	HydrogenMolecule=<int>	0 Number of hydrogen molecules to be created.
	HydrogenSpeciesA=<int>	0 Number of hydrogen species A to be removed.
	HydrogenSpeciesB=<int>	0 Number of hydrogen species B to be removed.
	HydrogenSpeciesC=<int>)	0 Number of hydrogen species C to be removed.

Table 324 IncompleteIonization() Incomplete Ionization

Dopants	=<string>	Restrict incomplete ionization to dopants named in <string>. Dopant names are separated by spaces.
Model(	<ident>(<string> [Table 380]))	Use PMI model <ident> for species named in <string>. Incomplete Ionization on page 1614
Split(	Doping=<string>	Dopant that is redistributed into multiple lattice sites. Multiple Lattice Sites on page 387
	Weights=(<float>...))	Occupation probabilities of the various lattice sites. Multiple Lattice Sites on page 387

Table 325 Mechanics() Mechanics Solver

binary	=<string>	Name of the Sentaurus Interconnect binary. Mechanics Solver on page 1119
command	=<string>	Sentaurus Interconnect Tcl commands. Mechanics Solver on page 1119
initial_structure	=<string>	Initial structure for first call of Sentaurus Interconnect. Mechanics Solver on page 1119
parameter	=<string>	Sentaurus Interconnect parameters. Mechanics Solver on page 1119

Table 326 MetalWorkfunction() Metal Workfunction

Randomize(	AtInsulatorInterface	off (r) Randomize workfunction at metal–insulator vertices only. Metal Workfunction Randomization on page 333
	AverageGrainSize=<float>	! (r) Average metal grain size. Metal Workfunction Randomization on page 333
	GrainProbability=(<float>...)	! (r) Probabilities that grains will have a certain workfunction value. Metal Workfunction Randomization on page 333
	GrainWorkfunction=(<float>...)	! eV (r) Workfunction values corresponding to the above probabilities. Metal Workfunction Randomization on page 333
	RandomSeed=<int>	Seed for the random number generator. Metal Workfunction Randomization on page 333

Table 326 MetalWorkfunction() Metal Workfunction (Continued)

	UniformDistribution)	off (r) Attempt to evenly distribute uniformly sized grains. Metal Workfunction Randomization on page 333
SFactor	=<string>	Dataset for the spatial distribution of metal workfunction. Metal Workfunction on page 332
	=<ident> [(Table 380)]	Name of model to be used by the PMI to compute the spatial distribution of metal workfunction. Space Factor on page 1624
	[Factor=<float>]	! Scale factor for normalized SFactor values. Metal Workfunction on page 332
	[Offset=<float>]	! Offset for raw or normalized SFactor values. Metal Workfunction on page 332
Workfunction	=<float>	eV Metal workfunction. Metal Workfunction on page 332
	=(<float> <float>*3) ...	eV, µm, µm, µm Metal workfunction–position quadruplets. Metal Workfunction on page 332

Table 327 Model(), options of stress-dependent models Mechanical Stress

Deformation Potential		Linear deformation potential model for computing band structure. Deformation of Band Structure on page 1046
(	ekp	k · p method for electron bands to account for shear strain components. Deformation of Band Structure on page 1046
	hkp	6 × 6 k · p method for hole bands. Deformation of Band Structure on page 1046
	Minimum	Compute conduction and valence band edges using minimum band energies. Deformation of Band Structure on page 1046
	multivalley)	Compute conduction and valance band edges using Multivalley band structure. Using Deformation Potential Model on page 1050
DOS(	eMass	Strained electron effective mass and DOS model. Strained Electron Effective Mass and DOS on page 1052

Feedback

Table 327 Model(), options of stress-dependent models Mechanical Stress (Continued)

	hMass	Strained hole effective mass and DOS model. Strained Hole Effective Mass and DOS on page 1055
	hMass(AnalyticLTFit)	Strained hole effective mass and DOS model with analytic lattice temperature fit. Strained Hole Effective Mass and DOS on page 1055
	hMass(NumericalIntegration))	Strained hole effective mass and DOS model with numeric integrations. Strained Hole Effective Mass and DOS on page 1055
Mobility(	eFactor[(Table 340)]	Piezoresistive factor for electrons. Isotropic Factor Models on page 1098
	eMinorityFactor=<float>	1.0 Factor to scale stress effect for minority electrons. Stress Mobility Model for Minority Carriers on page 1106
	eMinorityFactor(DopingThreshold=<float>) =<float>	1.0 Factor to scale stress effect for minority electrons with no doping dependency. Stress Mobility Model for Minority Carriers on page 1106
	eSaturationFactor=<float>	1.0 Saturation factor for electrons. Dependency of Saturation Velocity on Stress on page 1107
	eSubband(BalMob)	Stress-induced change of the electron ballistic mobility. Multivalley Ballistic Mobility Model on page 1078
	eSubband(Doping)	Strain-induced subband model for electrons with doping dependency. Multivalley Electron Mobility Model on page 1062
	eSubband(EffectiveMass))	Stress-induced change of the electron effective mass. Effective Mass on page 1065
	eSubband(EffectiveMass(-Transport))	Exclude 2D inverse transport mass tensor transformation. Effective Mass on page 1065
	eSubband(EffectiveMass(Transport<vector>))	Activate 1D inverse transport mass tensor transformation. Effective Mass on page 1065
	eSubband(EqRefMob)	Activate the equilibrium reference mobility computation. Using Multivalley Transferred Carrier Mobility Model on page 1080

Feedback

Table 327 Model(), options of stress-dependent models Mechanical Stress (Continued)

	eSubband(Fermi)	Strain-induced subband model for electrons with carrier concentration (Fermi statistics) dependency. Multivalley Electron Mobility Model on page 1062
	eSubband(NoStressInDDmob)	Account for only ballistic mobility. Using Multivalley Ballistic Mobility Model on page 1079
	eSubband(Scattering))	Strain-induced subband model for electrons with scattering. Intervalley Scattering on page 1063
Mobility(	eSubband(Scattering(MLDA))	Strain-induced subband model with interface scattering. Intervalley Scattering on page 1063
	eSubband(-RelChDir110)	Use <100> channel direction in reference mobility. Using Multivalley Electron Mobility Model on page 1069
	eSubband(-AutoOrientation)	Use full tensor reference mobility. Using Multivalley Electron Mobility Model on page 1069
	eTensor[(Table 340)]	Piezoresistive tensor for electrons. Piezoresistance Mobility Model on page 1086
	Factor[(Table 340)]	Piezoresistive factor for electrons and holes. Isotropic Factor Models on page 1098
	hFactor[(Table 340)]	Piezoresistive factor for holes. Isotropic Factor Models on page 1098
	hMinorityFactor=<float>	1.0 Factor to scale stress effect for minority holes. Stress Mobility Model for Minority Carriers on page 1106
	hMinorityFactor(DopingThreshold=<float>) =<float>	1.0 Factor to scale stress effect for minority holes with no doping dependency. Stress Mobility Model for Minority Carriers on page 1106
	hSaturationFactor=<float>	1.0 Saturation factor for holes. Dependency of Saturation Velocity on Stress on page 1107
	hSixband	Intel stress-induced model for holes. Intel Stress-Induced Hole Mobility Model on page 1081

Feedback

Table 327 Model(), options of stress-dependent models Mechanical Stress (Continued)

	hSixband(Doping)	Intel stress-induced model for holes with doping dependency. Intel Stress-Induced Hole Mobility Model on page 1081
	hSixband(Fermi)	Intel stress-induced model for holes with carrier concentration (Fermi statistics) dependency. Intel Stress-Induced Hole Mobility Model on page 1081
	hSubband(BalMob)	Stress-induced change of the hole ballistic mobility. Multivalley Ballistic Mobility Model on page 1078
Mobility(	hSubband(Doping)	Strain-induced subband model for holes with doping dependency. Multivalley Ballistic Mobility Model on page 1078
	hSubband(EffectiveMass)	Stress-induced change of the holes effective mass. Effective Mass on page 1072
	hSubband(EffectiveMass (Transport))	Activate 2D inverse transport mass tensor transformation. Effective Mass on page 1072
	hSubband(EffectiveMass (Transport<vector>))	Activate 1D inverse transport mass tensor transformation. Effective Mass on page 1072
	hSubband(EqRefMob)	Activate the equilibrium reference mobility computation. Using Multivalley Transferred Carrier Mobility Model on page 1080
	hSubband(Fermi)	Strain-induced subband model for holes with carrier concentration (Fermi statistics) dependency. Multivalley Hole Mobility Model on page 1072
	hSubband(NoStressInDDmob)	Account for only ballistic mobility. Using Multivalley Ballistic Mobility Model on page 1079
	hSubband(Scattering)	Strain-induced subband model for holes with bulk scattering. Scattering on page 1073
	hSubband(Scattering(MLDA))	Strain-induced subband model for holes with interface scattering. Scattering on page 1073
	hSubband(-RelChDir110)	Use <100> channel direction in reference mobility. Using Multivalley Hole Mobility Model on page 1076
	hSubband(-AutoOrientation)	Use full tensor reference mobility. Using Multivalley Hole Mobility Model on page 1076

Table 327 Model(), options of stress-dependent models Mechanical Stress (Continued)

	hTensor[(Table 340)]	Piezoresistive tensor for holes. Piezoresistance Mobility Model on page 1086
	SaturationFactor=<float>	1.0 Saturation factor for electrons and holes. Dependency of Saturation Velocity on Stress on page 1107
	Tensor[(Table 340)])	Piezoresistive tensor for electrons and holes. Piezoresistance Mobility Model on page 1086

Table 328 MoleFraction() Mole-Fraction Specification

Grading ((		Alternative way to specify grading; allows a nonzero mole fraction and different distance of grading from different parts of the boundaries.
	GrDistance=<float>	µm Distance in the direction normal to the specified interface, where linear interpolation of mole fractions from the constant value to the specified boundary mole fractions occurs.
	RegionInterface= (<string>*2)	Restrict this grading to the given interface (applied to all interfaces by default).
	xFraction=<[0,1]>	Boundary xMoleFraction.
	yFraction= <[0,1]>)...)	Boundary yMoleFraction.
GrDistance	=<float>	µm Distance in the direction normal to the boundaries of the specified regions where linear interpolation of mole fractions from the specified constant value to 0 occurs.
RegionName	=[<string>...]	List of regions where the mole-fraction specification will take effect.
	=<string>	Regions where the mole-fraction specification will take effect.
xFraction	=<[0,1]>	Constant value of xMoleFraction.
yFraction	=<[0,1]>	Constant value of yMoleFraction.

Table 329 MSConfig() Phase and State Transitions

BandEdgeShift(	<ident> [(Table 380)] [<int>])	Compute band-edge shifts for conduction and valence by PMI model <ident> with optional constructor argument <int>. Apparent Band-Edge Shift on page 666
Conc	=<float>	0 cm^{-d} Concentration of states ($d = 3$ for bulk; $d = 2$ for interface MSCs).
eBandEdgeShift(	<ident> [(Table 380)] [<int>])	Compute band-edge shift for conduction band by PMI model <ident> with optional constructor argument <int>. Apparent Band-Edge Shift on page 666
Elimination		+ Switch solving algorithm.
hBandEdgeShift(	<ident> [(Table 380)] [<int>])	Compute band-edge shift for valence band by PMI model <ident> with optional constructor argument <int>. Apparent Band-Edge Shift on page 666
Name	=<string>	! Identifier of MSConfig.
State(		! Define a state (at least two required).
	Charge=<int>	0 Number of positive elementary charges.
	Hydrogen=<int>	0 Number of hydrogen atoms.
	Name=<string>	! Identifier of state.
	TTOM)	– Activate trap occupancy for the given state.
Transition(	Table 343)	! Define a transition.

Table 330 NBTI() Two-Stage NBTI Degradation Model

Conc	=<float>	0 cm^{-2} Concentration of NBTI traps.
NumberOfSamples	=<int>	0 Number of random samples.
hSHEDistribution		– Use hole-energy distribution function.

Table 331 Noise() Noise, Fluctuations, and Sensitivity

BandEdgeFluctuations	<string> (Table 339)	Band edge fluctuations. Random Band Edge Fluctuations on page 883
ConductivityFluctuations	<string> (Table 339)	Metal conductivity fluctuations. Random Band Edge Fluctuations on page 883

Feedback

Table 331 Noise() Noise, Fluctuations, and Sensitivity (Continued)

DiffusionNoise(	Table 339	Diffusion noise, spatially restricted. Diffusion Noise on page 875
	e_h_Temperature	Noise temperatures are the respective carrier temperatures.
	eTemperature	Noise temperature is electron temperature for electrons, lattice temperature for holes.
	hTemperature	Noise temperature is lattice temperature for electrons, hole temperature for holes.
	LatticeTemperature)	* Noise temperature is lattice temperature.
Doping(	Table 302)	Random dopant fluctuations
EpsilonFluctuations	<string> (Table 339)	Dielectric constant fluctuations. Random Dielectric Constant Fluctuations on page 884
FlickerGRNoise(	Table 339)	Bulk flicker noise. Bulk Flicker Noise on page 877
GeometricFluctuations	<string> (Table 318)	Geometric fluctuations for surface <string>. Random Geometric Fluctuations on page 879
MonopolarGRNoise(	Table 339)	Equivalent monopolar noise. Equivalent Monopolar Generation–Recombination Noise on page 876
TrapConcentration(	Table 339)	Trap concentration fluctuations. Random Trap Concentration Fluctuations on page 881
Traps(	Table 339)	Trapping noise. Trapping Noise on page 877
WorkfunctionFluctuations	<string> (Table 339)	Workfunction fluctuations for surface <string>. Random Workfunction Fluctuations on page 882

Table 332 Optics() Transfer Matrix Method and Beam Propagation Method

Table 285		Optics standalone.
BPMScalar(	Table 250)	Scalar beam propagation method (BPM) solver. Beam Propagation Method on page 842
TMM(	Table 279)	Transfer matrix method (TMM) solver. Transfer Matrix Method on page 819

Feedback

Table 333 Optics() Specifying the Type of Optical Generation Computation

ComplexRefractive Index(	Table 299)	Complex refractive index models. Complex Refractive Index Model on page 777
Excitation		Specification of excitation parameters. Setting the Excitation Parameters on page 757
(	fromBottom | fromTop	Growth orientation from the bottom or the top of the device structure. Common Illumination Configurations on page 762
	Intensity=<float>	W cm^{-2}
	Phi=<float>	0 deg Angle between projection of propagation direction on xy plane and x-axis.
	Polarization=<float>	0 [0,1] Definition of polarization as in Using Transfer Matrix Method on page 825.
	Polarization=<ident>	TM Transverse electric (TE) or transverse magnetic (TM) polarized light.
	PolarizationAngle= <float>	0 deg Angle between H-field and $z \times \hat{k}$ used for vectorial solvers only.
	ReferenceSurface= IlluminationWindow | PropagationDirectionNormal)	PropagationDirectionNormal Reference surface with respect to which the intensity is defined (only applies to TMM and OptBeam solvers). Setting the Excitation Parameters on page 757
Excitation(	Theta=<float>	0 deg Angle between propagation direction and z-axis.
	Wavelength=<float>	µm
	Window(Table 345))	
OpticalGeneration(	Table 264)	Optical generation models. Specifying the Type of Optical Generation Computation on page 731
OpticalSolver		Specification of optical solver method. Specifying the Optical Solver on page 751
(	BPM(Table 250)	

Table 333 Optics() Specifying the Type of Optical Generation Computation (Continued)

	Composite	Composite Method on page 756
	FDTD(Table 256)	
	FromFile(Table 257)	
	OptBeam(Table 263)	
	RayTracing(Table 270)	Raytracer on page 785
	TMM(Table 279))	Note that Excitation section must not be defined in TMM section. Using Transfer Matrix Method on page 825
Verbosity	=<int>	1 Verbosity level for optical solver output. A verbosity level of 0 suppresses all optical solver output.

Table 334 ParameterVariation() in DeterministicVariation() Parameter Variations

Factor	=<float>	1 1 Multiplier for original parameter value.
Material	=<string>	Material location of varied parameter.
MaterialInterface	=<string>	Material interface location of varied parameter.
Model	=<string>	! Model to which varied parameter belongs.
Parameter	=<string>	! Name of the varied parameter.
Region	=<string>	Region location of varied parameter.
RegionInterface	=<string>	Region interface location of varied parameter.
Summand	=<float>	0 Summand to original parameter value.
Value	=<float>	Modified parameter value.

Table 335 Piezo() Mechanical Stress

Model(	Table 327)	Stress-dependent models. Deformation of Band Structure on page 1046 to Mobility Modeling on page 1061
OriKddX	=<vector>	(1 0 0) Miller indices of the stress system relative to the simulation system.

Table 335 Piezo() Mechanical Stress (Continued)

OriKddY	=<vector>	(0 1 0) Miller indices of the stress system relative to the simulation system.
Strain	=(<float>*6)	Components xx, yy, zz, yz, xz, xy of strain tensor.
	=Hooke	Use Hooke's law to compute strain tensor from stress tensor. Equation 1027 on page 1041
	=LoadFromFile	Load strain from Piezo file specified in File section.
Stress	=(<float>*6)	Pa Components xx, yy, zz, yz, xz, xy of stress tensor.
	=<ident> [(Table 380)]	Use PMI model <ident> to compute stress. Stress on page 1530

Table 336 PMIModel() Multistate Configuration–Dependent Bulk Mobility, High-Field Saturation With Two Driving Forces, Multistate Configuration–Dependent Thermal Conductivity, Multistate Configuration–Dependent Heat Capacity

Index	=<int>	0 Number passed to and interpreted by PMI model.
MSConfig	=<string>	"" Name of multistate configuration on which PMI depends.
Name	=<string>	! Name of PMI model.
String	=<string>	"" String passed to and interpreted by PMI model.
Table 380		PMI parameters.

Table 337 RandomizedVariation() in Physics{} Statistical Impedance Field Method

BandEdge	<string> (Table 297)	– Band Edge Variations on page 893
Conductivity	<string> (Table 300)	– Metal Conductivity Variations on page 894
Doping(	Table 303)	– Doping Variations on page 889
DopingVariation	<string> (Table 305)	– Doping Profile Variations on page 896
Epsilon	<string> (Table 310)	– Dielectric Constant Variations on page 895
Geometric	<string> (Table 317)	– Geometric Variations on page 892
TrapConcentration(	Table 339)	– Trap Concentration Variations on page 891

Table 337 RandomizedVariation() in Physics{} Statistical Impedance Field Method (Continued)

Workfunction	<string> (Table 346)	– Workfunction Variations on page 891

Table 338 Schroedinger() 1D Schrödinger Equation

DensityTail	=Extrapolate	* Use estimate for lowest noncomputed subband energy as $E_{\max,\nu}$. Equation 244
	=MaxEnergy	Use highest computed subband energy as $E_{\max,\nu}$. Equation 244
eDensityCorrection		+ Activate electron quantization model in the quantum well. 1D Schrödinger Equation
Electron		– Solve Schrödinger equation for electrons.
EnergyInterval	[(<carrier>)]=float>	0 eV Highest energy to which eigensolutions are computed, measured from the lowest interior potential point on the nonlocal line on which the Schrödinger equation is solved. When 0, only bound solutions are computed. Using the 1D Schrödinger Equation
Error	[(<carrier>)]=<(0,)>	1e-5 eV Precision target for eigenenergies. Using the 1D Schrödinger Equation
hDensityCorrection		+ Activate hole quantization model in the quantum well. 1D Schrödinger Equation
Hole		– Solve Schrödinger equation for holes.
MaxSolutions	[(<carrier>)]=<0..>	5 Maximum number of eigensolutions computed per ladder. Using the 1D Schrödinger Equation on page 394
Polarization	= TE | TM | Mixed	TE Set the polarization used for the computation of the optical transition matrix element. Optical Transition Matrix Element for Wurtzite Crystals on page 1206
PolarizationFactor	=<[0,1]>	1 Set the polarization factor used for the computation of the optical transition matrix element in mixed polarization simulations. Optical Transition Matrix Element for Wurtzite Crystals on page 1206

Table 338 Schroedinger() 1D Schrödinger Equation (Continued)

Smooth	=<float>	0 cm Length (measured from end of nonlocal line) over which to blend from classical to quantum density.

Table 339 Spatial restriction Energetic and Spatial Distribution of Traps, Options Common to sIFM Variations

SpaceMid	=<vector>	(0 0 0) µm Center of spatial distribution.
SpaceSig	=<vector>	(1e100 1e100 1e100) µm Width of spatial distribution.
SpatialShape	=Gaussian	Use Gaussian shape function.
	=Uniform	* Use Uniform shape function.

Table 340 Tensor(), eTensor(), hTensor() Piezoresistance Mobility Model, Factor(), eFactor(), hFactor() Isotropic Factor Models

<ident>	[(Table 380)]	Use PMI model <ident> to compute piezoresistive prefactors (first-order piezoresistance tensor model only). Piezoresistive Coefficients on page 1528
		Use PMI model <ident> to compute a mobility enhancement stress factor (isotropic factor model only). Mobility Stress Factor on page 1519
ApplyToMobility Components		– Apply the Factor model enhancement to individual mobility components. Factor Models Applied to Mobility Components on page 1105
AutoOrientation		off Use parameter set based on orientation of nearest interface. Auto-Orientation for Piezoresistance on page 1091 and Isotropic Factor Model Options on page 1104
ChannelDirection	=<1..3>	1 Channel direction (Factor models only). Isotropic Factor Models on page 1098
EffectiveStressModel		Use the effective stress Factor model. Effective Stress Model on page 1100
	(AxisAligned Normals)	off Use axis-aligned normals for the effective stress calculation. Effective Stress on page 1101

Table 340 Tensor(), eTensor(), hTensor() Piezoresistance Mobility Model, Factor(), eFactor(), hFactor() Isotropic Factor Models (Continued)

Enormal		off Use piezoresistive prefactors (first-order piezoresistance tensor model only). Enormal- and MoleFraction-Dependent Piezo Coefficients on page 1092
FirstOrder		* Use the first-order piezoresistance model. Piezoresistance Mobility Model on page 1086
Kanda		off Include temperature and doping dependency. Doping and Temperature Dependency on page 1088
ParameterSetName	=<string>	Name of Piezoresistance or EffectiveStressModel parameter set. Named Parameter Sets for Piezoresistance on page 1091 and Isotropic Factor Model Options on page 1104
SecondOrder		Use the second-order piezoresistance model. Piezoresistance Mobility Model on page 1086
SFactor	=<string>	Dataset for the spatial distribution of the mobility enhancement factor. SFactor Dataset or PMI Model on page 1104
	=<ident> [(Table 380)]	Name of model to be used by the PMI to compute the spatial distribution of the mobility enhancement factor. SFactor Dataset or PMI Model on page 1104 and Space Factor on page 1624

Table 341 ThermalConductivity() Thermal Conductivity

<ident>	[(Table 380)]	Use PMI model <ident> for thermal conductivity. Thermal Conductivity on page 1492
Constant	Conductivity	Use constant conductivity.
	Resistivity	Use constant resistivity.
Formula		Use the built-in strategy for thermal conductivity. Thermal Conductivity on page 1492
PMIModel(	Table 336)	Use multistate configuration–dependent model.
TempDep	Conductivity	Use Equation 1137 on page 1132.
	Resistivity	Use Equation 1136 on page 1132.

Table 342 Transition() in eTransitionRate() or hTransitionRate Semiconductor Band Structure

DrivingForce	=<string>	Define the driving force used in the transition rate model.
FromToValleyPairs	=<string> ...	List of valleys among which the defined transition is applied.
Model	=<string>	Define the model name.
ParameterSetName	=<string>	Define the named parameter set in the parameter file.

Table 343 Transition() in MSConfig() Phase and State Transitions

CEModel(	<ident> [(Table 380)] [<int>])	! Use PMI_TrapCaptureEmission model <ident> with optional constructor argument <int>.
FieldFromInsulator		– Use the insulator electric field instead of the semiconductor electric field for the MSCs defined at the semiconductor–insulator interface.
From	=<string>	! Interacting state.
Name	=<string>	! Identifier of transitions.
Reservoirs(		List of reservoirs for particle conservation.
	<string> (Particles=<int>)...)	Reservoir <string> and number of involved particles <int>.
To	=<string>	! Reference state.

Note:

In Table 344, $d = 3$ for bulk traps and $d = 2$ for interface traps.

Table 344 Traps(()...) Traps and Fixed Charges, Degradation

Table 339		Spatial restriction of trap distribution.
Acceptor		Acceptor trap type. Trap Types on page 609
ActEnergy	=<float>	eV Equilibrium activation energy of hydrogen on Si-H bonds, ε_A^0. Chapter 19, Degradation on page 672
ActEnergyDelta	=<float>	eV (0) Addition to phonon energy to account for an effect of optical carrier generation in generation of charged oxygen vacancies. Degradation

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

Add2TotalDoping(		Include the trap concentration in the acceptor, donor, and total doping concentrations used to compute mobility, lifetimes, and so on. Specifying Doping Species on page 63
	ChargedTraps)	Include the charged trap concentration in the acceptor or donor doping concentrations used to compute mobility. Specifying Doping Species on page 63
BarrierTunneling Enhan	=(<float>*4)	(1 1 1 1) 1, 1, 1, 1 Barrier tunneling enhancement parameters for the Degradation model depassivation constant, δ_{BTe}, ρ_{BTe}, δ_{BTh}, and ρ_{BTh}. Reaction Enhancement Factors on page 678
BondConc	=<float>	cm^{-d} Total silicon dangling bond concentration N. Chapter 19, Degradation on page 672
BondConcSFactor	=<ident> [(Table 380)]	Name of model to be used by the PMI to compute the spatial distribution of bond concentration. Space Factor on page 1624
	=<string>	Dataset for the spatial distribution of bond concentration. Using the Trap Degradation Model on page 679
CBRate	=<ident> [(Table 380)]	Use PMI <ident> to compute electron capture and emission rate for conduction band.
	=(<ident> [(Table 380)] <int>)	Use PMI <ident> with constructor argument <int> to compute electron capture and emission rate for conduction band. Trap Capture and Emission Rates on page 1534
Conc	=<float>	cm^{-d} or $\mathrm{eV}^{-1}\mathrm{cm}^{-d}$ Concentration or the peak density of the trap distribution. For KMC-MIM simulation, sets defect density. Energetic and Spatial Distribution of Traps on page 610, Specifying Defects on page 1231
Coupled	=Off	* Deactivate trap coupling.
	=Tunneling	Couple traps by tunneling. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
CritConc	=<float>	cm^{-d} Critical concentration N_{crit}, default $0.1N$. Chapter 19, Degradation on page 672

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

CurrentEnhan	=(<float>*4)	(0 1 0 1) 1, 1, 1, 1 Parameters of the tunneling and hot carrier–dependent terms of the depassivation constant, δ_{Tun}, ρ_{Tun}, δ_{HC}, and ρ_{HC}. Reaction Enhancement Factors on page 678
Cutoff	=BandGap	Truncate trap energy distribution to plain band gap at 300 K. Energetic and Spatial Distribution of Traps on page 610
	=EffectiveBandgap	* Truncate trap energy distribution to effective bandgap at 300 K.
	=None	Do not truncate trap energy distribution.
	=Simple	Use a legacy truncation of trap energy distribution.
Degradation		Use degradation model based on kinetic equation. Chapter 19, Degradation on page 672
	(PowerLaw)	Use degradation model based on power law. Chapter 19, Degradation on page 672
DePasCoef	=<float>	s^{-1} Depassivation coefficient ν_0 at the passivation conditions (the equilibrium at the passivation temperature T_0). Chapter 19, Degradation on page 672
DiffusionEnhan	=(6*<float>)	cm cm^2 s^{-1} eV cm s^{-1} cm^{-3} 1 Parameters of hydrogen diffusion in oxide, x_P, D_0, ε_H, k_P, N_H^0, and N_{ox} in Equation 622 on page 676, Chapter 19, Degradation on page 672.
DirectTunneling Enhan	=(<float>*4)	(1 1 1 1) 1, 1, 1, 1 Direct tunneling enhancement parameters for the Degradation model depassivation constant, δ_{DTe}, ρ_{DTe}, δ_{DTh}, and ρ_{DTh}. Reaction Enhancement Factors on page 678
-Discontinuity		Switch off automatic activation of Discontinuity. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
DiscreteTrapT2T		Specify trap-to-trap tunneling entry. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
Donor		Donor trap type. Trap Types on page 609
eBarrierTunneling(	NonLocal=<string>	Use nonlocal tunneling from the conduction band at reference surface of all connected unnamed, and all listed, connected named nonlocal meshes. Nonlocal Tunneling for Traps on page 645

Feedback

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

	TwoBand)	Use two-band dispersion. WKB Tunneling Probability on page 922
eConstEmissionRate	=<float>	s^{-1} Constant electron emission rate term to the conduction band e^{n}_{const}. Local Capture and Emission Rates on page 621
eGfactor	=<float>	Electron degeneracy factor g_n. Trap Occupation Dynamics on page 619
eHCSDegradation		Use the hot-carrier stress (HCS) degradation model for electrons. Hot-Carrier Stress Degradation Model on page 715
(	BDEnergyIntervals=<int>	20 Number of energy intervals to use in the bond dispersion integral. Bond Dispersion on page 719
	BondDispersion	+ Use bond dispersion. Bond Dispersion on page 719
	SHE	- Use the carrier distribution function from SHE. Spherical Harmonics Expansion Option on page 718
	TimeDependentRates)	- Use the more general time-dependent HCS model formulation. Time-Dependent Hot-Carrier Stress Degradation Model on page 720
eJfactor	=<float>	Electron J-model factor g^{J}_{n}. Local Capture and Emission Rates on page 621
ElectricField	[(<carrier>)]	Field-dependent model for cross sections. J-Model for Cross Sections on page 622
EmptyCharge	=<int>	1 Set the charge state of an empty defect for KMC-MIM simulation. Applies only when IncludeMIMChargeInPoisson is specified. Specifying Defects on page 1231, Including the MIM Charge in the Poisson Equation on page 1248
EnergyDepeXsectionCoef	=<float>	Use trap energy dependent electron capture cross section area. Trap Energy Dependent Model for Cross Sections
EnergyDephXsectionCoef	=<float>	Use trap energy dependent hole capture cross section area. Trap Energy Dependent Model for Cross Sections
EnergyMid	=<float>	eV Central energy E_0 of the trap distribution. Equation 547 on page 610

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

EnergyShift	=<ident> [(Table 380)]	Use PMI <ident> to compute trap energy shift.
	=(<ident> [(Table 380)] <int>)	Use PMI <ident> with constructor argument <int> to compute trap energy shift. Energetic and Spatial Distribution of Traps on page 610, Trap Energy Shift on page 1538
EnergySig	=<(0,)>	eV Width E_S of the trap distribution. Equation 547 on page 610
eNeutral		Electron trap type. Trap Types on page 609
eSHEDistribution	=(<float>*6)	(0 0 0 0 1 1) $\left(\mathrm{cm}^2\mathrm{A}^{-1}\right)^{\rho_{\mathrm{SHE}}}$, eV, eV, eV, 1, 1 Parameters of the electron energy–dependent terms of the depassivation constant, δ_{SHE}, $\varepsilon_{\mathrm{th}}$, ε_{a}, $\delta_{\perp}$, $\rho_{\perp}$, and ρ_{SHE}. Trap Degradation Model on page 673
EtEmpty	=<float>	eV Set the empty trap level measured from the conduction band for KMC-MIM simulation. Specifying Defects on page 1231
EtFill	=<float>	eV Set filled trap level measured from the conduction band for KMC-MIM simulation. Specifying Defects on page 1231
Exponential		Exponential energetic distribution. Energetic and Spatial Distribution of Traps on page 610, Equation 547 on page 610
eXsection	=<[0,)>	cm^2 Electron capture cross section σ_n^0. Local Capture and Emission Rates on page 621
FieldEnhan	=(<float>*4)	(0 1 0 1) $\mathrm{eVcm}^{\rho_{//}}\mathrm{V}^{-\rho_{//}}$, 1, $\mathrm{eVcm}^{\rho_{\perp}}\mathrm{V}^{-\rho_{\perp}}$, 1 Parameters of the electric field–dependent terms of the Si-H bond energy and the activation energy, $\delta_{//}$, $\rho_{//}$, $\delta_{\perp}$, and $\rho_{\perp}$. Chapter 19, Degradation on page 672
FillFrac	=<float>	Initial fraction of defects that are filled with electrons in KMC-MIM simulations. Specifying Defects on page 1231
FixedCharge		Fixed charge. Trap Types on page 609, Insulator Fixed Charges on page 653
FluenceDependence	=(<float>*3)	(0 0 0) 1, 1, 1 Activate fluence degradation model with the given parameters β_{F}, f_{cb}, and f_{vb}. Fluence Model on page 684

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

FowlerNordheimEnhan	=(<float>*4)	(1 1) 1, 1 Fowler–Nordheim tunneling enhancement parameters for the Degradation model depassivation constant, δ_{FN}, and ρ_{FN}. Reaction Enhancement Factors on page 678
fromCondBand		Zero point for E_0 is conduction band. Equation 548 on page 611
fromMidBandGap		Zero point for E_0 is intrinsic energy. Equation 548 on page 611
fromValBand		Zero point for E_0 is valence band. Equation 548 on page 611
gamma		1 1 Prefactor for elastic tunneling. Electron Capture Rate for the Elastic Transition on page 646
Gaussian		Gaussian energetic distribution. Energetic and Spatial Distribution of Traps on page 610, Equation 547 on page 610
GrainBoundary		Generate traps on the grain boundary. A prerequisite is that the grain field must be generated with GrainFieldGenerator specified in the Math section. Specifying Defects on page 1231
(	Density=<float>	cm^{-2} for 3D device, cm^{-1} for 2D device Defect density on the grain boundary. The calculated number of defects is then randomized over the insulator volume inside the KMC simulation space.
	EtEmpty=<float>	eV Set the empty grain-boundary trap level measured from the conduction band.
	EtFill=<float>	eV Set the filled grain-boundary trap level measured from the conduction band.
	FillFrac=<float>	Set the initial fraction of defects that are filled with electrons.
	TrapSigma=<float>)	eV Randomizes the grain-boundary trap energy levels by using a Gaussian distribution with this standard deviation.
hBarrierTunneling(	NonLocal=<string>	Use nonlocal tunneling from the valence band at reference surface of all connected unnamed, and all listed, connected named nonlocal meshes. Nonlocal Tunneling for Traps on page 645

Feedback

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

	TwoBand)	Use two-band dispersion. WKB Tunneling Probability on page 922
HCIDepeCaptureRate	Table 108	Use HCI current based capture rate for electron (Traps and Fixed Charges and Hot-Carrier Injection)
HCIDephCaptureRate	Table 108	Use HCI current based capture rate for holes (Traps and Fixed Charges and Hot-Carrier Injection).
hConstEmissionRate	=<float>	s^{-1} Constant hole emission rate to the valence band e^{p}_{const}. Local Capture and Emission Rates on page 621
hGfactor	=<float>	Hole degeneracy factor g_p. Trap Occupation Dynamics on page 619
hHCSDegradation		Use the HCS degradation model for holes. Hot-Carrier Stress Degradation Model on page 715
(	BDEnergyIntervals=<int>	20 Number of energy intervals to use in the bond dispersion integral. Bond Dispersion on page 719
	BondDispersion	+ Use bond dispersion. Bond Dispersion on page 719
	SHE	- Use the carrier distribution function from SHE. Spherical Harmonics Expansion Option on page 718
	TimeDependentRates)	- Use the more general time-dependent HCS model formulation. Time-Dependent Hot-Carrier Stress Degradation Model on page 720
hJfactor	=<float>	Hole J-model factor g^{J}_{p}. Local Capture and Emission Rates on page 621
hNeutral		Hole trap type. Trap Types on page 609
hSHEDistribution	=(<float>*6)	(0 0 0 0 1 1) $\left(\mathrm{cm}^2\mathrm{A}^{-1}\right)^{\rho_{\mathrm{SHE}}}$, eV, eV, eV, 1, 1 Parameters of the hole energy–dependent terms of the depassivation constant, δ_{SHE}, $\varepsilon_{\mathrm{th}}$, ε_{a}, $\delta_{\perp}$, $\rho_{\perp}$, and ρ_{SHE}. Trap Degradation Model on page 673
HuangRhys	=<[0,)>	Huang–Rhys factor. Nonlocal Tunneling for Traps on page 645
hXsection	=<[0,)>	cm^2 Hole capture cross section σ^{0}_{p}. Local Capture and Emission Rates on page 621

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

KmcMim1 | KmcMim2		Specify KMC-MIM trap type. Specifying Defects on page 1231
KMCReRAM		Activate KMC ReRAM simulation.
Level		Trap with single energy level. Energetic and Spatial Distribution of Traps on page 610, Equation 547 on page 610
Location	=(<vector>...)	µm Locations of traps coupled by tunneling. For KMC-MIM simulations, locations of specifically placed defects. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629, Specifying Defects on page 1231
	=(smesh)	Select all vertices in the region for trap-to-trap tunneling. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
Makram-Ebeid	[(<carrier> <simpleCapt>)]	Use Makram–Ebeid–Lannoo model. Makram-Ebeid–Lannoo Model on page 625
Material	=<string>	(i) Material to which energy specification refers. Energetic and Spatial Distribution of Traps on page 610
MultipleRandomTraps		+ Allow multiple traps to occupy the same vertex. By default, this option is switched on and can be deactivated as outlined in Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629.
Name	=<string>	Identifier for trap.
-NormalizeSfactor		Switch off normalization of Sfactor for DiscreteTrapT2T. Default is on. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
NumberOfRandomTraps	=<int>	Input number of randomly distributed traps for trap-to-trap tunneling. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
OutputT2TPaths	(<string>)	Name of input file to output trap-to-trap tunneling locations and linkages. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
PasCoef	=<float>	s^{-1} Passivation coefficient γ_0. By default, computed automatically to provide the equilibrium, $v_0 N_{hb}^0 / \left(N - N_{hb}^0\right)$. Chapter 19, Degradation on page 672

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

PasTemp	=<float>	300 K Passivation temperature T_0. Chapter 19, Degradation on page 672
PasVolume	=<float>	0 cm^2 | cm^3 Passivation volume Ω. Chapter 19, Degradation on page 672
PasWeight	=<float>	(1) Weighting factor for passivation or thermal stress temperature in degradation model.Chapter 19, Degradation on page 672
PhononEnergy	=<[0,)>	eV Phonon energy for inelastic tunneling. Nonlocal Tunneling for Traps on page 645
PooleFrenkel		Use Poole–Frenkel model for exchange with conduction or valence band, depending on trap type. Poole–Frenkel Model for Cross Sections on page 623
	(Electron)	Use Poole–Frenkel model for exchange with conduction band, irrespective of trap type.
	(Hole)	Use Poole–Frenkel model for exchange with valence band, irrespective of trap type.
PowerEnhan	=(<float>*3)	(0 0 0) 1, V^{-1} cm, V^{-1} cm Parameters in the chemical potential for kinetic equation degradation and the power for degradation by power law, β_0, $\beta_{//}$, and $\beta_{\perp}$. Chapter 19, Degradation on page 672
Randomize	[=<int>]	Randomize traps. Trap Randomization on page 615
RandomizeEnergy	[=<int>]	Randomize energy at each vertex for the Uniform, Gaussian, or Table trap energy distributions. Energetic and Spatial Distribution of Traps on page 610
Reference	=BandGap	Refer trap energies to band edges excluding bandgap narrowing. Energetic and Spatial Distribution of Traps on page 610
	=EffectiveBandgap	* Refer trap energies to effective band edges.
ReferencePoint	=<vector>	Coordinate where to take the data for EnergyShift.
Region	=<string>	(i) Region to which energy specification refers. Energetic and Spatial Distribution of Traps on page 610
SFactor	=<string>	Dataset for the spatial distribution of the traps. Energetic and Spatial Distribution of Traps on page 610

Feedback

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

	=<ident> [(Table 380)]	Name of model to be used by the PMI to compute the spatial trap distribution. Space Factor on page 1624
SimpleCapt		– Restrict field enhancement of trapping to the emission rates. Local Capture and Emission Rates on page 621
SingleTrap		* Use a single trap. Specifying Single Traps on page 614
	(RTN)	Specify as RTN trap. Specifying Single Traps on page 614
T2TconcScale	=<float>	1 Scaling factor for DiscreteTrapT2T concentration. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
T2TfixedConc	=<float>	0 Fix the same concentration at every trap-to-trap vertex in the region. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
T2Trecombination		* Switch on T2T recombination. Trap-to-Trap Recombination on page 640
Table	=(<float>*2...)	eV $eV^{-1}cm^{-d}$ Pairs of energy and concentration of a tabular approximation to a trap distribution. Energetic and Spatial Distribution of Traps on page 610, Equation 547 on page 610
TailCutoff	=<float>	0 When positive, truncation energy of for Gaussian and exponential trap energy distributions (in units of EnergySig).
TDDB		Account for time-dependent dielectric breakdown. Time-Dependent Dielectric Breakdown Simulations on page 1253
TempDepeXsection		Use temperature dependent model of capture cross section for electron. Temperature Dependent Model for Cross Sections
	(exponential = <float>)	Use exponential temperature dependence to compute electron capture cross section. The default value for model parameter is set to 300K.
	(powerlaw = <float>)	Use powerlaw temperature dependence to compute the electron capture cross section.
TempDephXsection		Use the temperature dependent model of capture cross section for holes. Temperature Dependent Model for Cross Sections

Table 344 Traps(()...) Traps and Fixed Charges, Degradation (Continued)

	(exponential = <float>)	Use the exponential temperature dependence to compute the hole capture cross section. The default value for model parameter is set to 300K.
	(powerlaw = <float>)	Use the powerlaw temperature dependence to compute hole capture cross section.
Transition	= InelasticPhonon | ModeOscillator	ModeOscillator Choose the tunneling transition for trap-to-trap tunneling. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629
TrapSigma	=<float>	eV Randomize trap levels using a Gaussian distribution with this standard deviation for KMC-MIM simulation. Specifying Defects on page 1231
TrapVolume	=<[0,)>	μm^3 Interaction volume for nonlocal tunneling. Nonlocal Tunneling for Traps on page 645
Tunneling(	Hurkx[(<carrier>)])	Use Hurkx trap-assisted tunneling model. Hurkx Model for Cross Sections on page 623
Uniform		Uniform energetic distribution. Energetic and Spatial Distribution of Traps on page 610, Equation 547 on page 610
VBRate	=<ident> [(Table 380)]	Use PMI <ident> to compute hole capture and emission rate for valence band.
	=(<ident> [(Table 380)] <int>)	Use PMI <ident> with constructor argument <int> to compute hole capture and emission rate for valence band. Trap Capture and Emission Rates on page 1534
Xsection	=<float>	For KMC-MIM simulations, use to calculate defect localization radius used in inelastic electrode-to-trap rate calculations. Specifying Defects on page 1231
XsectionT2T	=<float>	For KMC-MIM simulations, use to calculate defect localization radius used in multiphonon trap-to-trap calculations. Specifying Defects on page 1231

Table 345 Window Illumination Window

Circle(	Radius=<float>)	
Intensity Distribution(	Gaussian(Table 316))	Specify spatial intensity profile. Spatial Intensity Function Excitation on page 767
Line(	Dx=<float>	

Table 345 Window Illumination Window (Continued)

	X1=<float>	
	X2=<float>)	
Origin	=(<float>*3)	Global location of window origin.
OriginAnchor	=<ident>	Center Origin anchor of illumination window in terms of cardinal direction (North, South, East, West, NorthEast, SouthEast, NorthWest, SouthWest).
Polygon(	(<float>*2)*n	Specification of simple polygon using several vertices.
	((<float>*2)*n)*m)	Specification of complex polygon using several loops of vertices.
Rectangle(	Dx=<float>	
	Dy=<float>	
	Corner1=(<float>*2)	
	Corner2=(<float>*2))	
RotationAngles	=(<float>*3)	Angles specifying global orientation of window coordinate system.
WeightedAPD integration		– Switch on weighted absorbed photon density (APD) integration. Accurate Absorbed Photon Density for 1D Optical Solvers on page 774
(	Mode= Auto | Full	Auto Set integration mode of APD. Options are: • Auto: Mixed wireframe and 1D integration • Full: Wireframe integration.
	NumberOfCellsPerLayer=<int>	100 Set the number of wireframe cells per layer.
	NumberOfTransverse CellsPerDirection=<int>	200 Wireframe discretization parameter of the illumination window plane.
	PrintInfo)	Print detailed information about wireframe integration.
XDirection	=(<float>*3)	(1, 0, 0) Direction of x-axis.
YDirection	=(<float>*3)	(0, 1, 0) Direction of y-axis.

Feedback

Table 346 Workfunction() in RandomizedVariation() Workfunction Variations

Table 240		Statistical properties of the correlation.
Table 339		Spatial restriction of variation.
Amplitude	=<float>	eV Amplitude of workfunction variation for exponential and Gaussian correlation.
GrainProbability	=(<(0,)>...)	Probability for grain orientation.
GrainWorkfunction	=(<float>...)	eV Grain workfunctions.
Surface	=<string>	! Surface of variation. Workfunction Variations on page 891

Plot

Table 347 Plot{} Device Plots

Table 202		Scalar plot data. Device Plots on page 195
Table 202/Element		Scalar plot datasets defined on elements.
Table 202/RegionInterface		Scalar plot data on region interfaces; supported for just a few datasets. Interface Plots on page 199
Table 202/Tensor		Tensorial plot data. Device Plots on page 195
Table 203/Vector		Vectorial plot data. Device Plots on page 195
Table 204/SpecialVector		Special vectorial plot data; supported for the SHE distribution data. SHE Distribution Hot-Carrier Injection on page 945
Table 206		MIM particle data. Particle Plots on page 1268
Table 349		Occupation rates of MSConfig states. Specifying Multistate Configurations on page 657
DatasetsFromGrid		Copy datasets from TDR grid file. What to Plot on page 196
	(<ident>...)	Datasets to be copied.
HBPlot(	Table 348)...	Select harmonic balance plot variables. Device Plots on page 176

Table 347 Plot{} Device Plots (Continued)

T2TTrappingRates		Trap capture and emission rates in Trap-to-Trap tunneling using the discrete Trap formulation. Trap-to-Trap Tunneling (DiscreteTrapT2T)
TrapCaptureCrossSection		e/h trap capture cross section area. Visualizing Traps on page 649
TrappingRates		Trap capture and emission rates. Visualizing Traps on page 649
TrappingRatesPerNLMesh		Trap capture and emission rates along each nonlocal mesh. Visualizing Traps on page 649
TTOMFraction		Fraction of occupied traps. Transient Trap Occupancy Model on page 689

Table 348 HBPlot() Device Plots

DataName[	Table 202 ...]	List of vertex-based DC data names for harmonic balance variables.
NumberOfPoints	=<int>	0 Number of sampling points to plot.
Order	=<int>	The maximum spectrum order to plot.
<int>		Same as Order.

Table 349 MSConfig in Plot{} Specifying Multistate Configurations

MSConfig		All state occupations of all MSConfig.
	(<string>)	All state occupations of MSConfig <string>.
	(<string1> <string2>)	Occupation of state <string2> of MSConfig <string1>.

RayTraceBC

Table 350 RayTraceBC{} Boundary Condition for Raytracing

AngularRange	= all | refraction	all Specify how the user-specified values for reflectivity and transmittivity are applied. Constant Reflectivity and Transmittivity Boundary Condition on page 799

Feedback

Table 350 RayTraceBC{} Boundary Condition for Raytracing (Continued)

Fresnel		Fresnel boundary condition.
Name	=<string>	Name of the reflectivity contact or photon-recycling contact.
LayerStructure{	<float> <string> [; <float> <string> ...}]	Definition of multilayer structure used for TMM calculation. First column contains thickness of layer in µm. Second column contains material name of layer.
MapOptGenToRegions(	<string> <string> ...)	Specify list of regions to map the lumped TMM BC optical generation to.
PMIModel	=<ident> [(Table 380)]	Name of the PMI model associated with this BC contact.
QuantumEfficiency	=<float>	Define the quantum efficiency of the TMM BC optical generation.
ReferenceMaterial	=<string>	Definition of LayerStructure orientation. The topmost layer in the LayerStructure specification is connected to the region with material ReferenceMaterial.
ReferenceRegion	=<string>	Definition of LayerStructure orientation. The topmost layer in the LayerStructure specification is connected to the region with name ReferenceRegion.
Reflectivity	=<[0,1]>	0 Reflectivity.
{Side	="X" Periodic}	Define periodic BC at opposing x-surfaces. Periodic Boundary Condition on page 807
	="Y" Periodic}	Define periodic BC at opposing y-surfaces.
	="Z" Periodic}	Define periodic BC at opposing z-surfaces.
Transmittivity	=<[0,1]>	0 Transmittivity.

Solve

Table 351 Solve{}

Table 365		Solve selected standalone problem.

Feedback

Table 351 Solve{} (Continued)

[<ident>.] Table 356		Solve selected equation [for instance <ident>].
ACCoupled(	Table 352)	Perform small-signal and noise analysis. Small-Signal AC Analysis on page 162
{	[<ident>.] Table 356...}	Equations to be solved consistently.
Continuation(	Table 353)	Continuation options.
{	Table 351}	Problem to be solved.
Coupled(	Table 354)	Solve coupled equations consistently by Newton iteration.
{	[<ident>.] Table 356...}	Equations to be solved.
CurrentPlot	as Load	Control current output. When to Write to the Current File on page 180
HBCoupled(	Table 359)	Perform harmonic balance analysis. Harmonic Balance Analysis Output on page 175
{	[<ident>.] Table 356...}	Equations to be solved.
Load(	Table 360) [{	Load and continue with solution stored by Save. Save and Load Statements on page 238
	Circuit	Load circuit.
	<ident>...}]	Devices to be loaded.
NewCurrentPrefix	=<string>	Use prefix <string> for current file. NewCurrentPrefix Statement on page 182
Plot(	Table 360)	Plot current solution. When to Plot on page 197
	[{<ident>...}]	Devices to be plotted.
PlotCarrierPath(	Table 360)	Plot carrier paths for Saddle Point Analysis in CIS Devices
Plugin(	Table 362)	Solve equations consistently by Gummel iteration.
{	Table 365	Solve selected standalone problem.

Table 351 Solve{} (Continued)

	[<ident>.] Table 356...	Solve selected equation (for example, <ident>).
	Coupled (Table 354) {[<ident>.] Table 356...}...}	Solve coupled equations.
Quasistationary(	Table 363)	Quasistationary simulation options.
{	Table 351}	Problem to be solved.
Save	as Load	Save current solution for retrieval with Load. Save and Load Statements on page 238
Set(	Table 364)	Set status according to Table 364.
SpectralPlot(	Table 360)	Plot spectral simulation results of current solution. When to Plot on page 197, Illumination Spectrum on page 733
	[{<ident>...}]	Devices to be plotted.
System	(<string>)	Execute UNIX command <string> (ignore return status). System Command on page 252
+System	(<string>)	Execute UNIX command <string>; use its return status to decide whether it was successful. System Command on page 252
Transient(	Table 367)	Transient simulation options.
{	Table 351}	Problem to be solved.
Unset	(<ident>...)	Unset nodes. Mixed-Mode Electrical Boundary Conditions on page 128
	(TrapFilling)	Use the standard trap equations. Explicit Trap Occupation on page 616

Table 352 ACCoupled() Small-Signal AC Analysis

Table 354		Coupled() parameters.
ACCompute(	Table 360)	Restrict AC or noise analysis to selected points in a Quasistationary command.

Feedback

Table 352 ACCoupled() Small-Signal AC Analysis (Continued)

ACExtract	=<string>	Prefix for the name of the file to which the results of AC analysis are written (overrides the specification from the File section).
ACMethod	=Blocked	Use Blocked solver.
ACPlot	=<string>	Plot the responses of the solution variables to the AC signals. <string> is a prefix for the names of the files to which the responses are written.
ACSubMethod	(<ident>)= Table 231	Super (1D and 2D), ILS (3D) Inner solver for device <ident>.
CircuitNoise		Compute noise from circuit elements. Noise From SPICE Circuit Elements on page 885
Decade		Use logarithmic intervals between the frequencies.
EndFrequency	=<(0,)>	! Hz Upper frequency.
Exclude	(<ident>...)	Devices that will be removed from the circuit for AC analysis.
Extraction{	Table 22}	List of frequency-dependent extraction curves. Extraction File on page 205
Frequency	=<0..>	! List of frequencies for which the AC analysis is to be performed. For example, you can specify it as: Frequency=(1.0e+06;2e+08)
Linear		Use linear intervals between the frequencies.
Node	(<ident>...)	Nodes for which to perform AC analysis.
NoisePlot	=<string>	Prefix for a file name. Chapter 23, Noise, Fluctuations, and Sensitivity on page 869
NumberOfPoints	=<0..>	! Number of frequencies for which to perform the analysis.
ObservationNode	(<ident>...)	Nodes for which to perform noise analysis; subset of those in Node. Chapter 23, Noise, Fluctuations, and Sensitivity on page 869
Optical		Perform optical AC analysis. Optical AC Analysis on page 167
SolvePrecision	=Table Table 29	off (g) Specify floating-point precision for Solve. Mixed Precision Solve

Feedback

Table 352 ACCoupled() Small-Signal AC Analysis (Continued)

StartFrequency	=<(0,)>	! Hz Lower frequency.
VoltageGreen Functions		– Compute voltage responses for frequency zero. Analysis at Frequency Zero on page 872

Table 353 Continuation() Continuation Command

AcceptNewtonParameter		Apply relaxed Newton parameters. Relaxed Newton Method on page 223
BreakCriteria{	Table 220}	Break criteria. Break Criteria: Conditionally Stopping the Simulation on page 130
Decrement	=<float>	1.5 (2.0 with NewArc) Divisor for step size on failure to solve.
Iadapt	=<float>	0.0 A Lower current limit for adaptive algorithm.
Increment	=<float>	2.0 (1.5 with NewArc) Multiplier for step size on successful solve.
InitialVstep	=<float>	! Initial voltage step.
IscaleMin	=<float>	1e-10 A Value added to scale factor for current when using the NewArc option.
MaxCurrent	=<float>	! Upper current limit.
MaxIfactor	=<float>	1e35 Maximum-allowed current relative to previous point as a multiplication factor.
MaxIstep	=<float>	1e35 Maximum-allowed current step.
MaxRload	=<float>	1e14 Ω (1e8 with NewArc) Maximum value for adaptive load resistor.
MaxStep	=<float>	1e35 Maximum step for the internal arc length variable.
MaxVoltage	=<float>	! Upper voltage limit.
MaxVstep	=<float>	1e35 Maximum-allowed voltage step.
MinCurrent	=<float>	! Lower current limit.
MinRload	=<float>	1 W Minimum value for adaptive load resistor.
MinStep	=<float>	1e-35 Minimum step for the internal arc length variable.

Feedback

Table 353 Continuation() Continuation Command (Continued)

MinVoltage	=<float>	! Lower voltage limit.
Name	=<ident>	Electrode to bias in continuation. It must be specified.
NewArc		– Use a new arc length calculation for continuation.
NewGdiff		– (+ with NewArc) Use a new differential conductance calculation.
Rfixed	=<float>	0.001 Fixed resistor value.
RloadAdjust		– (+ with NewArc) Adjust the adaptive load resistance at low values of slope.
Vadapt	=<float>	0.0 V Lower voltage limit for adaptive algorithm.

Table 354 Coupled() Coupled Statement

CheckRhsAfterUpdate		Check whether the RHS can be reduced further after the update error has converged. If CheckRhsAfterUpdate is specified in the Math section, then -CheckRhsAfterUpdate can be used to deactivate it for this Coupled statement.
Digits	=<float>	Relative error target.
IncompleteNewton		Incomplete Newton. Incomplete Newton Algorithm on page 219
(	RhsFactor=<float>	Maximum change in RHS to allow old Jacobian to be reused.
	UpdateFactor=<float>)	Maximum change in update to allow old Jacobian to be reused.
Iterations	=<0..>	Maximum number of iterations of the Newton algorithm.
LineSearchDamping	=<(0,1]>	Minimal coefficient for line search damping, where 1 deactivates damping. Damped Newton Iterations on page 218
Method	=Table 231	Linear solver.
	=Blocked	Block decomposition solver.
NotDamped	=<0..>	Number of Newton iterations before Bank–Rose damping is applied. Damped Newton Iterations on page 218

Feedback

Table 354 Coupled() Coupled Statement (Continued)

PMI_NewtonStep	(<ident>)	Use PMI model <ident> for Newton step computation. Preprocessing for Newton Iterations and Newton Step Control on page 1588
RhsAndUpdate Convergence		Require both RHS and update error convergence.
RhsMin	=<float>	Upper bound for RHS norm convergence.
RhsMinFactor	=<float>	Factor for RhsMin affecting convergence. Convergence and Error Control on page 215
SolvePrecision	=TableTable 29	off (g) Specify floating-point precision for Solve. Mixed Precision Solve
SubMethod	[(<ident>)]= Table 231	Super (1D and 2D), ILS (3D) Inner solver (for device <ident>).
UpdateIncrease	=<float>	Maximum-allowed increase of update error per Newton step.
UpdateMax	=<float>	Maximum-allowed update error in Newton algorithm.

Table 355 Cyclic() Large-Signal Cyclic Analysis

Accuracy	=<float>	Tolerance ε_{cyc}.
CyclicExtrapolation Method	=<int>	0 Cyclic extrapolation method. Description of Method on page 158
Extrapolate(	Average	+ Use averaged factor r^{o}_{av} for each object. Factor r is defined separately for each vertex.
	CMaxVal=<float>	25 Maximum-allowed value of extrapolation factor (r_{max}) for circuit node voltages.
	Factor=<float>	1 Coefficient f for r^{o}_{av} estimation.
	Forward	– Proceed as in a standard transient, without cyclic extrapolation.
	MaxVal=<float>	25 Value of r_{max}.
	MinVal=<float>	1 Value of r_{min}.
	PMaxVal=<float>	25 Maximum-allowed value of extrapolation factor (r_{max}) for electrostatic potential.

Table 355 Cyclic() Large-Signal Cyclic Analysis (Continued)

	`Print`	+ Print averaged factors r_{av}^{o} for each object.
	`QFtraps`	– Apply extrapolation to 'trap quasi-Fermi level' Φ_T instead of trap occupation probabilities f_T.
	`TMaxVal=<float>)`	`25` Maximum-allowed value of extrapolation factor (r_{max}) for carrier or lattice temperature.
`IgnoreCircuit`		– Ignore circuit node voltages from extrapolation.
`IgnoreContact`		– Ignore contact nodes from extrapolation.
`IgnorecTemperature PDE`		– Exclude carrier temperature from extrapolation.
`IgnoreDGPDE`		– Exclude quantum potential from extrapolation.
`IgnoreElectronPDE`		– Exclude electron concentration from extrapolation.
`IgnoreHolePDE`		– Exclude hole concentration from extrapolation.
`IgnorelTemperature PDE`		– Exclude lattice temperature from extrapolation.
`IgnorePotentialPDE`		– Exclude electrostatic potential from extrapolation.
`IgnoreRegion`		– Ignore a particular region for extrapolation.
`MaximumCycles`	`=<int>`	`100000` Number of cyclic cycles after which transient simulation methodology is switched on.
`Period`	`=<float>`	s Period of the cycle.
`RegionName`	`=<string>`	Name of region to be excluded from extrapolation.
`RelFactor`	`=<float>`	Relaxation factor Υ.
`StartingPeriod`	`=<2..>`	`2` Period from which the extrapolation procedure starts.

In the description column of Table 356, the default for `ErrRef` (first number; see Table 218 on page 1720) and its unit, and the default for `Error` (second number, see Table 218) are given.

Table 356 Equations

`Charge`	`1.602192e-19` C `1e-3` Floating gate charge, available for `TransientErrRef` and `TransientError` only. Floating Gates on page 1343

Table 356 Equations (Continued)

`Circuit`	`0.0258` V `1e-3` Circuit equations.
`CondInsulator`	`0.0258` V `1e-3` Conductive insulator equations.
`Contact`	`0.0258` V `1e-3` Contact equations.
`Electron`	`1e10` cm^{-3} `1e-5` Electron continuity equation. Chapter 8, Carrier Transport Equations in Semiconductors on page 266
`eMVTransitionRate`	`1e10` cm^{-3} `1e-5` Intervalley transition rate equation. Chapter 12, Semiconductor Band Structure on page 339
`eQuantumPotential`	`0.0258` V `1e-3` Electron quantum-potential equation. Density Gradient Model on page 404
`eSHEDistribution`	`1` V `1e-3` Electron SHE distribution equation. Using Spherical Harmonics Expansion Method on page 951
`eTemperature`	`300` K `1e-4` Electron temperature equation. Hydrodynamic Model for Temperatures on page 282
`FEPolarization`	`1e-9` C/cm^2 `1e-5` Ferroelectric polarization equation. Chapter 29, Ferroelectric Materials on page 1003
`hMVTransitionRate`	`1e10` cm^{-3} `1e-5` Intervalley transition rate equation. Chapter 12, Semiconductor Band Structure on page 339
`Hole`	`1e10` cm^{-3} `1e-5` Hole continuity equation. Chapter 8, Carrier Transport Equations in Semiconductors on page 266
`hQuantumPotential`	`0.0258` V `1e-3` Hole quantum potential equation. Density Gradient Model on page 404
`hSHEDistribution`	`1` V `1e-3` Hole SHE distribution equation. Using Spherical Harmonics Expansion Method on page 951
`hTemperature`	`300` K `1e-4` Hole temperature equation. Hydrodynamic Model for Temperatures on page 282
`HydrogenAtom`	`1e10` cm^{-3} `1e-3` Hydrogen atom transport equation. Hydrogen Transport on page 692
`HydrogenIon`	`1e10` cm^{-3} `1e-3` Hydrogen ion transport equation. Hydrogen Transport on page 692
`HydrogenMolecule`	`1e10` cm^{-3} `1e-3` Hydrogen molecule transport equation. Hydrogen Transport on page 692
`HydrogenSpeciesA`	`1e10` cm^{-3} `1e-3` Hydrogen species A transport equation. Hydrogen Transport on page 692

Table 356 Equations (Continued)

HydrogenSpeciesB	1e10 cm^{-3} 1e-3 Hydrogen species B transport equation. Hydrogen Transport on page 692
HydrogenSpeciesC	1e10 cm^{-3} 1e-3 Hydrogen species C transport equation. Hydrogen Transport on page 692
LLG	Landau–Lifshitz–Gilbert equation. Chapter 30, Ferromagnetism and Spin Transport on page 1022
Mechanics	Mechanical stress equation. Mechanics Solver on page 1119
Poisson	0.0258 V 1e-3 Poisson equation. Electrostatic Potential on page 255
SingletExciton	Singlet exciton equation. Singlet Exciton Equation on page 325
TCircuit	Thermal circuit equations.
TContact	Thermal contact equations.
Temperature	300 K 1e-3 Temperature equation. Chapter 9, Lattice and Carrier Temperature Equations on page 274
Traps	1e-5 1 1e-3 Explicit trap equation. Trap-to-Trap Tunneling (DiscreteTrapT2T) on page 629

Table 357 Goal{} Quasistationary Ramps

Charge	=<float>	C Target charge for contact.
Contact	=<string1>.<string2>	Name of instance and contact to be ramped.
Current	=<float>	$A\ \mu\ m^{d-3}$ Target current for contact.
Device	=<string>	Name of the device where the parameter will be ramped.
DopingWell	(<vector>)	Semiconductor well defined by point <vector> where quasi-Fermi potential will be ramped.
DopingWells	(Material=<string>)	Semiconductor wells in material <string> where quasi-Fermi potential will be ramped.
	(Region=<string>)	Semiconductor wells in the region <string> where quasi-Fermi potential will be ramped.
	(Semiconductor)	All device semiconductor wells where Fermi potential will be ramped.

Table 357 Goal{} Quasistationary Ramps (Continued)

eQuasiFermi	=<float>	V Target electron quasi-Fermi potential for semiconductor wells. Ramping Quasi-Fermi Potentials in Doping Wells on page 135
hQuasiFermi	=<float>	V Target hole quasi-Fermi potential for semiconductor wells. Ramping Quasi-Fermi Potentials in Doping Wells on page 135
Material	=<string>	Name of material where the parameter will be ramped.
MaterialInterface	=<string>	Name of material interface where parameter will be ramped.
Model	=<string>	Name of model for which the parameter will be ramped.
ModelParameter	=<string>	Parameter path or name of parameter that is ramped when using Parameter Ramping on page 772.
Name	=<string>	Name of contact to be ramped.
	=Regexp(<string>)	Regular expression for matching contacts.
Node	=<string>	Name of node for which the voltage will be ramped.
Parameter	=<string>	Name of parameter that will be ramped.
	=<string1>.<string2>	Name of instance and parameter that will be ramped. This syntax only applies to circuit parameters, but not device parameters.
Power	=<float>	Target heat for contact.
Region	=<string>	Name of region where the parameter will be ramped.
RegionInterface	=<string>	Name of region interface where parameter will be ramped.
Temperature	=<float>	K Target temperature for contact.
Value	=<float>	Target value for parameter.
Voltage	=<float>	V Target voltage for node or contact.
WellContactName	=<string>	Name of contact defining the well where quasi-Fermi potential will be ramped.

Table 358 Goal{} Transient Ramps

Charge	=<float>	C Target charge for contact.
Contact	=<string1>.<string2>	Name of instance and contact to be ramped.
Current	=<float>	$A\ \mu\ m^{d-3}$ Target current for contact.
Name	=<string>	Name of contact to be ramped.
	=Regexp(<string>)	Regular expression for matching contacts.
Voltage	=<float>	V Target voltage for node or contact.

Table 359 HBCoupled() Harmonic Balance

Table 354		Coupled() parameters.
CNormPrint		+ Print instance equation errors per Newton step (MDFT and SDFT modes).
Derivative		– Use complete Jacobian for HB Newton.
ForceCloseToSolution		– Force the application of the Derivative flag. Convergence Parameters on page 170
GMRES		Use GMRES linear solver; requires Method=ILS.
(	MaxIterations=<int>	200 Maximal number of iterations.
	Restart=<int>	20 Size of minimization subspace.
	Tolerance=<float>)	1e-4 Required residuum reduction.
Initialize	=DCMode	0-th harmonic from DC solution; others, zero.
	=HBMode	All harmonics from previous harmonic balance (HB) solution.
	=MixedMode	0-th harmonic from DC; others, from previous HB solution.
Method	=ILS	Required for GMRES.
	=Pardiso	Use PARDISO to solve linear system (SDFT mode only).

Feedback

Table 359 HBCoupled() Harmonic Balance (Continued)

Name	=<string>	Name component of plot files. Default is Coupled_<number>, where <number> is the global index of all Coupled, ACCoupled, and HBCoupled solve entries.
NewtonConvergence	=<string>	Select named NewtonConvergence statement.
NonlinearSolver	=<string>	Select named NonlinearSolver statement.
RhsScale	(Table 356)=<float>	Scaling of RHS in Newton (MDFT and SDFT modes).
SolveSpectrum	=<string>	Referenced solve spectrum (MDFT mode only).
Tone		Multiple specification for multitone (MDFT mode only).
(	Frequency=<freqspec>	! Hz Base frequency. Performing Harmonic Balance Analysis on page 169
	NumberOfHarmonics=<int>)	! Number of harmonics H. Equation 25 on page 168
UpdateScale	(Table 356)=<float>	Scaling of update in Newton (MDFT and SDFT modes).
ValueMin	(Table 356)=<float>	Lower bound for quantity in time domain (MDFT mode only).
ValueVariation	(Table 356)=<float>	Allowed variation of quantity in time domain (MDFT mode only).

Table 360 Load(), Plot(), PlotCarrierPath(), Save(), SpectralPlot() in Solve{} When to Plot

Current	(Difference=<float>)	A Only for Continuation simulations. Write save or plot files when the difference between the actual and previous plotting current values on the continuation contact is greater than the specified Difference.
	(Intervals=<int>)	A Only for Continuation simulations. Divide the range specified with MinCurrent and MaxCurrent of Continuation into <int> intervals, and write save or plot files every time the current enters one of these intervals. When LogCurrent is specified in Continuation section, logarithmic (asinh(10^{100} current)) current range is used. Write save or plot file when the current enters a new logarithmic interval.

Table 360 Load(), Plot(), PlotCarrierPath(), Save(), SpectralPlot() in Solve{} When to Plot (Continued)

	(LogDifference=<(0,)>)	A Only for Continuation simulations. LogCurrent must be specified in Continuation section. Write save or plot files when the difference between the actual and previous logarithmic plotting current values is greater than the specified Difference.
Explicit		– Write datasets specified in Plot section only. Explicit is ignored by SpectralPlot and PlotCarrierPath.
FilePrefix	=<string>	Prefix of file name. File names consist of the prefix, the instance name, an optional local number (depending on Overwrite and noOverwrite), and an extension. The default file prefix is save<globalsaveindex> for Save and plot<globalplotindex> for Plot.
Iterations	=(<int>;...)	(0;1;...) Save or plot at certain Plugin iterations. Iterations is ignored by PlotCarrierPath.
IterationStep	=<int>	Save or plot all <int> steps of plugins, quasistationaries, continuations, or transients.
Loadable		+ Write additional information required to load a simulation from a .tdr file. Loadable is not supported for SpectralPlot and PlotCarrierPath.
noOverwrite		off Give each new save or plot file a new name by numbering.
Number	=<int>	Number of the solution to be retrieved by Load.
Overwrite		on Rewrite the same file name at each loop.
Time=(	<float>	Time point for plot. When to Plot on page 197
	decade	Use logarithmic range subdivision. When to Plot on page 197
	intervals=<int>	Number of subdivisions. When to Plot on page 197
	range=(<float>*2); ...)	Plotting range. When to Plot on page 197
Voltage	(Difference=<float>)	V Only for Continuation simulations. Write save or plot files when the difference between current and previous plotting voltages is greater than the specified Difference.

Feedback

Table 360 Load(), Plot(), PlotCarrierPath(), Save(), SpectralPlot() in Solve{} When to Plot (Continued)

	(Intervals=<int>)	Only for Continuation simulations. Divide the range specified with MinVoltage and MaxVoltage of Continuation into <int> intervals, and write save or plot files every time the voltage enters one of these intervals.
When		Generate output whenever the target was crossed the between the previous and the current iteration i, $X_{i-1} < X_T \leq X_i$ or $X_{i-1} > X_T \geq X_i$. Available for Plot, SpectralPlot, Save, and CurrentPlot inside a Quasistationary, Transient, or Continuation.
(	Contact=[<string>.] <string>	[Instance and] contact name for target. The instance name defaults to " ", appropriate for single-device simulation.
	Current=<float>	A Target current X_T.
	Node=<string>	Node name for target.
	Voltage=<float>)	V Target voltage X_T.

Table 361 MSConfigs() in Set() in Solve{} Manipulating MSCs During Solve

Frozen		+ Freeze MSCs.
MSConfig		Set occupations for the specified MSC.
(	Device=<string>	Device instance name to where the MSC belongs.
	Name=<string>	Name of MSC.
	State(Name=<string> Value=<float>)...)	Set occupation of MSC state <string> to given value.

Table 362 Plugin() Plugin Command

BreakOnFailure		Stop when an inner Coupled fails.
Digits	=<float>	Relative precision target.
Iterations	=<int>	Maximum number of iterations.0 is used to perform one loop without error testing.

Table 363 Quasistationary() Quasistationary Ramps

AcceptNewtonParameter		Apply relaxed Newton parameters. Relaxed Newton Method on page 223
(	ReferenceStep= <float>)	1.e-9 Apply relaxed Newton parameters for step size < ReferenceStep.
BreakCriteria{	Table 220}	Break criteria. Break Criteria: Conditionally Stopping the Simulation on page 130
Decrement	=<float>	2 Divisor for the step size when last step failed.
DoZero		+| – The equations are solved for $t = 0$.
Extraction{	Table 22}	List of voltage-dependent extraction curves. Extraction File on page 205
Extrapolate	(Table 225)	– Use extrapolation. Extrapolation on page 143
Goal{	Table 357}	Goal for ramping.
Increment	=<float>	2 Multiplier for the step size when last step was successful.
InitialStep	=<float>	0.1 Initial step size.
MaxStep	=<float>	1 Maximum step size.
MinStep	=<float>	0.001 Minimum step size.
NewtonPlotStep	=<float>	Upper limit for step size for which to write Newton plot files. NewtonPlot on page 235
NonlocalPath	(Table 236)	Dynamic Nonlocal Path Band-to-Band Tunneling Model on page 592, Handling Derivatives on page 599
Plot{	Intervals=<int>	! Number of intervals in Range. Saving and Plotting During a Quasistationary on page 143
	Range= (<float>*2)}	! Range of t for which to generate plots.
PlotBandstructure	as Plot	Plot band structure data in an LED simulation.
PlotCarrierPath	as Plot	Plot carrier paths for Saddle Point Analysis in CIS Devices
PlotGain	as Plot	Plot stimulated and spontaneous emission data in LED simulation.
PlotLEDRadiation	as Plot	Optical far field of LED.

Table 363 Quasistationary() Quasistationary Ramps (Continued)

ReadExtrapolation		+ Try to use the extrapolation information from a previous Quasistationary if it is available and compatible. Extrapolation on page 143
SaveOptField	as Plot	Save optical field data in an LED simulation.
SpectralPlot	as Plot	Plot spectral simulation results. Illumination Spectrum on page 733
StoreExtrapolation		+ Store the extrapolation information internally at the end of the Quasistationary. Extrapolation on page 143

Note:

Multiple entries from Table 364 must be specified in multiple Set() definitions, because Set() only takes a single option.

Table 364 Set() in Solve{}

<ident>	=<float>	Set node. Mixed-Mode Electrical Boundary Conditions on page 128
<ident>.<string>	=<float>	Set parameter <string> of compact circuit instance <ident>. Mixed-Mode Electrical Boundary Conditions on page 128
<ident>	mode Charge	Use charge boundary condition for contact <ident>. Changing Boundary Condition Type During Simulation on page 127
<ident>	mode Current	Use current boundary condition for contact <ident>. Changing Boundary Condition Type During Simulation on page 127
<ident>	mode Voltage	Use voltage boundary condition for contact <ident>. Changing Boundary Condition Type During Simulation on page 127
MSConfigs(	Table 361)	Set MSCs. Manipulating MSCs During Solve on page 669
(PreFactor	=<float> [Device=<string>] [MSConfig=<string>] [Transition=<string>])	Set MSC transition prefactors for emission, capture, or both using EPreFactor, CPreFactor, or PreFactor, respectively. Manipulating Transition Dynamics on page 670

Table 364 Set() in Solve{} (Continued)

eSHEDistributions(	Frozen)	Freeze or unfreeze (-Frozen) the electron distribution function. Using Spherical Harmonics Expansion Method on page 951
hSHEDistribution(	Frozen)	Freeze or unfreeze (-Frozen) the hole distribution function. Using Spherical Harmonics Expansion Method on page 951
HydrogenAtom	=<string>	Set the concentration of hydrogen atom. Hydrogen Transport on page 692
HydrogenIon	=<string>	Set the concentration of hydrogen ion. Hydrogen Transport on page 692
HydrogenMolecule	=<string>	Set the concentration of hydrogen molecule. Hydrogen Transport on page 692
HydrogenSpeciesA	=<string>	Set the concentration of hydrogen atom. Hydrogen Transport on page 692
HydrogenSpeciesB	=<string>	Set the concentration of hydrogen atom. Hydrogen Transport on page 692
HydrogenSpeciesC	=<string>	Set the concentration of hydrogen atom. Hydrogen Transport on page 692
TrapFilling	=Table 369	Set trap filling. Explicit Trap Occupation on page 616
Traps(	Table 369)	Set traps. Explicit Trap Occupation on page 616

Table 365 Standalone

Optics	Optical problem.
Wavelength	Update wavelength according to optical problem.

Table 366 Time conditions Time-Stepping, When to Write to the Current File, When to Plot

<float>	s A time point.
Range=(<float>*2)	Interval on time axis.

Feedback

Table 366 Time conditions Time-Stepping, When to Write to the Current File, When to Plot (Continued)

`Range=(<float>*2) Intervals=<int> [Decade | Linear*]`	Subdivided interval on time axis. Subdivision on logarithmic or linear scale.

Table 367 Transient() Transient Command

`Table 242`		Time step control.
`AcceptNewtonParameter`		Apply relaxed Newton parameters. Relaxed Newton Method on page 223
`(`	`ReferenceStep=<float>)`	`1.e-9` s Apply relaxed Newton parameters for step size < `ReferenceStep`.
`Bias{`	`Table 370}`	Ramping physical parameters. Ramping Physical Parameter Values
`BreakCriteria{`	`Table 220}`	Break criteria. Break Criteria: Conditionally Stopping the Simulation on page 130
`Cyclic(`	`Table 355)`	Cyclic analysis. Large-Signal Cyclic Analysis on page 158
`Decrement`	`=<float>`	`2` Divisor for the step size when last step failed.
`Extrapolate`	`(Table 225)`	– Use extrapolation. Extrapolation on page 143
`FinalTime`	`=<float>`	s Final time.
`Goal{`	`Table 358}`	Goal for ramping.
`Increment`	`=<float>`	`2` Multiplier for the step size when last step was successful.
`InitialStep`	`=<float>`	`0.1` s Initial step size.
`InitialTime`	`=<float>`	`0` s Start time.
`MaxStep`	`=<float>`	`1` s Maximum step size.
`MinStep`	`=<float>`	`0.001` s Minimum step size.
`NewtonPlotStep`	`=<float>`	Upper limit for step size for which to write Newton plot files. NewtonPlot on page 235
`NonlocalPath`	`(Table 236)`	Dynamic Nonlocal Path Band-to-Band Tunneling Model on page 592, Handling Derivatives on page 599

Feedback

Table 367 Transient() Transient Command (Continued)

`Plot{`	`Intervals=<int>`	Number of intervals in `Range`.
	`Range=(<float>*2)}`	s Range of t for which to generate plots. Saving and Plotting During a Quasistationary on page 143
`PlotCarrierPath`	as `Plot`	Plot carrier paths for Saddle Point Analysis in CIS Devices
`ReadExtrapolation`		+ Try to use the extrapolation information from a previous Transient if it is available and compatible. Extrapolation on page 143
`SpectralPlot`	as `Plot`	Plot spectral simulation results. Illumination Spectrum on page 733
`StoreExtrapolation`		+ Store the extrapolation information internally at the end of the `Transient`. Extrapolation on page 143
`Transient`	`=BE`	Backward Euler method. Backward Euler Method on page 1340
	`=TRBDF`	* TRBDF method. TRBDF Composite Method on page 1341
`TurningPoints(`	`(<float1> <float2>)...`	Time point `<float1>` and associated advancing time-step limit `<float2>`.
	`(Condition(Time(` Table 366 `[; Table 366]...)) Value=<float> Factor=,int>)... )`	Time point conditions and associated advancing time-step limit `Value`.

Table 368 TrapFilling= in Set() in Solve {} Explicit Trap Occupation

`0`	Set trap occupation to be in equilibrium with zero electron and hole concentration.
`-Degradation`	Return trap concentrations to initial values. Device Lifetime and Simulation on page 680
`Empty`	(deprecated) Set all traps to empty.
`Frozen`	Keep the current trap occupation unchanged until the next `Set` or `UnSet`.
`Full`	(deprecated) Set all traps to fully occupied.

Feedback

Table 368 TrapFilling= in Set() in Solve {} Explicit Trap Occupation (Continued)

n	(deprecated) Set trap occupation to be in equilibrium with a very high electron and zero hole concentration.
p	(deprecated) Set trap occupation to be in equilibrium with a very high hole and zero electron concentration.

Table 369 Traps() in Set() in Solve {} Explicit Trap Occupation

[<string1>.]<string2>=<float>...	Set occupation of trap <string2> of device <string1> to specified value.
[<string1>.]value=<float>...	Set occupation of traps of device <string1> to specified value.
Frozen	+ Freeze traps.

Table 370 Bias{}

Device	=<device-name>	Name of device for which parameter is ramped.
Material	=<string>	Name of a material that must exist in the device.
MaterialInterface	=<string>	Material interface of the form <ident1>/<ident2>, with <ident1> and <ident2> as the names of materials that must exist in the device and must have a common interface.
Region	=<string>	Name of a region that must exist in the device.
RegionInterface	=<string>	Region interface of the form <ident1>/<ident2>, with <ident1> and <ident2> as the names of regions that must exist in the device and must have a common interface.
Model	=<string>	! Name of a model that must be specified for the device.
Parameter	=<string>	! Name of the parameter to be ramped for the specified model.
Value	=(<float>[<value unit>] at <float><time unit>, ...)	! List of parameter values at given times. Note, square brackets are required for value units.

Feedback

Table 370 Bias{} (Continued)

	=(<float><time unit> <float>[<value unit>])	

System

Table 371 System{} System Section

<device-type>	<instance-name> (<string>=<node>...) [{Table 212}]	Create device instance <instance-name> from device <device-type> and connect contact <string> to node <node>.
<parameter-set-name>	<instance-name> (<node>...) {<param>=<pvalue>...}	Create circuit instance <instance-name> from parameter set <parameter-set-name>, connect terminals to nodes <node>, and override parameter <param> by <pvalue>, the type of which depends on the parameter. Table 9 Table 10 Table 11
<device-type>	<instance-name> (OpticalDevice=["<optdev1>"...])	Create device instance <instance-name> from device <device-type>, and connect instance to optical device instances <optdev1>.
ACPlot(	Table 372)	Define circuit quantities for AC analysis output. Modifying Plot Output on page 119
Electrical	(<ident>|Regexp(<string>)...)	List of nodes or regular expressions for matching nodes to be declared as electrical. System Section on page 112
HBPlot	[<string>](Table 372)	Define circuit quantities for HB analysis output. Harmonic Balance on page 167
Hint(	<ident>=<float>)	Set node only for the first solve. Initializing Nodes on page 117
Initialize(	<ident>=<float>)	Set node until first transient. Initializing Nodes on page 117.
Netlist	=<string>	PrimeSim HSPICE netlist file. PrimeSim HSPICE Netlist Files on page 103
Plot	[<string>] (Table 372)	Plot circuit quantities to file named <string>.
Set(	<ident>=<float>	Set node. Initializing Nodes on page 117

Table 371 System{} System Section (Continued)

	<ident>.<string>=<float>)	Set parameter <string> of compact circuit instance <ident>.
SystemTopology (	Print)	Provide description and analysis of system. Additional Remarks on page 172
Thermal	(<ident>|Regexp(<string>)...)	List of nodes or regular expressions for matching nodes to be declared as thermal. System Section on page 112
Unset(	<ident>)	Unset node. Initializing Nodes on page 117

Table 372 Plot(), ACPlot(), and HBPlot() in System{} Plotting QuantitiesModifying Plot OutputHarmonic Balance

<ident>		Print the voltage at node <ident>.
freq	()	Print the current AC analysis frequency.
h	(<ident1> <ident2>)	Print the heat that exits device <ident1> through node <ident2>.
i	(<ident1> <ident2>)	Print the current that exits device <ident1> through node <ident2>.
p	(<ident1> <ident2>)	Print attribute <ident2> of circuit element <ident1>.
t	(<ident>)	Print the temperature at node <ident>.
	(<ident1> <ident2>)	Print the temperature difference between two given nodes.
time	()	Print the current time in transient analysis, or quasistationary t parameter for frequency-domain analysis.
v	(<ident>)	Print the voltage at node <ident>.
	(<ident1> <ident2>)	Print the voltage difference between two given nodes.

TensorPlot

Table 373 TensorPlot(){} Visualizing Results on Native Tensor Grid

Name	=<string>	! Name of tensor plot. The file name for the TensorPlot given in the File section is appended by this name.

Feedback

Table 373 TensorPlot(){} Visualizing Results on Native Tensor Grid (Continued)

OutputLevel	=maximum	For bidirectional BPM, tensor plots are generated not only for the final solution, but also after every iteration if maximum is specified.
Xconst	=<float>	µm X-coordinate.
Xmax	=<float>	µm Maximum x-coordinate.
Xmin	=<float>	µm Minimum x-coordinate.
Yconst	=<float>	µm Y-coordinate.
Ymax	=<float>	µm Maximum y-coordinate.
Ymin	=<float>	µm Minimum y-coordinate.
Zconst	=<float>	µm Z-coordinate.
Zmax	=<float>	µm Maximum z-coordinate.
Zmin	=<float>	µm Minimum z-coordinate.

Thermode

Table 374 Thermode{} Boundary Conditions for Lattice Temperature

AreaFactor	=<float>	1 Multiplier for heat fluxes. Reading a Structure on page 59
Name	=<string>	Name of thermode.
	=Regexp(<string>)	Regular expression for matching structure contacts.
Power	=<float>	W cm^{-2} Heat flux boundary condition.
SurfaceConductance	=<float>	cm^{-2} K^{-1} W Contact thermal conductance.
SurfaceResistance	=<float>	cm^{2} K W^{-1} Contact thermal resistivity.
Temperature	=<float>	K Contact temperature.

Feedback

Various

Table 375 Locations, in Physics/Math section location specifiers

Material	=<string>	Name of a material.
Region	=<string>	Name of a region that must exist in the device.
	=Regexp(<string>)	Regular expression for the matching region.
MaterialInterface	=<string>	Material interface of the form "<ident1>/<ident2>", with <ident1> and <ident2> as the names of materials.
Region Interface	=<string>	Region interface of the form "<ident1>/<ident2>", with <ident1> and <ident2> as the names of regions that must exist in the device and must have a common interface.
	=Regexp(<string>)	Regular expression for matching region interface names. The <string> should be of the form "<regexp1>/<regexp2>", with <regexp1> and <regexp2> as separate regular expressions that are used to match region names.
Electrode	=<string>	Name of an electrode that must exist in the device.

Table 376 Locations, bulk (dimension is that of the device)

Material	=<string>	Name of a material.
Region	=<string>	Name of a region that must exist in the device.

Table 377 Locations, interface (dimension is one less than that of the device)

MaterialInterface	=<string>	Material interface of the form "<ident1>/<ident2>", with <ident1> and <ident2> as the names of materials.
RegionInterface	=<string>	Region interface of the form "<ident1>/<ident2>", with <ident1> and <ident2> as the names of regions that must exist in the device and must have a common interface.

Table 378 Locations, noncontact

Table 376	Bulk location.

Table 378 Locations, noncontact (Continued)

Table 377	Interface location.

Table 379 Optics() in Physics{}

Table 285		Optics standalone.
BPMScalar(	Table 250)	Scalar beam propagation method (BPM) solver. Beam Propagation Method on page 842
TMM(	Table 250)	Transfer matrix method (TMM) solver. Transfer Matrix Method on page 819

Table 380 PMI parameters Command File of Sentaurus Device

<ident>	=<float>	Scalar floating-point parameter.
	=<string>	Scalar string parameter.
	=(<float>...)	Vector of floating-point parameters.
	=(<string>...)	Vector of string parameters.